72-§. Sonlarni tublikka tekshirish algoritmi. va katta tub sonlarni qurish.

$$[n = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot \ldots \cdot p_k^{\alpha_k}]$$

n natural sonning kanonik yoyilmasi

$p_1 < p_2 < p_3 < \ldots < p_k$ — har xil tub son.

$\alpha_1, \alpha_2, \ldots, \alpha_k \in N$.

Natural sonni tublikka tekshirish va katta tub sonlarni qurish kriptografiyada juda muhim h-di.

Bölish usuli
(Метод пробных делений)

Agar n natural soni $2, 3, \ldots, \sqrt{n}$ sonlaridan biriga bölinsa murakkab son, bölinmasa tub son boladi. Qiyinlik darajasi $O(n^{1/2})$

Agar n natural sonini $1+6j$, $5+6j$, $j=1,2,\ldots$ sonlarga bölib chiqib tub yoki tub emaslikka tekshirilsa, yuqoridagi algoritmning qiyinlik darajasi 6 paradi (kamayadi) $O(\cdot)$.

Eratosfen g'alviri (59-sahifaga javob)

(2)	(3)	~~4~~	(5)	~~6~~	(7)	~~8~~	~~9~~	~~10~~	(11)	~~12~~
(13)	~~14~~	~~15~~	~~16~~	(17)	~~18~~	(19)	~~20~~	~~21~~	~~22~~	(23)
~~24~~	~~25~~	~~26~~	~~27~~	~~28~~	(29)	~~30~~	(31)	~~32~~	~~33~~	~~34~~
(35)	~~36~~	(37)	~~38~~	~~39~~	~~40~~	(41)	~~42~~	(43)	~~44~~	~~45~~

Eratosfen g'alviri [1; n] oraliqdagi barcha tub sonlarni aniqlash algoritmi hisoblanadi. Algoritmni bajarish u-n O(n loglogn) ta operatsiya amalga oshiriladi.

Realizatsiya

```
int  n;
vector<char>  prime (n+1, true);
prime[0] = prime[1] = false;
for (int  i=2;  i <=n;  ++i)
    if (prime[i])
        if (i * 1ll * i <= n)
            for (int  j= i*i;  j<=n;  j+=i)
                prime[j] = false;
```

for(int i=2; i*i<=n; ++i)

Xotiradan joy olishi — O(n) (space complexity)
Bajaradigan amallar soni — O(nloglogn) (time complexity)

Complexities of an Algoritm

The complexity of an algorithm computes the amount of time and spaces required by an algorithm for an input of size (n). The complexity of an algorithm can be divided into two types. The time complexity and the space complexity.

Time Complexity of an Algorithm

The time complexity is defined as the process of determining a formula for total time required towards the execution of that algorithm. This calculation is totally independent of implementation and programming language.

Space Complexity of an Algorithm.

Space complexity is defining as the process of defining a formula for prediction of how much memory space is required for the successful execution of the algorithm. The memory space is generally considered as the primary memory.

There are *five types of Time complexity Cases:*

1. Constant Time Complexity - $O(1)$
2. Logarithmic Time Complexity - $O(\log n)$
3. Linear Time Complexity - $O(n)$
4. $O(n \log n)$ Time Complexity.
5. Quadratic Time Complexity - $O(n^2)$

Space complexity using big-O notation

1. $O(1)$ - constant complexity
2. $O(\log n)$ - logarithmic complexity
3. $O(n)$ - linear complexity
4. $O(n \log n)$ - log-linear/quasilinear complexity (also called "linearithmic".
5. $O(n^2)$ - square/polynomial complexity.

4

Fermaning kichik teoremasiga asoslanib sonning tubligini tekshirish

(Тест на основе Малой теоремы Ферма)

10^{30}-10^{40} kabi katta sonlarni tubligini tekshirish u-n bölish usulidan foydalanish optimal usul emas, bunday sonlarni Fermaning kichik teoremasiga asoslanib testlash k-k.

Fermaning kichik teoremasi:

Agar n-tub son ($n \in P$) va $a \in Z$ bölsa ~~hamda $(a, n) = 1$ (a va n sonlari özaro tub bölsa)~~ quyidagi örinli böladi:

$$a^n \equiv a \pmod n;$$

Agar bunda $(a, n) = 1$ (ya'ni **a** va **n** özaro tub bolsa, (EKUB$(a, n) = 1$)), u holda quyidagi örinli böladi:

$$a^{n-1} \equiv 1 \pmod n$$

n sonining tubligini tekshirish u-n qandaydir $a \in Z$ soni olinadi va Fermaning kichik teoremasiga köra $O(\log n)$ ta operatsiya bajariladi. ((mod n) ga köra binar darajaga kötariladi).

Agar Fermaning kichik teoremasi bajarilmasa, demak **n**-murakkab son. Agar teorema bajarilsa n sonini tub deb atashga shoshilmaymiz, chunki Fermaning kichik teoremasi faqat zaruriylik shartini beradi xolos (yetarlilik shartini bermaydi). Fermaning kichik teoremasi murakkab sonlarni topishda samarali hisoblanadi.

100 xonali sonlarni ($n=10^{99}+\delta$, $\delta=1,3,5,7,\dots$) tublikka tekshirishda $13^{n-1}\equiv 1 \pmod n$ testini o'tkazganda dastlabki 10 ta son tub bo'lib chiqqan.

Lekin shunday n murakkab sonlari borki, ular $a\in\mathbb{Z}$ bo'lganda $a^n\equiv a \pmod n$ shartini qanoatlantiradi. Bu sonlar **Karmaykl sonlari** deb ataladi.

Eng kichik Karmaykl soni 561 bo'ladi. ya'ni $a^{561}\equiv a \pmod{561}$.

Shunday qilib, n sonining tubligini tekshirish uchun Fermaning kichik teoremasi yetarli emas.

Ta'rif: $F_k=2^{2^k}+1$, $k=0,1,2,\dots$, sonlari Ferma sonlari deyiladi.

Teorema: $n=F_k$ soni ($k>0$) tub son bo'ladi, qachonki

$$3^{\frac{n-1}{2}}\equiv -1 \pmod n$$

shart bajarilsa.

Ta'rif: Agar p - tub son bo'lsa, u holda $M_p=2^p-1$ soni ham tub son bo'ladi. Ushbu M_p sonlari **Mersen sonlari** deyiladi.

Sonni tublikka tekshirishning $(N\pm 1)$ usuli va katta tub sonlarni qurish.

Teorema: Agar $n\in\mathbb{N}$, $n>1$, n-toq son bo'lsa va $n-1=\prod_{i=1}^{k} p_i^{\alpha_i}$ (ya'ni $n-1$ sonining tub ko'paytuvchilari) ma'lum bo'lsa hamda

har bir $i=1,2,\dots,k$ u-n $a_i \in N$ mavjud bolsinki, unda

$$a_i^{n-1} \equiv 1 \pmod n, \quad a_i^{\frac{n-1}{p_i}} \not\equiv 1 \pmod n$$

sharti bajarilsa, demak **n-tub son**.

Misol u-n: $n=7$. $n-1=6=2^1 \cdot 3^1$

$p_1=2$,
$p_2=3$.

$a_1=5$
$a_2=4$.

$a_1=5 \rightarrow a_1^{n-1} \equiv 1 \pmod n$

$5^{7-1} \equiv 1 \pmod 7$

$5^6=25^3=(21+4)^3=\dots 64=63+1 \rightarrow 5^6 \equiv 1 \pmod 7$ ✓

$a_1^{\frac{n-1}{p_1}} \not\equiv 1 \pmod n$

$5^{\frac{7-1}{2}} \not\equiv 1 \pmod 7$

$5^3 \not\equiv 1 \pmod 7$

$5^3 \equiv 6 \pmod 7$ ✓

$a_2=4 \rightarrow a_2^{n-1} \equiv 1 \pmod n$

$4^{7-1} \equiv 1 \pmod 7$

$4^6=64^2=(63+1)^2 \rightarrow 4^6 \equiv 1 \pmod 7$ ✓

$$a_2^{\frac{n-1}{p_i}} \not\equiv 1 \pmod n.$$

$$4^{\frac{7-1}{3}} \not\equiv 1 \pmod 7$$

$$4^2 \equiv 2 \not\equiv 1 \pmod 7 \quad \checkmark$$

Demak 7 - tub son.

Ta'rif: Lyuka sonlar ketma-ketligi deb shunday $u_0, u_1, u_2, u_3, \dots$ va $v_0, v_1, \dots$ ketma-ketlikka aytiladiki, bunda

$$u_0 = 0, \quad u_1 = 1, \quad v_0 = 2, \quad v_1 = 4$$

$$x_{j+1} = 4x_j - x_{j-1}$$

shartlar orinli boladi.

Teorema: $P, Q \in \mathbb{Z}$, $D = P^2 - 4Q \neq 0$ bolsin. Lyuka sonlar ketma-ketligini quyidagi qonuniyat boyicha aniqlaymiz:

$$u_0 = 0, \quad u_1 = 1, \quad u_{j+2} = Pu_{j+1} - Qu_j \quad (j \geq 0).$$

Agar n - toq natural son bolsa, $n > 1$, $n+1 = \prod_{i=1}^{k} q_i^{\beta_i}$ ($n+1$ sonining tub boluvchisi)

$\left(\frac{D}{n}\right) = -1$. Agar har bir $i = 1, \dots, k$ uchun $P_i, Q_i \in \mathbb{Z}$, $D = P_i^2 - 4Q_i$ topilsa, qaysiki u bilan quyidagi

$$n \mid u_{n+1}^{(i)}, \quad n \nmid u_{(n+1)/q_i}^{(i)}$$

shartni qanoatlantiradigan Lyuka ketma-ketligi $(u_0^{(i)}, u_1^{(i)}, \dots)$ topilsa demak n - tub.

Agar $n \nmid U_{n+1}$ bolsa, u holda n-murakkab.

Yuqoridagi algoritm (teoremalar) bo'yicha katta tub sonlarni qurish amalga oshiriladi.

1-§ Faktorizatsiyaning oddiy algoritmlari.

1. Ferma usuli (bo'sa solga javob)

$$n = A^2 - B^2$$

1) $m = [\sqrt{n}]$

2) $x = 1, 2, \ldots$

$$q(x) = (m+x)^2 - n$$

bu ifoda $q(x)$ to'la kvadratga kelguncha davom ettiriladi.

3) $q(x) = B^2$ bolganda

$A = m + x$

$A^2 - n = B^2$

$n = A^2 - B^2$

$n = (A-B)(A+B)$ (ko'paytuvchilarga ajraladi)

Misol: $n = 19691$

$m = [\sqrt{n}] = 140$

Jadvalning oxirgi qator va oxirgi ustuniga e'tibor bersak: $(140+10)^2 - n = 53^2$

Demak:

$n = 150^2 - 53^2 = 203 \cdot 97$

$19691 = 203 \cdot 97$

x	y	$\sqrt{y}$
1	190	13,78
2	473	21,75
3	758	27,53
4	1046	32,33
5	1334	36,52
6	1625	40,31
7	1918	43,79
8	2213	47,04
9	2510	50,10
10	2809	53

Bu hisoblash 10 ta iteratsiya, har bir iteratsiyada 1 marta darajaga ko'tarish, 1 marta ayirish, 1 marta ildiz olish amallarini bajarmoqda

Ferma usulining samaradorligini baholash.

$n = p \cdot q$ bo'lsa, q 1 ga, p n ga yaqin bo'lganda, usul samaradorligi eng yomon bo'ladi, shu holda iteratsiyalar soni:

$$Iter(n) = \frac{p+q}{2} - \lfloor n^{1/2} \rfloor \approx \frac{n}{2} - \lfloor n^{1/2} \rfloor$$

ya'ni $O(n)$ - time complexity

Ferma usuli bo'lish usulidan yaxshi ishlashi uchun $Iter(n) < n^{1/2}$ va katta bo'luvchi $p < 4n^{1/2}$ bo'lishi k-k.

Ferma usuli bo'lish usuliga o'xshab (метод пробных делений) eksponensial bahoga ega va katta sonlar u-n samarasiz hisoblanadi.

Ferma usulini samarasini oshirish u-n n sonini 2 dan qandaydir konstanta B sonigacha bo'lish usulidan foydalanib bo'lish k-k, keyin Ferma usuli b-n qidirish k-k.

2. Pollardning $(p-1)$ usuli.

Bu usul 1974-yilda Jon Pollard (ingliz matematigi tomonidan ishlab chiqilgan.

Agar p va q sonlari tub, $N=p\cdot q$ bolsa, N sonini tub boluvchilarga ajratish (factoring) haqida fikr yuritamiz.

Agar a va p sonlari ozaro tub bolsa, Fermaning kichik teoremasiga kora:

$$a^{p-1} \equiv 1 \pmod p$$

Agar L soni $p-1$ ga karrali bolsa, ya'ni $L=(p-1)\cdot k$ bolsa, u holda quyidagi tenglik orinli boladi:

$$a^L = a^{(p-1)\cdot k}$$

$$a^{(p-1)k} \equiv 1^k = 1 \pmod p$$

demak: $a^L \equiv 1 \pmod p$.

Demak $a^L - 1$ soni p ga bolinadi, oz navbatida N soni ham p ga karrali. Demak $EKUB(a^L-1, N)$ soni p ga bolinadi.

Pollard $(p-1)$ usulining ketma-ketligi:

N sonini faktorlarga ajratish u-n N soniga ozaro tub bolgan a soni tanlanadi (yoki $a \in P$ bolishi ham yetarli)

- $k = 1, 2, 3, \ldots$ uchun $a^{k!}$ h-di.
- $EKUB\,((a^{k!}-1) \bmod N,\ N)$ topiladi.
- 1 ga teng bolmagan EKUB N sonining faktori boladi.

Misol. 1) $N = 1403$

$a = 2$

$k = 2, 3, 4$.

$k=2 \rightarrow 2^{2!} \equiv 4 \pmod{1403}$

$EkuB((2^{2!}-1) \bmod (1403), 1403) = 1$ (demak davom etamiz)

$\bmod (1403) = 3$

$k=3 \rightarrow 2^{3!} \equiv 64 \pmod{1403}$

$EkuB(63, 1403) = 1$

$k=4 \rightarrow 2^{4!} \equiv 142 \pmod{1403}$

$EkuB(141, 1403) = 1$.

$k=5 \rightarrow 2^{5!} \equiv 794 \pmod{1403}$

$EkuB(794, 1403) = 61$

Demak $\rightarrow$ $1403 = 61 \times 23$

Misol. 2) $N = 2993$

$a = 2$, $k = 2, 3, 4, \ldots$

$k=2 \rightarrow 2^{2!} \equiv 4 \pmod{2993}$

$EkuB(3, 2993) = 1$.

$k=3 \rightarrow 2^{3!} \equiv 64 \pmod{2993}$

$EkuB(63, 2993) = 1$

$k=4 \rightarrow 2^{4!} \equiv 1451 \pmod{2993}$

$EkuB(1450, 2993) = 1$

$k=5 \rightarrow 2^{5!} \equiv 1395 \pmod{2993}$

$EkuB(1394, 2993) = 41$

Demak $2993 = 41 \times 73$

Pollardning $(p-1)$ usuli $O(\sqrt{n}\log^2 n)$ qiyinlik darajasida ishlaydi. Bu usul n sonining katta bo'lmagan tub bo'luvchilarini topishda samarali h-di, faktorizatsiyaning eksponensial algoritmlaridan biri h-di.

Faktorizatsiyaning eksponensial algoritmlari

1. Bo'lish usuli (Перебор возможных делителей)

$O(\sqrt{n}\log^2 n)$ yoki $O(\sqrt{n}\log n)$

Faktorizatsiyaning eng oddiy usullaridan biri. $2, \dots, \sqrt{n}$ sonigacha bo'lib chiqishga asoslangan. $[2, \sqrt{n}]$ oraliqdagi tub sonlarga bo'lib chiqsa ham yetarli, lekin shu oraliqdagi tub sonlar to'plami tayyor bo'lishi k-k.

Odatda bu usul 2^{16} gacha bo'lgan sonlar uchun qo'llaniladi. Undan katta sonlar u-n samarasiz h-di.

2. Pollardning ρ-algoritmi

$O(n^{1/4})$

~~[Lenstra algoritmi $O(n^{1/3}\cdot\log^2 n)$]~~

Algoritm ketma-ketligi:

1-qadam: Katta bo'lmagan x_0 soni tanlanadi va $x_{k+1} \equiv x_k^2 - 1 \pmod n$, $k = 0, 1, 2, \dots$

ketma-ketlik tuziladi.

2-qadam: Har qadamda $d = EKUB(n, |x_i - x_j|)$ $(j<i)$ topiladi.

3-qadam: Qachonki $d = EKUB(n, |x_i - x_j|) \neq 1$ bo'lsa hisoblash tugatiladi. Topilgan d soni n ning bo'luvchisi bo'ladi. Agar n/d soni tub bo'lmasa n soni n/d bilan almashtiriladi va algoritm yana qo'llaniladi.

3. Lenstra algoritmi: $O(n^{1/3}\log^2 n)$

4. Pollard-Shtrassen algoritmi: $O(n^{1/4}\log^4 n)$

5. Shenksning kvadrat shakl metodi: $O(n^{1/4+\varepsilon})$

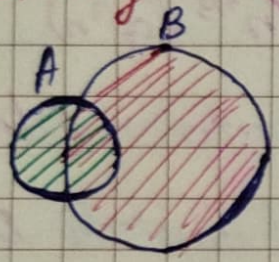
6. Pollardning P-1 algoritmi: $O(n^{1/2}\log^6 n)$

7. Leman usuli: $O(n^{1/3})$

2-savol: Toplamlar va ular ustida amallar

Toplam bu – obyektlar, elementlar yigʻindisidir.

① Toplamlar birlashmasi: Har qanday 2 ta toplamning barcha elementlaridan, ularni takrorlamasdan tuzilgan toplamga shu toplamlarning birlashmasi (yoki yigʻindisi) d-di.



→ $A \cup B$

1-misol: $A=\{a, b\}$, $B=\{a, b, c\}$, $C=\{e, f, k\}$

$E = A \cup B = \{a, b, c\}$

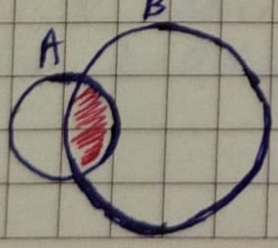
$E \cup C = \{a, b, c, e, f, k\}$

$C \cup B = \{a, b, c, e, f, k\}$

$A \cup C = \{a, b, e, f, k\}$

2-misol: $N \cup R = R$

② Toplamlar kesishmasi. Har qanday 2 ta toplamning barcha umumiy elementlaridan tuzilgan toplamga toplamlarning kesishmasi (yoki kopaytmasi) d-di.



– $A \cap B$

1-misol. $A=\{a, b, c\}$, $B=\{a, b, c, d\}$, $C=\{e, f, k\}$

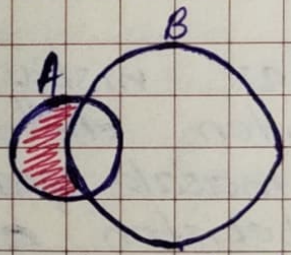
$D = A \cap B = \{a, b, c\}$ $\quad B \cap C = \emptyset$

$D \cap C = \emptyset$ $\quad D \cap B = \{a, b, c\}$

$A \cap C = \emptyset$

2-misol. $N \cap R = N$

③ Toplamlar ayirmasi. Ixtiyoriy A va B toplamlar berilgan bolsin. A toplamning B toplamda bolmagan barcha elementlaridan tuziladigan toplamni hosil qilish A toplamdan B toplamni ayirish deb, tuzilgan toplam esa shu A va B toplamning ayirmasi deb ataladi.



→ $A \setminus B$ yoki $A - B$

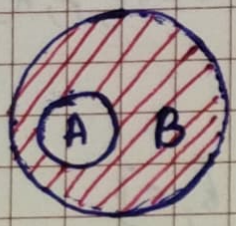
1-misol: $A=\{a, b\}$, $B=\{a, b, c\}$, $C=\{e, f, k\}$

$A \setminus B = \emptyset$, $B \setminus A = \{c\}$, $B \setminus C = \emptyset$

2-misol: $R \setminus N$ – tarkibida natural sonlar qatnashmagan barcha haqiqiy sonlar toplami.

$N \setminus R = \emptyset$

④ Toldiruvchi toplam. Faraz qilaylik, A va B toplamlar berilgan va $A \subseteq B$ bolsin. Bu holda B toplamning A toplamga kirmagan barcha elementlaridan tashkil topgan $B \setminus A$ toplam A toplamning B toplamgacha toldiruvchi toplami deb



$\rightarrow \bar{A}_B$

1-misol. Barcha juft sonlar toplamini $A=\{2, 4, 6, \dots, 2n, \dots\}$ $(n \in N)$ deb belgilasak, A toplamni N toplamgacha to'ldirish amalini qo'llab, $\bar{A}_N=\{1, 3, \dots, 2n-1, \dots\}$ toplamni, ya'ni barcha tog sonlar to'plamini hosil qilamiz.

⑤ Universal toplam va bulean tushunchalari.

Qaralayotgan barcha to'plamlarni o'zida qism toplam sifatida saqlovchi toplamga universal toplam d-di. U deb belgilanadi, universum deb ham atashadi.

Universal toplam tushunchasi nisbiy tushunchadir. Masalan, O'zbekiston sharoitida aholi b-n bog'liq qandaydir masala qaralayotgan bolsa, u holda O'zbekiston aholisi toplamini universal toplam, deb qarash m-n. O'z navbatida O'zbekiston aholisi toplami dunyo aholisi toplamining qism toplamidir.

Universal toplamning 2 ta xosmas qism toplamlari:

1. Universal toplamning o'zi
2. $\emptyset$

Qolgan barcha toplamlari xos qism toplamlardir.

Berilgan A to'plamning barcha qism to'plamlaridan tuzilgan to'plam A to'plamning buleani deb ataladi. 2^A ko'rinishida ifodalanadi.

1-misol: $A=\{a,b,c,d\}$

$2^A=\{\varnothing,\{a\},\{b\},\{c\},\{d\},\{a,b\},\{a,c\},\{a,d\},\{b,c\},\{b,d\},\{c,d\},\{a,b,c\},\{a,b,d\},\{a,c,d\},\{b,c,d\},\{a,b,c,d\}\}$.

$|A|=4 \quad |2^A|=16$.

To'plamlar algebrasi

1-teorema: Universal to'plam U va uning ixtiyoriy qism to'plami A uchun quyidagi tengliklar o'rinli.

1. (Nolning xossalari). $A\cup\varnothing=A$, $\varnothing\cup A=A$, $A\cap\varnothing=\varnothing$, $\varnothing\cap A=\varnothing$

2. (Birning xossalari). $A\cap U=A$, $U\cap A=A$, $A\cup U=U$, $U\cup A=U$

3. (Idempotentlik qonuni). $A\cup A=A$, $A\cap A=A$

4. (Nol va birning bog'liqligi xossasi). $\bar{\varnothing}=U$, $\bar{U}=\varnothing$

5. (Involyutivlik qonuni) $\bar{\bar{A}}=A$

2-teorema (birlashmaga nisbatan kommutativlik qonuni).

$$A\cup B=B\cup A$$

3-teorema (birlashmaga nisbatan assosiativlik qonuni).

$$(A \cup B) \cup C = A \cup (B \cup C)$$

4-teorema (kesishmaga nisbatan kommutativlik qonuni)

$$A \cap B = B \cap A$$

5-teorema. (kesishmaga nisbatan assosiativlik qonuni)

$$(A \cap B) \cap C = A \cap (B \cap C)$$

6-teorema. (birlashmaga nisbatan distributivlik qonuni).

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

7-teorema (kesishmaga nisbatan distributivlik qonuni)

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

8-teorema. Universal to'plamning ixtiyoriy A va B qism to'plamlari u-n.

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

9-teorema: U universal to'plamning ixtiyoriy A va B qism to'plamlari uchun

$$\overline{A \cap B} = \bar{A} \cap \bar{B}$$

8- va 9- teoremalardagi $\overline{A \cap B} = \bar{A} \cup \bar{B}$ va $\overline{A \cup B} = \bar{A} \cap \bar{B}$ tengliklar de-Morgan qonunlari deb ataladi.

10-teorema. $A \cup (A \cap B) = A$

11-teorema. $A \cap (A \cap B) = A$

10- va 11- teoremalarda isbotlangan $A \cup (A \cap B) = A$ va $A \cap (A \cup B) = A$ tengliklar yutilish qonunlari deb ataladi.

12-teorema: Ixtiyoriy A va B to'plamlar uchun quyidagi tasdiqlar ekvivalentdir.

1) $A \subseteq B$ 2) $A \cap B = A$ 3) $A \cup B = B$

5-savol: Determinant va uning xossalari.

$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ → 2-tartibli kvadrat matritsa

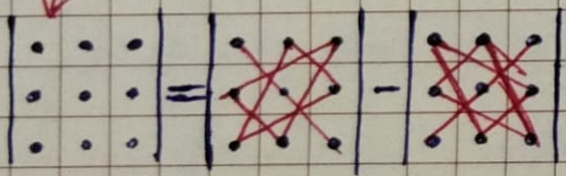
$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11} \cdot a_{22} - a_{12} \cdot a_{21}$ → 2-tartibli determinant

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} a_{22} a_{33} + a_{12} a_{23} a_{31} + a_{21} a_{32} a_{13} - a_{31} a_{22} a_{13} - a_{21} a_{12} a_{33} - a_{32} a_{23} a_{11}$$

↓

3-tartibli determinant.

Uchburchaklar qoidasi



Determinant ~~[illegible]~~ a_{ik} elementining M_{ik} minori deb, shu determinantdan bu element turgan qator va ustunni o'chirish natijasida hosil bo'lgan determinantga aytiladi.

Determinant a_{ik} elementining algebraik to'ldiruvchisi:

$$A_{ik} = (-1)^{i+k} \cdot M_{ik}$$

Determinantning xossalari.

1) Agar determinantning barcha satrlari mos ustunlari o'rin almashtirilsa, uning qiymati o'zgarmaydi.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{vmatrix}$$

2) Agar determinant nollardan iborat qatorga ega bo'lsa, uning qiymati nol bo'ladi.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 0$$

3) Agar determinant 2 ta bir xil parallel qatorga ega bo'lsa, uning qiymati nol bo'ladi.

$$\begin{vmatrix} a_{11} & a_{12} & a_{12} \\ a_{21} & a_{22} & a_{22} \\ a_{31} & a_{32} & a_{32} \end{vmatrix} = 0$$

4) Agar determinant 2 ta parallel qatorining mos elementlari mutanosib bo'lsa (proporsional) uning qiymati nolga teng.

$$\begin{vmatrix} a_{11} & a_{12} & ka_{11} \\ a_{21} & a_{22} & k \cdot a_{21} \\ a_{31} & a_{32} & k \cdot a_{31} \end{vmatrix} = 0$$

qator = ustun yoki satr

5) Biror qator elementlarining umumiy ko'paytuvchisini determinant belgisidan tashqariga chiqarish m-n.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ ka_{21} & ka_{22} & ka_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k \cdot \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k \cdot \Delta$$

6) Agar determinant 2 ta parallel qatorining o'rinlari almashtirilsa, determinant ishorasini qarama-qarshisiga o'zgartiradi.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = - \begin{vmatrix} a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \end{vmatrix}$$

7) Determinantning qiymati biror qator elementlari bilan, shu elementlarga teg-ishli algebraik to'ldiruvchilari ko'paytmalari yig'indisiga teng.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{21} \cdot A_{21} + a_{22} \cdot A_{22} + a_{23} \cdot A_{23}$$

Bu xossa determinantni qator elementlari bo'yicha yoyish d-di.

8) Biror qator elementlari bilan parallel qator mos elementlari algebraik to'ldiruvchilari ko'paytmasining yig'indisi nolga teng.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \Rightarrow a_{11}\cdot A_{21} + a_{12}\cdot A_{22} + a_{13}\cdot A_{23} = 0$$

9) Agar determinant biror qatorining har bir elementi 2 qoshiluvchining yigindisidan iborat bolsa, u holda determinant 2 determinant yigindisiga teng bolib, ularning biri tegishli qator birinchi qoshiluvchilardan, ikkinchisi esa ikkinchi qoshiluvchilardan iborat boladi.

$$\begin{vmatrix} a_{11} & a_{12}+b_1 & a_{13} \\ a_{21} & a_{22}+b_2 & a_{23} \\ a_{31} & a_{32}+b_3 & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}$$

10) Agar determinantning biror qatori elementlariga parallel qatorning mos elementlarini biror ozgarmas songa kopaytirib qoshilsa, determinantning qiymati ozgarmaydi.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31}+\lambda a_{11} & a_{32}+\lambda a_{12} & a_{33}+\lambda a_{13} \end{vmatrix}$$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

$(n\times n)$ ta sondan iborat ushbu jadval n-tartibli kvadrat matritsa deyiladi.

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} = a_{11}A_{11} + a_{12}A_{12} + \dots + a_{1n} \cdot A_{1n}$$

↓

n-tartibli determinant.

Ixtiyoriy tartibli determinantni hisoblashning usullari.

1-usul. Determinant tartibini pasaytirish usuli — determinant biror qatori elementlarining bittasidan boshqasini oldindan nolga aylantirib olib, shu qator bo'yicha yoyish usuli.

$$\Delta = \begin{vmatrix} 3 & -1 & 12 & 8 \\ -5 & 3 & -34 & -23 \\ 1 & 1 & -3 & -7 \\ -9 & 2 & 8 & -15 \end{vmatrix} \Big| \cdot 3, 1, 2 + = \begin{vmatrix} 3 & -1 & 12 & 8 \\ 4 & 0 & 2 & 1 \\ 4 & 0 & 15 & 1 \\ -3 & 0 & 32 & 1 \end{vmatrix} =$$

$$= -(-1)^3 \cdot \begin{vmatrix} 4 & 2 & 1 \\ 4 & 15 & 1 \\ -3 & 32 & 1 \end{vmatrix} \Big| \cdot (-1) + = \begin{vmatrix} -4 & -2 & 1 \\ 0 & 13 & 0 \\ -7 & 30 & 0 \end{vmatrix} = (-1)^{1+3} \cdot \begin{vmatrix} 0 & 13 \\ -7 & 30 \end{vmatrix} =$$

$$= 91$$

2-usul. Determinantni uchburchak ko'rinishga keltirish usuli. — determinantni shunday almashtirishdan iboratki, u = bosh diagonalidan bir tomonida yotuvchi hamma elementlari nolga aylantiriladi. va uchburchaksimon shaklga keltiriladi.

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{vmatrix} = a_{11} \cdot a_{22} \cdot a_{33} \cdot \dots a_{nn}$$

Misol:

$$\Delta = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 5 & 9 \\ 0 & 0 & 3 & 7 \\ -2 & -4 & -6 & 0 \end{vmatrix} \begin{matrix} \\ \\ \\ \cdot 2 \end{matrix} = \begin{matrix} \\ \\ \\ + \end{matrix} = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 5 & 9 \\ 0 & 0 & 3 & 7 \\ 0 & 0 & 0 & 8 \end{vmatrix} =$$

$$= 1 \cdot 2 \cdot 3 \cdot 8 = 48.$$

Misollar: 1) Tartibini pasaytirish.

① $$\Delta = \begin{vmatrix} 1 & -2 & 3 \\ -2 & 1 & -5 \\ 3 & 2 & 7 \end{vmatrix} \begin{matrix} \\ \cdot 2, -3 \\ \\ \end{matrix} = \begin{matrix} \\ \swarrow + \\ \\ \end{matrix} = \begin{vmatrix} 1 & -2 & 3 \\ 0 & -3 & 1 \\ 0 & 8 & -2 \end{vmatrix} = (-1)^{1+1} \cdot \begin{vmatrix} -3 & 1 \\ 8 & -2 \end{vmatrix} =$$

$$= 6 - 8 = -2$$

② $$\Delta = \begin{vmatrix} 2 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 \\ 3 & -1 & 2 & 3 \\ 3 & 1 & 6 & 1 \end{vmatrix} \begin{matrix} \\ \\ \\ \\ \end{matrix} \begin{matrix} \cdot 1, -1, 1, \\ \swarrow \\ \swarrow \\ \swarrow \end{matrix} + = \begin{vmatrix} 2 & -1 & 1 & 0 \\ 2 & 0 & 3 & -1 \\ 1 & 0 & 1 & 3 \\ 5 & 0 & 7 & 1 \end{vmatrix} =$$

$$= (-1) \cdot (-1)^{1+2} \cdot \begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 3 \\ 5 & 7 & 1 \end{vmatrix} \begin{matrix} \\ \\ \\ \end{matrix} \begin{matrix} \cdot 3, 1 \\ \swarrow \\ \swarrow \end{matrix} + = \begin{vmatrix} 2 & 3 & -1 \\ 7 & 10 & 0 \\ 7 & 10 & 0 \end{vmatrix} =$$

$$= (-1) \cdot (-1)^{1+3} \cdot \begin{vmatrix} 7 & 10 \\ 7 & 10 \end{vmatrix} = 0$$

③ $\Delta = \begin{vmatrix} 2 & 4 & -1 & 2 \\ -1 & 2 & 3 & 1 \\ 2 & 5 & 1 & 4 \\ 1 & 2 & 0 & 3 \end{vmatrix} = \begin{matrix} \cdot 3, 1 \\ \\ \\ \\ + \end{matrix} = \begin{vmatrix} 2 & 4 & -1 & 2 \\ 5 & 14 & 0 & 7 \\ 4 & 9 & 0 & 6 \\ 1 & 2 & 0 & 3 \end{vmatrix} =$

$= (-1) \cdot (-1)^{1+3} \cdot \begin{vmatrix} 5 & 14 & 7 \\ 4 & 9 & 6 \\ 1 & 2 & 3 \end{vmatrix} = \begin{matrix} \\ \\ \cdot(-4), (-5) \end{matrix} + = \begin{vmatrix} 0 & 4 & -8 \\ 0 & 1 & -6 \\ 1 & 2 & 3 \end{vmatrix} =$

$= -1 \cdot (-1)^{3+1} \cdot \begin{vmatrix} 4 & -8 \\ 1 & -6 \end{vmatrix} = -(-24+8) = 16$

2) Uchburchak shaklga keltirib hisoblash.

a) $\Delta = \begin{vmatrix} 2 & 3 & -3 & 4 \\ 2 & 1 & -1 & 2 \\ 6 & 2 & 1 & 0 \\ 2 & 3 & 0 & -5 \end{vmatrix} = \begin{matrix} \cdot -1, -3, -1 \\ \\ \\ \\ + \end{matrix} = \begin{vmatrix} 2 & 3 & -3 & 4 \\ 0 & -2 & 2 & -2 \\ 0 & -7 & 10 & -12 \\ 0 & 0 & 3 & -9 \end{vmatrix} \begin{matrix} \cdot(-\frac{7}{2}) + \\ \cdot 2 \\ \\ \end{matrix}$

$= \begin{vmatrix} 2 & 3 & -3 & 4 \\ 0 & -2 & 2 & -2 \\ 0 & 0 & 6 & -10 \\ 0 & 0 & 3 & -9 \end{vmatrix} \begin{matrix} \\ \\ \cdot(-3) + \\ \cdot 2 + \end{matrix} = \begin{vmatrix} 2 & 3 & -3 & 4 \\ 0 & -2 & 2 & -2 \\ 0 & 0 & 6 & -10 \\ 0 & 0 & 3 & -9 \end{vmatrix} \begin{matrix} \\ \\ \\ \cdot(-\frac{1}{2}) + \end{matrix} =$

$= \begin{vmatrix} 2 & 3 & -3 & 4 \\ 0 & -2 & 2 & -2 \\ 0 & 0 & 6 & -10 \\ 0 & 0 & 0 & -4 \end{vmatrix}$

a) $\Delta = \begin{vmatrix} 2 & 3 & -3 & 4 \\ 2 & 1 & -1 & 2 \\ 6 & 2 & 1 & 0 \\ 2 & 3 & 0 & -5 \end{vmatrix} = \begin{matrix} \cdot -1, -3, -1 \\ \\ \\ \\ + \end{matrix} = \begin{vmatrix} 2 & 3 & -3 & 4 \\ 0 & -2 & 2 & -2 \\ 0 & -7 & 10 & -12 \\ 0 & 0 & 3 & -9 \end{vmatrix} \begin{matrix} \cdot(-\frac{7}{2}) + = \\ \\ \\ \\ \end{matrix}$

$$\begin{vmatrix} 2 & 3 & -3 & 4 \\ 0 & -2 & 2 & -2 \\ 0 & 0 & 3 & -5 \\ 0 & 0 & 3 & -9 \end{vmatrix} \begin{matrix} \\ \\ \\ |\cdot(-1) \end{matrix} + = \begin{vmatrix} 2 & 3 & -3 & 4 \\ 0 & -2 & 2 & -2 \\ 0 & 0 & 3 & -5 \\ 0 & 0 & 0 & -4 \end{vmatrix} =$$

$$= 2 \cdot (-2) \cdot 3 \cdot (-4) = 48$$

в) $$\begin{vmatrix} 1 & -2 & 5 & 9 \\ 1 & -1 & 7 & 4 \\ 1 & 3 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{vmatrix} \begin{matrix} |\cdot(-1) \\ \\ \\ \\ \end{matrix} + = \begin{vmatrix} 1 & -2 & 5 & 9 \\ 0 & 1 & 2 & -5 \\ 0 & 5 & -2 & -5 \\ 0 & 4 & -2 & -5 \end{vmatrix} =$$

$$= -\begin{vmatrix} 1 & 9 & 5 & -2 \\ 0 & -5 & 2 & 1 \\ 0 & -5 & -2 & 5 \\ 0 & -5 & -2 & 4 \end{vmatrix} = \begin{matrix} \\ |\cdot(-1) \\ \\ \\ \end{matrix} + = -\begin{vmatrix} 1 & 9 & 5 & -2 \\ 0 & -5 & 2 & 1 \\ 0 & 0 & -4 & 4 \\ 0 & 0 & -4 & 3 \end{vmatrix} = \begin{matrix} \\ \\ |\cdot(-1) \\ \\ \end{matrix} +$$

$$= -\begin{vmatrix} 1 & 9 & 5 & -2 \\ 0 & -5 & 2 & 1 \\ 0 & 0 & -4 & 4 \\ 0 & 0 & 0 & -1 \end{vmatrix} = -1 \cdot (-5) \cdot (-4) \cdot (-1) = +20$$

3) Determinantlarni avval soddalashtirib, keyin hisoblang.

a) $$\Delta = \begin{vmatrix} x^2+a^2 & ax & 1 \\ y^2+a^2 & ay & 1 \\ z^2+a^2 & az & 1 \end{vmatrix} = a \cdot \begin{vmatrix} x^2+a^2 & x & 1 \\ y^2+a^2 & y & 1 \\ z^2+a^2 & z & 1 \end{vmatrix} =$$

$$= a \cdot \left(\begin{vmatrix} x^2 & x & 1 \\ y^2 & y & 1 \\ z^2 & z & 1 \end{vmatrix} \begin{matrix} |\cdot(-1) \\ \\ \\ \end{matrix} + \begin{vmatrix} a^2 & x & 1 \\ a^2 & y & 1 \\ a^2 & z & 1 \end{vmatrix} \right) = a \cdot \left(\begin{vmatrix} x^2-x^2 & x-x & 0 \\ y^2-x^2 & y-x & 0 \\ z^2-x^2 & z-x & 0 \end{vmatrix} + a^2 \begin{vmatrix} 1 & x & 1 \\ 1 & y & 1 \\ 1 & z & 1 \end{vmatrix} \right) =$$

$$= a \cdot \left((-1)^{1+3} \cdot \begin{vmatrix} y^2-x^2 & y-x \\ z^2-x^2 & z-x \end{vmatrix} \right) = a \cdot \left((y^2-x^2)(z-x) - (y-x)(z^2-x^2) \right) =$$

$$= a \cdot (y-x)(z-x)(y+x-z-x) = a \cdot (y-x)(z-x)(y-z)$$

в)
$$\begin{vmatrix} 1 & 2 & 0 & 0 & 0 \\ 3 & 2 & 3 & 0 & 0 \\ 0 & 4 & 3 & 4 & 0 \\ 0 & 0 & 5 & 4 & 5 \\ 0 & 0 & 0 & 6 & 5 \end{vmatrix} \overset{(-3)+}{=} \begin{vmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & -4 & 3 & 0 & 0 \\ 0 & 4 & 3 & 4 & 0 \\ 0 & 0 & 5 & 4 & 5 \\ 0 & 0 & 0 & 6 & 5 \end{vmatrix} \overset{2+}{=}$$

$$= \begin{vmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & -4 & 3 & 0 & 0 \\ 0 & 0 & 6 & 4 & 0 \\ 0 & 0 & 5 & 4 & 5 \\ 0 & 0 & 0 & 6 & 5 \end{vmatrix} \overset{(-\frac{5}{6})+}{=} \begin{vmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & -4 & 3 & 0 & 0 \\ 0 & 0 & 6 & 4 & 0 \\ 0 & 0 & 0 & +\frac{2}{3} & 5 \\ 0 & 0 & 0 & 6 & 5 \end{vmatrix} \overset{4+}{=}$$

$$= \begin{vmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & -4 & 3 & 0 & 0 \\ 0 & 0 & 6 & 4 & 0 \\ 0 & 0 & 0 & \frac{2}{3} & 5 \\ 0 & 0 & 0 & 0 & -40 \end{vmatrix} = 1 \cdot (-4) \cdot 6 \cdot \frac{2}{3} \cdot (-40) = 640$$

c)
$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ a & b-a & c-a & d-a \\ a^2 & b^2-a^2 & c^2-a^2 & d^2-a^2 \\ a^3 & b^3-a^3 & c^3-a^3 & d^3-a^3 \end{vmatrix} =$$

$$= 1 \cdot (-1)^{1+1} \cdot \begin{vmatrix} b-a & c-a & d-a \\ b^2-a^2 & c^2-a^2 & d^2-a^2 \\ b^3-a^3 & c^3-a^3 & d^3-a^3 \end{vmatrix} = (b-a)(c-a)(d-a) \cdot$$

$$\cdot \begin{vmatrix} 1 & 1 & 1 \\ b+a & c+a & d+a \\ \frac{b^3-a^3}{b-a} & \frac{c^3-a^3}{c-a} & \frac{d^3-a^3}{d-a} \end{vmatrix} = (b-a)(c-a)(d-a) \cdot \left((c+a)(d^2+ad+d^2)+\right.$$

$$\left.+(d+a)(a^2+ab+b^2)+(b+a)(a^2+ac+c^2)-\right.$$

$-(c+a)(a^2+ab+b^2)-(b+a)(a^2+ad+d^2)-(d+a)(a^2+ac+c^2)=$

$=(b-a)(c-a)(d-a)\cdot((c+a)(ad-ab+d^2-b^2)+$

$+(b+a)(ac-ad+c^2-d^2)+(d+a)(ab-ac+b^2-c^2))=$

$=(b-a)(c-a)(d-a)(acd+a^2d-abc-a^2b+cd^2+ad^2-b^2c-ab^2+$

$+abc+a^2c-abd-a^2d+bc^2+ac^2-bd^2-ad^2+abd+a^2b-$

$-acd-a^2c+b^2d+ab^2-c^2d-ac^2)=(b-a)(c-d)(d-a)\cdot$

$(cd^2-b^2c+bc^2-bd^2+b^2d-c^2d)=(b-a)(c-d)(d-a)\cdot$

$(cd\cdot(d-c)+b^2(d-c)-b(d-c)(d+c))=(b-a)(c-d)(d-a)\cdot$

$\cdot(d-c)(cd+b^2-bd-bc)=(b-a)(c-d)(d-a)\cdot(d-c)\cdot$

$\cdot(b(b-d)-c(b-d))=(b-a)(c-d)(d-a)(d-c)(b-d)(b-c)$

6-2-savollar. Chiziqli tenglamalar sistemasini yechishning Kramer va Gauss usuli.

Kramer usuli:

2 noma'lumli, 2 ta chiziqli tenglamalar sistemasi

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$$

ning bosh determinanti $\Delta = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \neq 0$

① bolganda, yagona yechimga ega va u Kramer qoidasi bo'yicha quyidagi formulalar b-n h-di.

$$\left[x_1 = \frac{\Delta_{x_1}}{\Delta}, \quad x_2 = \frac{\Delta_{x_2}}{\Delta} \right]$$

bu yerda: $\Delta_{x_1} = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}$, $\Delta_{x_2} = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}$

Agar $\Delta = 0$ va shu bilan birga $\Delta_{x_1}, \Delta_{x_2}$ lardan ayalli bittasi nolga teng bo'lmasa, ② sistema yechimga ega emas.

Agar $\Delta = \Delta_{x_1} = \Delta_{x_2} = 0$ bo'lsa, u holda berilgan sistema cheksiz ko'p yechimga ega. ③

① $\frac{a_{11}}{a_{21}} \neq \frac{a_{12}}{a_{22}} \rightarrow a_{11} \cdot a_{22} - a_{12}a_{21} \neq 0 \rightarrow \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \neq 0 \quad (\Delta \neq 0)$

② $\frac{a_{11}}{a_{21}} = \frac{a_{12}}{a_{22}} \neq \frac{b_1}{b_2} \rightarrow \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = 0 \quad (\Delta = 0)$

$\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix} \neq 0 \quad (\Delta_{x_1} \neq 0)$ $\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix} \neq 0 \quad (\Delta_{x_2} \neq 0)$

③ $\frac{a_{11}}{a_{21}} = \frac{a_{12}}{a_{22}} = \frac{b_1}{b_2} \rightarrow \Delta = \Delta_{x_1} = \Delta_{x_2} = 0$

3 noma'lumli 3 ta chiziqli tenglamalar sistemasi

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases}$$

ning bosh determinanti

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \neq 0$$

bolganda yagona yechimga ega boladi va uning ildizlari kramer qoidasiga ko'ra

$$x_1 = \frac{\Delta x_1}{\Delta}, \quad x_2 = \frac{\Delta x_2}{\Delta}, \quad x_3 = \frac{\Delta x_3}{\Delta}$$

$$\Delta x_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix} \qquad \Delta x_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}$$

$$\Delta x_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$$

Misol:

$$\begin{cases} x_1 - 2x_2 + x_3 = -4 \\ 3x_1 + 2x_2 - x_3 = 8 \\ 2x_1 - 3x_2 + 2x_3 = -6 \end{cases}$$

$$\Delta = \begin{vmatrix} 1 & -2 & 1 \\ 3 & 2 & -1 \\ 2 & -3 & 2 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 1 \\ 0 & 8 & -4 \\ 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 8 & -4 \\ 1 & 0 \end{vmatrix} = 4$$

$\Delta = 4 \neq 0$

$$\Delta_{x_1} = \begin{vmatrix} -4 & -2 & 1 \\ 8 & 2 & -1 \\ -6 & -3 & 2 \end{vmatrix} = \begin{vmatrix} -4 & -2 & 1 \\ 4 & 0 & 0 \\ 2 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 4 & 0 \\ 2 & 1 \end{vmatrix} = 4$$

$$\Delta_{x_2} = \begin{vmatrix} 1 & -4 & 1 \\ 3 & 8 & -1 \\ 2 & -6 & 2 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 1 \\ 0 & 20 & -4 \\ 0 & 2 & 0 \end{vmatrix} = \begin{vmatrix} 20 & -4 \\ 2 & 0 \end{vmatrix} = 8$$

$$\Delta_{x_3} = \begin{vmatrix} 1 & -2 & -4 \\ 3 & 2 & 8 \\ 2 & -3 & -6 \end{vmatrix} = \begin{vmatrix} 1 & -2 & -4 \\ 0 & 8 & 20 \\ 0 & 1 & 2 \end{vmatrix} = \begin{vmatrix} 8 & 20 \\ 1 & 2 \end{vmatrix} = -4$$

$$x_1 = \frac{\Delta x_1}{\Delta} = \frac{4}{4} = 1$$

$$x_2 = \frac{\Delta x_2}{\Delta} = \frac{8}{4} = 2$$

$$x_3 = \frac{\Delta x_3}{\Delta} = \frac{-4}{4} = -1$$

Gauss usuli.

n ta noma'lumli n ta chiziqli tenglamalar sistemasini n ning katta ($n \geq 4$) qiymatlarida Kramer qoidasi b-n yechish bir nechta yuqori tartibli determinantlarni hisoblashni talab qiladi. Shu sababli, bunday sistemalarni Gauss usulidan foydalanib yechgan ma'qul.

Bu usulning mohiyati shundan iboratki, unda noma'lum ketma-ket

yoqotilib, sistema uchburchaksimon shaklga keltiriladi. Agar sistema uchburchaksimon shaklga kelsa, u yagona yechimga ega, boladi va uning noma'lumlari oxirgi tenglamadan boshlab topib boriladi. (Sistema cheksiz ko'p yechimga ega bolsa, noma'lumlar ketma-ket yoqotilgach, u trapetsiyasimon shaklga keladi).

Misol:

$$\begin{cases} x_1 + x_2 + 5x_3 + 2x_4 = 1 \\ x_1 + x_2 + 3x_3 + 4x_4 = -3 \\ 2x_1 + 3x_2 + 11x_3 + 5x_4 = 2 \\ 2x_1 + x_2 + 3x_3 + 2x_4 = -3 \end{cases} \quad \begin{matrix} \cdot(-1) \\ \cdot(-2) \\ \cdot(-2) \end{matrix}$$

$$\begin{cases} x_1 + x_2 + 5x_3 + 2x_4 = 1 \\ -2x_3 + 2x_4 = -4 \\ x_2 + x_3 + x_4 = 0 \\ -x_2 - 7x_3 - 2x_4 = -5 \end{cases}$$

$$\begin{cases} x_1 + x_2 + 5x_3 + 2x_4 = 1 \\ x_2 + x_3 + x_4 = 0 \\ x_2 + 7x_3 + 2x_4 = 5 \\ x_3 - x_4 = 2 \end{cases} \quad \cdot(-1)$$

$$\begin{cases} x_1 + x_2 + 5x_3 + 2x_4 = 1 \\ x_2 + x_3 + x_4 = 0 \\ 6x_3 + x_4 = 5 \\ x_3 - x_4 = 2 \end{cases} \quad \cdot(-6)$$

$$\begin{cases} x_1 + x_2 + 5x_3 + 2x_4 = 1 \\ x_2 + x_3 + x_4 = 0 \\ x_3 - x_4 = 2 \\ 7x_4 = -7 \end{cases} \rightarrow \begin{matrix} x_4 = -1 & x_3 = 1 \\ x_2 = 0 & x_1 = -2 \end{matrix}$$

Misollar:

1) $\begin{cases} 2x_1 + x_2 = 3 \\ 3x_1 + 2x_2 = 4 \end{cases}$

$$\Delta = \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} = 1 \neq 0$$

$$\Delta_{x_1} = \begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix} = 2 \qquad x_1 = \frac{\Delta x_1}{\Delta} = 2$$

$$\Delta_{x_2} = \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} = -1 \qquad x_2 = \frac{\Delta x_2}{\Delta} = -1$$

2) $\begin{cases} 3x_1 + 4x_2 - x_3 = 8 \\ 2x_1 + x_2 + x_3 = 2 \\ 3x_1 - x_2 + 2x_3 = 0 \end{cases}$

$$\Delta = \begin{vmatrix} 3 & 4 & -1 \\ 2 & 1 & 1 \\ 3 & -1 & 2 \end{vmatrix} = \begin{vmatrix} 3 & 4 & -1 \\ 5 & 5 & 0 \\ 9 & 7 & 0 \end{vmatrix} = (-1) \cdot (-1)^{1+3} \cdot \begin{vmatrix} 5 & 5 \\ 9 & 7 \end{vmatrix} = 10 \neq 0$$

$$\Delta_{x_1} = \begin{vmatrix} 8 & 4 & -1 \\ 2 & 1 & 1 \\ 0 & -1 & 2 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -5 \\ 2 & 1 & 1 \\ 0 & -1 & 2 \end{vmatrix} = 2 \cdot (-1)^{2+1} \cdot \begin{vmatrix} 0 & -5 \\ -1 & 2 \end{vmatrix} = 10$$

$$\Delta_{x_2} = \begin{vmatrix} 3 & 8 & -1 \\ 2 & 2 & 1 \\ 3 & 0 & 2 \end{vmatrix} = \begin{vmatrix} -5 & 0 & -5 \\ 2 & 2 & 1 \\ 3 & 0 & 2 \end{vmatrix} = 2 \cdot (-1)^{2+2} \cdot \begin{vmatrix} -5 & -5 \\ 3 & 2 \end{vmatrix} = 10$$

$$\Delta_{x_3} = \begin{vmatrix} 3 & 4 & 8 \\ 2 & 1 & 2 \\ 3 & -1 & 0 \end{vmatrix} = \begin{vmatrix} -5 & 0 & 0 \\ 2 & 1 & 2 \\ 3 & -1 & 0 \end{vmatrix} = (-5) \cdot (-1)^{1+1} \cdot \begin{vmatrix} 1 & 2 \\ -1 & 0 \end{vmatrix} = -10$$

$$x_1 = \frac{\Delta x_1}{\Delta} = \frac{10}{10} = 1$$

$$x_2 = \frac{\Delta x_2}{\Delta} = \frac{10}{10} = 1$$

$$x_3 = \frac{\Delta x_3}{\Delta} = \frac{-10}{10} = -1$$

3) Gauss usuliga oid misol.

$$\begin{cases} x_1 + x_2 - 3x_3 + 2x_4 = 6 \\ x_1 - 2x_2 - x_4 = -6 \\ x_2 + x_3 + 3x_4 = 16 \\ 2x_1 - 3x_2 + 2x_3 = 6 \end{cases} \quad \begin{matrix} \cdot(-1) \\ \cdot(-2) \end{matrix}$$

$$\begin{cases} x_1 + x_2 - 3x_3 + 2x_4 = 6 \\ -3x_2 + 3x_3 - 3x_4 = -12 \\ x_2 + x_3 + 3x_4 = 16 \\ -5x_2 + 8x_3 - 4x_4 = -6 \end{cases} \quad \begin{matrix} \cdot 3 \\ \cdot 5 \end{matrix}$$

$$\begin{cases} x_1 + x_2 - 3x_3 + 2x_4 = 6 \\ x_2 + x_3 + 3x_4 = 16 \\ 6x_3 + 6x_4 = 36 \\ 13x_3 + 11x_4 = 74 \end{cases} \quad /:6 \quad \cdot(-13)$$

$$\begin{cases} x_1 + x_2 - 3x_3 + 2x_4 = 6 \\ x_2 + x_3 + 3x_4 = 16 \\ x_3 + x_4 = 6 \\ -2x_4 = -4 \end{cases} \Rightarrow \begin{matrix} x_4 = 2; \quad x_3 = 4 \\ x_2 = 6, \quad x_1 = 8 \end{matrix}$$

4) $$\begin{cases} 2x_1 + x_3 + x_4 = 9 \\ 3x_2 - 2x_3 + 3x_4 = 12 \\ -2x_1 + x_2 - x_3 + x_4 = 1 \\ 5x_1 + 2x_2 - 3x_4 = -3 \end{cases} \quad \begin{matrix} \cdot 5 \\ \cdot 1 \\ /:2 \end{matrix}$$

$$\begin{cases} 2x_1 + x_3 + x_4 = 9 \\ 3x_2 - 2x_3 + 3x_4 = 12 \\ x_2 + 2x_4 = 10 \\ 4x_2 - 5x_3 - 11x_4 = -51 \end{cases} \Rightarrow \begin{cases} 2x_1 + x_3 + x_4 = 9 \\ x_2 + 2x_4 = 10 \\ 3x_2 - 2x_3 + 3x_4 = 12 \\ 4x_2 - 5x_3 - 11x_4 = -51 \end{cases} \quad \cdot(-3)$$

$$\begin{cases} 2x_1 + x_3 + x_4 = 9 \\ x_2 + 2x_4 = 10 \\ -2x_3 - 3x_4 = -18 \quad /\cdot(-5) \\ -5x_3 - 19x_4 = -91 \quad /\cdot 2 \end{cases} +$$

$$\begin{cases} 2x_1 + 0x_2 + x_3 + x_4 = 9 \\ x_2 + 0x_3 + 2x_4 = 10 \\ -2x_3 - 3x_4 = -18 \\ -23x_4 = -92 \end{cases} \Rightarrow \begin{array}{l} x_4 = 4, \quad x_3 = 3 \\ x_2 = 2, \quad x_1 = 1 \end{array}$$

4-savol. Matritsalar va ular ustida amallar.

Sonlarning m ta satr va n ta qatordan iborat to'g'ri to'rtburchakli jadvali $m \times n$ o'lchamli matritsa deyiladi.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & & a_{mn} \end{pmatrix}$$

$m=1 \rightarrow$ satr matritsa
$n=1 \rightarrow$ ustun matritsa
$m=n \rightarrow$ kvadrat matritsa

n-tartibli determinantni hisoblash.

$$\det A = |A| = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & & a_{nn} \end{vmatrix}$$

$\det A = 0 \rightarrow$ A-maxsus matritsa
$\det A \neq 0 \rightarrow$ A-maxsusmas matritsa

O'lchamlari bir xil $m \times n$ bo'lgan ikki matritsaning barcha mos elementlari o'zaro teng bo'lsa, bu matritsalar teng deyiladi.

Matritsalar yig'indisi.

Bir xil $m \times n$ o'lchamli A va B matritsaning yig'indisi deb o'sha o'lchamli shunday $C = A + B$ matritsaga aytiladiki, uning har bir elementi A va B matritsa-

larning mos elementlari yig'indisidan iborat bo'ladi.

Matritsaning λ songa ko'paytmasi.

m×n o'lchamli A matritsaning λ songa ko'paytmasi deb, o'sha o'lchamdagi $B=\lambda \cdot A$ matritsaga aytiladiki. bu matritsa elementlarini λ ga ko'paytirishdan hosil bo'ladi.

Matritsalar ko'paytmasi.

m×k o'lchamli A matritsaning k×n o'lchamli B matritsaga ko'paytmasi deb, m×n o'lchamli shunday $C=A \cdot B$ matritsaga aytiladiki, uning c_{ij} elementi A matritsaning i-satri elementlarini B matritsaning j-ustunidagi mos elementlariga ko'paytmalari yig'indisiga teng ya'ni

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}$$

Agar $AB=BA$ bo'lsa u holda A va B matritsalar o'rni almashinadigan yoki kommutativ matritsalar d-di.

Misol. $A=\begin{pmatrix} 3 & 1 & -2 \\ 2 & -4 & 5 \end{pmatrix}$ $B=\begin{pmatrix} 2 & -1 \\ -1 & 3 \\ 4 & -5 \end{pmatrix}$ $AB=?$ $BA=?$

AB matritsa 2×2 o'lchamga ega bo'ladi.

$$A \cdot B = \begin{pmatrix} 3 & 1 & -2 \\ 2 & -4 & 5 \end{pmatrix} \cdot \begin{pmatrix} 2 & -1 \\ -1 & 3 \\ 4 & -5 \end{pmatrix} = \begin{pmatrix} 6+(-1)-2\cdot 4 & -3+3+10 \\ 4+4+20 & -2-12-25 \end{pmatrix} =$$

$$= \begin{pmatrix} -3 & 10 \\ 28 & -39 \end{pmatrix}$$

BA matritsa 3×3 o'lchamga ega bo'ladi.

$$BA=\begin{pmatrix}2 & -1\\ -1 & 3\\ 4 & -5\end{pmatrix}\begin{pmatrix}3 & 1 & -2\\ 2 & -4 & 5\end{pmatrix}=\begin{pmatrix}6-2 & 2+4 & -4-5\\ -3+6 & -1-12 & 2+15\\ 12-10 & 4+20 & -8-25\end{pmatrix}=$$

$$=\begin{pmatrix}4 & 6 & -9\\ 3 & -13 & 17\\ 2 & 24 & -33\end{pmatrix}$$

$AB \neq BA$

Teskari matritsa

Agar kvadrat matritsa maxsusmas bo'lsa, u holda $AA^{-1}=A^{-1}A=E$ tenglikni qanoatlantiruvchi yagona A^{-1} matritsa mavjud bo'ladi va u A matritsaga teskari matritsa d-di.

$$A^{-1}=\frac{1}{\det A}\cdot\begin{pmatrix}A_{11} & A_{21} & \dots & A_{n1}\\ A_{12} & A_{22} & \dots & A_{n2}\\ \dots & \dots & \dots & \dots\\ A_{1n} & A_{2n} & \dots & A_{nn}\end{pmatrix}$$

bu yerda A_{ij} – A matritsa determinanti a_{ij} elementining algebraik to'ldiruvchisi.

Misol.

$$A=\begin{pmatrix}1 & 2 & -1\\ 3 & 0 & 2\\ 4 & -2 & 5\end{pmatrix} \qquad A^{-1}=?$$

$$\det A=\begin{vmatrix}1 & 2 & -1\\ 3 & 0 & 2\\ 4 & -2 & 5\end{vmatrix}=0+16+6-(0+30-4)=22-26=-4\neq 0$$

$\det A \neq 0$ demak A-maxsusmas.

$A_{11} = \begin{vmatrix} 0 & 2 \\ -2 & 5 \end{vmatrix} = 4$ $A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} 2 & -1 \\ -2 & 5 \end{vmatrix} = -8$ $A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} 2 & -1 \\ 0 & 2 \end{vmatrix} = 4$

$A_{12} = -\begin{vmatrix} 3 & 2 \\ 4 & 5 \end{vmatrix} = -7$ $A_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 1 & -1 \\ 4 & 5 \end{vmatrix} = 9$ $A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 1 & -1 \\ 3 & 2 \end{vmatrix} = -5$

$A_{13} = (-1)^{1+3} \begin{vmatrix} 3 & 0 \\ 4 & -2 \end{vmatrix} = -6$ $A_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 4 & -2 \end{vmatrix} = 10$ $A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix} = -6$

$$A^{-1} = \frac{1}{-4} \cdot \begin{pmatrix} 4 & -8 & 4 \\ -7 & 9 & -5 \\ -6 & 10 & -6 \end{pmatrix} = -\frac{1}{4} \begin{pmatrix} -1 & 2 & -1 \\ 9/4 & -9/4 & 5/4 \\ 3/2 & -5/2 & 3/2 \end{pmatrix}$$

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n ta noma'lumli n ta chiziqli tenglamalar sistemasi

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n \end{cases}$$

ni matritsa ko'rinishida $AX = B$ kabi yozish m-n, bunda

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}, \quad X = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix}, \quad B = \begin{pmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{pmatrix}$$

Agar A maxsusmas, ya'ni $\det A \neq 0$ bolsa, u holda bu sistemaning matritsa shaklidagi yechimi ushbu ko'rinishga ega boladi.

$$X = A^{-1} \cdot B$$

Matritsaning rangi.

A matritsaning rangi deb, uning noldan farqli minorlarining eng katta tartibiga aytiladi va u rang(A) kabi belgilanadi.

Elementar almashtirishlar:

a) faqat noldan iborat satr (ustun)ni o'chirish,
b) 2 ta satrning (ustunning) o'rinlarini almashtirish
c) bir satr (ustun)ning barcha elementlarini biror k≠0 songa ko'paytiruvchiga ko'paytirib, boshqa satr (ustun)ning mos elementlariga qo'shish,
d) satr (ustun)ning barcha elementlarini noldan farqli bir xil songa ko'paytirish.

Elementar almashtirishlar matritsa rangini o'zgartirmaydi. Shu sababli, elementar almashtirishlardan foydalanib, matritsani diagonal elementlaridan tashqari barcha elementlari nolga teng bo'ladigan ko'rinishga keltirish m-n. Bu holda matritsa rangi diagonaldagi nolga teng bo'lmagan elementlari soniga teng bo'ladi.

Misol. Matritsa rangini toping.

$$A=\begin{pmatrix}0&2&-4\\-1&-4&5\\3&1&7\\0&5&-10\\2&3&0\end{pmatrix}\overset{1\leftrightarrow 2}{\sim}\begin{pmatrix}1&4&-5\\2&3&0\\3&1&7\\0&2&-4\\0&5&-10\end{pmatrix}\overset{\begin{matrix}-2\\-3\end{matrix}}{\sim}\begin{pmatrix}1&4&-5\\0&-5&10\\0&-11&22\\0&1&-2\\0&1&-2\end{pmatrix}\overset{\begin{matrix}:(-5)\\:(-11)\end{matrix}}{\sim}$$

$$\sim\begin{pmatrix}1&4&-5\\0&1&-2\\0&1&-2\\0&1&-2\\0&1&-2\end{pmatrix}\overset{\begin{matrix}-1\\-1\\-1\end{matrix}}{\sim}\begin{pmatrix}1&4&-5\\0&1&-2\\0&0&0\\0&0&0\\0&0&0\end{pmatrix}\overset{-4}{\sim}\begin{pmatrix}1&0&3\\0&1&-2\\0&0&0\\0&0&0\\0&0&0\end{pmatrix}\sim$$

$\times 2 \rightarrow$

$$\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

bu matritsaning rangi 2 ga teng demak $\operatorname{rang} A = 2$

Kroneker - Kapelli teoremasi

n ta noma'lumli m ta chiziqli tenglamalar sistemasi

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \dots\dots\dots\dots\dots\dots\dots\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases} \quad (1)$$

Birgalikda bolishi uchun $\operatorname{rang} A = \operatorname{rang} B$ bolishi zarur va yetarlidir.
Bu yerda:

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$ – sistemaning asosiy matritsasi.

$$B = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{pmatrix}$$ – sistemaning kengaytirilgan matritsasi.

Birgalikdagi sistema → yechimlar soni ≥ 1
Birgalikda bolmagan sistema → yechimlar soni $= 0$ $(\emptyset)$

- Agar $\operatorname{rang} A = n$ → sistema yagona yechimga ega.
- Agar $\operatorname{rang} A < n$ → sistema $(n - \operatorname{rang} A)$ ta ixtiyoriy parametrga boglig bolgan cheksiz kop yechimga ega.

9-savol. Bir jinsli tenglamalar sistemasini yechish.

Agar (1) tenglamalar sistemasida barcha b_i ozod hadlar nolga teng bolsa, u holda tenglamalar sistemasi bir jinsli deyiladi. Bunday tenglamalar sistemasida har doim $\text{rang}A = \text{rang}B$, shu sababli bir jinsli sistema birgalikda boladi. Bir jinsli tenglamalar sistemasini $x_1=0, x_2=0, \dots, x_n=0$ qiymatlar qanoatlantiradi, lekin A matritsaning rangi noma'lumlar soni n dan kichik bolganda uning determinanti nolga teng bolib, sistema nolmas yechimga ega bolladi.

Misol. Ushbu bir jinsli sistemani yeching.

$$\begin{cases} 3x_1 + 4x_2 - x_3 = 0 \\ x_1 - 3x_2 + 5x_3 = 0 \\ 4x_1 + x_2 + 4x_3 = 0 \end{cases}$$

$\text{rang}A$ ni topamiz.

$$A = \begin{pmatrix} 3 & 4 & -1 \\ 1 & -3 & 5 \\ 4 & 1 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & 5 \\ 3 & 4 & -1 \\ 4 & 1 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & 5 \\ 0 & 13 & -16 \\ 0 & 13 & -16 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & 5 \\ 0 & 13 & -16 \\ 0 & 0 & 0 \end{pmatrix}$$

demak $\text{rang}A = 2 < 3$ (3-noma'lumlar soni), chunki:

$$\Delta = \begin{vmatrix} 1 & -3 \\ 0 & 13 \end{vmatrix} = 13 \neq 0$$

ya'ni matritsaning eng kichik nolga teng bolmagan minori 2-darajali ekan.

Demak sistema yechimga ega.

● Sistemaning determinanti:

$$\begin{vmatrix} 3 & 4 & -1 \\ 1 & -3 & 5 \\ 4 & 1 & 4 \end{vmatrix} = -36 + 80 + (-1) - (12 + 16 + 15) = 0$$

Demak sistema cheksiz kop yechimga ega.

Sistemaning dastlabki ikki tenglamasini yechamiz:

$$\begin{cases} 3x_1 + 4x_2 - x_3 = 0 \\ x_1 - 3x_2 + 5x_3 = 0 \end{cases} \rightarrow \begin{cases} 3x_1 + 4x_2 = x_3 \\ x_1 - 3x_2 = -5x_3 \end{cases}$$

Kramer qoidasidan foydalanamiz:

$$\Delta = \begin{vmatrix} 3 & 4 \\ 1 & -3 \end{vmatrix} = -9 - 4 = -13$$

$$\Delta_{x_1} = \begin{vmatrix} x_3 & 4 \\ -5x_3 & -3 \end{vmatrix} = -3x_3 + 20x_3 = 17x_3$$

$$\Delta_{x_2} = \begin{vmatrix} 3 & x_3 \\ 1 & -5x_3 \end{vmatrix} = -15x_3 - x_3 = -16x_3$$

$$x_1 = \frac{\Delta x_1}{\Delta} = \frac{17x_3}{-13} \qquad x_2 = \frac{\Delta_{x_2}}{\Delta} = \frac{-16x^3}{-13} = \frac{16x^3}{13}$$

Agar $x_3 = 13t$ deb olsak:

$x_1 = -17t$, $x_2 = 16t$, $x_3 = 13t$ boladi. t ga ixtiyoriy qiymatlarni berib cheksiz kop yechimlarni hosil qilamiz.

Misollar.

① A^{-1} ni toping.

$$A=\begin{pmatrix}3 & 1 & 2\\ -1 & 0 & 2\\ 1 & 2 & 1\end{pmatrix}$$

$$A^{-1}=\frac{1}{\det A}\cdot\begin{pmatrix}A_{11} & A_{21} & A_{31}\\ A_{12} & A_{22} & A_{32}\\ A_{13} & A_{23} & A_{33}\end{pmatrix}$$

$$\det A=\begin{vmatrix}3 & 1 & 2\\ -1 & 0 & 2\\ 1 & 2 & 1\end{vmatrix}=0+2+(-4)-(0-1+12)=-13\neq 0$$

$$A_{11}=\begin{vmatrix}0 & 2\\ 2 & 1\end{vmatrix}=-4 \qquad A_{21}=-\begin{vmatrix}1 & 2\\ 2 & 1\end{vmatrix}=3 \qquad A_{31}=\begin{vmatrix}1 & 2\\ 0 & 2\end{vmatrix}=2$$

$$A_{12}=-\begin{vmatrix}-1 & 2\\ 1 & 1\end{vmatrix}=3 \qquad A_{22}=\begin{vmatrix}3 & 2\\ 1 & 1\end{vmatrix}=1 \qquad A_{32}=-\begin{vmatrix}3 & 2\\ -1 & 2\end{vmatrix}=-8$$

$$A_{13}=\begin{vmatrix}-1 & 0\\ 1 & 2\end{vmatrix}=-2 \qquad A_{23}=-\begin{vmatrix}3 & 1\\ 1 & 2\end{vmatrix}=-5 \qquad A_{33}=\begin{vmatrix}3 & 1\\ -1 & 0\end{vmatrix}=1$$

$$A^{-1}=-\frac{1}{13}\begin{pmatrix}-4 & 3 & 2\\ 3 & 1 & -8\\ -2 & -5 & 1\end{pmatrix}$$

② $\operatorname{rang} A=?$

$$A=\begin{pmatrix}4 & -1 & 2 & 0\\ -1 & 1 & 1 & 1\\ 1 & 3 & 2 & -1\\ 0 & 4 & 3 & 0\end{pmatrix}\sim\begin{pmatrix}1 & 1 & 1 & 1\\ 1 & 3 & 2 & -1\\ 4 & -1 & 2 & 0\\ 0 & 4 & 3 & 0\end{pmatrix}\sim\begin{pmatrix}1 & 1 & 1 & 1\\ 0 & 4 & 3 & 0\\ 0 & 3 & 6 & 4\\ 0 & 4 & 3 & 0\end{pmatrix}\sim$$

$$\sim\begin{pmatrix}-1 & 1 & 1 & 1\\ 0 & 3 & 6 & 4\\ 0 & 4 & 3 & 0\\ 0 & 4 & 3 & 0\end{pmatrix}\sim\left(\begin{array}{ccc|c}-1 & 1 & 1 & 1\\ 0 & 3 & 6 & 4\\ 0 & 4 & 3 & 0\\ \hline 0 & 0 & 0 & 0\end{array}\right)$$

$\det=-9+24=15\neq 0$

$\operatorname{rang} A=3$

③ Tenglamalar sistemasining birgalikda bo'lish-bo'lmasligini tekshiring. Agar sistema birgalikda bo'lsa, uni matritsa usuli b-n yeching.

a) $\begin{cases} 4x+2y-z=-9 \\ x+2z=5 \\ x-3y+z=5 \end{cases}$

$$A=\begin{pmatrix} 4 & 2 & -1 \\ 1 & 0 & 2 \\ 1 & -3 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & 1 \\ 1 & 0 & 2 \\ 4 & 2 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & 1 \\ 0 & 3 & 1 \\ 0 & 14 & -3 \end{pmatrix} \to \det A \neq 0 \text{ demak } rang=3$$

$$B=\begin{pmatrix} 4 & 2 & -1 & -9 \\ 1 & 0 & 2 & 5 \\ 1 & -3 & 1 & 5 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 2 & 5 \\ 1 & -3 & 1 & 5 \\ 4 & 2 & -1 & -9 \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} 1 & 0 & 2 & 5 \\ 0 & -3 & -1 & 0 \\ 0 & 2 & -9 & -29 \end{pmatrix} \cdot\left(\frac{2}{3}\right) \sim \begin{pmatrix} 1 & 0 & 2 & 5 \\ 0 & -3 & -1 & 0 \\ 0 & 0 & -9\frac{2}{3} & -29 \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -3 & -1 & 0 \\ 0 & 0 & -9\frac{2}{3} & -29 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & -9\frac{2}{3} & -29 \end{pmatrix} \to rang B=3$$

$rang A = rang B \to$ sistema birgalikda.

$AX=B \to X=A^{-1}\cdot B$

$$A=\begin{pmatrix} 4 & 2 & -1 \\ 1 & 0 & 2 \\ 1 & -3 & 1 \end{pmatrix} \quad X=\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad B=\begin{pmatrix} -9 \\ 5 \\ 5 \end{pmatrix}$$

$$A^{-1}=\frac{1}{\det A}\cdot A^T$$

$\det A = 0 + (+3) + 4 - (0 + 2 + (-24)) = 7 + 22 = 29 \neq 0.$

$A_{11} = \begin{vmatrix} 0 & 2 \\ -3 & 1 \end{vmatrix} = 6 \quad A_{21} = -\begin{vmatrix} 2 & -1 \\ -3 & 1 \end{vmatrix} = -1 \quad A_{31} = \begin{vmatrix} 2 & -1 \\ 0 & 2 \end{vmatrix} = 4$

$A_{12} = -\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1 \quad A_{22} = \begin{vmatrix} 4 & -1 \\ 1 & 1 \end{vmatrix} = 5 \quad A_{32} = -\begin{vmatrix} 4 & -1 \\ 1 & 2 \end{vmatrix} = -9$

$A_{13} = \begin{vmatrix} 1 & 0 \\ 1 & -3 \end{vmatrix} = -3 \quad A_{23} = -\begin{vmatrix} 4 & 2 \\ 1 & -3 \end{vmatrix} = 14 \quad A_{33} = \begin{vmatrix} 4 & 2 \\ 1 & 0 \end{vmatrix} = -2$

$A^{-1} = \frac{1}{29}\begin{pmatrix} 6 & 1 & 4 \\ 1 & 5 & -2 \\ -3 & 14 & -2 \end{pmatrix} = \begin{pmatrix} \frac{6}{29} & \frac{1}{29} & \frac{4}{29} \\ \frac{1}{29} & \frac{5}{29} & -\frac{2}{29} \\ -\frac{3}{29} & \frac{14}{29} & -\frac{2}{29} \end{pmatrix}$

$X = A^{-1} \cdot B = \frac{1}{29}\begin{pmatrix} 6 & 1 & 4 \\ 1 & 5 & -2 \\ -3 & 14 & -2 \end{pmatrix} \cdot \begin{pmatrix} -9 \\ 5 \\ 5 \end{pmatrix} = \frac{1}{29}\begin{pmatrix} -54+5+20 \\ -9+25-10 \\ 27+70-10 \end{pmatrix} = \frac{1}{29} \cdot \begin{pmatrix} -29 \\ -29 \\ 87 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix}$

$x_1 = -1, \quad x_2 = -1, \quad x_3 = 3.$

b) $\begin{cases} 2x - 3y + z = 2 \\ 3x + y - 3z = 1 \\ 5x - 2y - 2z = 4 \end{cases} \quad A = \begin{pmatrix} 2 & -3 & 1 \\ 3 & 1 & -3 \\ 5 & -2 & -2 \end{pmatrix} \quad B = \begin{pmatrix} 2 & -3 & 1 & 2 \\ 3 & 1 & -3 & 1 \\ 5 & -2 & -2 & 4 \end{pmatrix}$

$A = \begin{pmatrix} 2 & -3 & 1 \\ 3 & 1 & -3 \\ 5 & -2 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & -3 \\ -3 & 3 & 1 \\ -2 & 5 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & -3 \\ 0 & 9 & -8 \\ 0 & 9 & -8 \end{pmatrix} \sim \left(\begin{array}{ccc} 1 & 2 & -3 \\ 0 & 9 & -8 \\ \hline 0 & 0 & 0 \end{array}\right)$

$\begin{vmatrix} 2 & -3 \\ 9 & -8 \end{vmatrix} = -16 + 27 = 11 \neq 0$ demak rang $A = 2$

$B = \begin{pmatrix} 2 & -3 & 1 & 2 \\ 3 & 1 & -3 & 1 \\ 5 & -2 & -2 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & -3 & 2 \\ -3 & 3 & 1 & 1 \\ -2 & 5 & -2 & 4 \end{pmatrix} \sim$

$$\sim\begin{pmatrix}1 & 2 & -3 & 2\\ 0 & 9 & -8 & 7\\ 0 & 9 & -8 & 8\end{pmatrix}\sim\left(\begin{array}{c|ccc}1 & 2 & -3 & 2\\ \hline 0 & 9 & -8 & 7\\ 0 & 0 & 0 & 1\end{array}\right)$$

$$\begin{vmatrix}2 & -3 & 2\\ 9 & -8 & 7\\ 0 & 0 & 1\end{vmatrix} = -16+27 = 11 \neq 0$$ demak $rang\,B = 3$

$rang\,A \neq rang\,B$

sistema birgalikda emas.

④ Bir jinsli sistemani yeching.

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 0\\ 2x_1 - 3x_2 + 4x_3 = 0\\ 3x_1 - x_2 + 7x_3 = 0\end{cases} \qquad A = \begin{pmatrix}1 & 2 & 3\\ 2 & -3 & 4\\ 3 & -1 & 7\end{pmatrix}$$

$$A = \begin{pmatrix}1 & 2 & 3\\ 2 & -3 & 4\\ 3 & -1 & 7\end{pmatrix}\sim\begin{pmatrix}1 & 2 & 3\\ 0 & -7 & -2\\ 0 & -7 & -2\end{pmatrix}\sim\begin{pmatrix}1 & 2 & 3\\ 0 & -7 & -2\\ 0 & 0 & 0\end{pmatrix} \to rang\,A = 2$$

$$\begin{vmatrix}2 & 3\\ -7 & -2\end{vmatrix} = -4+21 = 17 \neq 0$$

$n - rang\,A = 3 - 2 = 1$

Yana 1 ta parametrga bog'liq cheksiz ko'p yech. epa.

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 0\\ 2x_1 - 3x_2 + 4x_3 = 0\end{cases} \Rightarrow \begin{cases} x_1 + 2x_2 = -3x_3\\ 2x_1 - 3x_2 = -4x_3\end{cases}$$

Kramer qoidasi:

$$\Delta = \begin{vmatrix}1 & 2\\ 2 & -3\end{vmatrix} = -3-4 = -7 \neq 0$$

$$\Delta_{x_1} = \begin{vmatrix}-3x_3 & 2\\ -4x_3 & -3\end{vmatrix} = (9+8)^{x_3} = 17x^3$$

$$\Delta_{x_2} = \begin{vmatrix} 1 & -3x_3 \\ 2 & -4x_3 \end{vmatrix} = -4x_3 + 6x_3 = 2x_3$$

$$x_1 = \frac{\Delta_{x_1}}{\Delta} = \frac{17x_3}{-7} \qquad x_2 = \frac{\Delta_{x_2}}{\Delta} = \frac{2x_3}{-7} \qquad x_3 = -7t$$

deb olamiz.

$$x_1 = 17t \quad x_2 = 2t \quad x_3 = -7t$$

⑤ $\begin{cases} 3x - y + z = 12 \\ x + 2y + 4z = 6 \\ 5x + y + 2z = 3 \end{cases}$ tenglamalar sistemasining birgalikda bolish-bolmaslipini tekshirip. Agar u birgalikda bolsa, uni matritsa usuli b-n yeching.

$$\begin{cases} 3x - y + z = 12 \\ x + 2y + 4z = 6 \\ 5x + y + 2z = 3 \end{cases} \rightarrow A = \begin{pmatrix} 3 & -1 & 1 \\ 1 & 2 & 4 \\ 5 & 1 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 4 \\ 3 & -1 & 1 \\ 5 & 1 & 2 \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} 1 & 2 & 4 \\ 0 & -7 & -11 \\ 0 & -9 & -18 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 \\ 0 & -7 & 3 \\ 0 & -9 & 0 \end{pmatrix} \rightarrow \det \neq 0 \ (27)$$

$\text{rang}A = 3$

$\text{rang}B = 3$ $\rightarrow \text{rang}A = \text{rang}B$

$$A = \begin{pmatrix} 3 & -1 & 1 \\ 1 & 2 & 4 \\ 5 & 1 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 12 \\ 6 \\ 3 \end{pmatrix} \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\det A = \begin{vmatrix} 3 & -1 & 1 \\ 1 & 2 & 4 \\ 5 & 1 & 2 \end{vmatrix} = 12 - 20 + 1 - (10 - 2 + 12) = -7 - 20 = -27$$

$$A_{11} = \begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix} = 0 \quad A_{21} = -\begin{vmatrix} -1 & 1 \\ 1 & 2 \end{vmatrix} = 3 \quad A_{31} = \begin{vmatrix} -1 & 1 \\ 2 & 4 \end{vmatrix} = -6$$

$$A_{12} = -\begin{vmatrix} 1 & 4 \\ 5 & 2 \end{vmatrix} = 18 \quad A_{22} = \begin{vmatrix} 3 & 1 \\ 5 & 2 \end{vmatrix} = 1 \quad A_{32} = -\begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} = -11$$

$$A_{13} = \begin{vmatrix} 1 & 2 \\ 5 & 1 \end{vmatrix} = -9 \quad A_{23} = -\begin{vmatrix} 3 & -1 \\ 5 & 1 \end{vmatrix} = -8 \quad A_{33} = \begin{vmatrix} 3 & -1 \\ 1 & 2 \end{vmatrix} = 7$$

$$A^{-1} = -\frac{1}{27}\cdot\begin{pmatrix} 0 & 3 & -6 \\ 18 & 1 & -11 \\ -9 & -8 & 7 \end{pmatrix} \qquad B = \begin{pmatrix} 12 \\ 6 \\ 3 \end{pmatrix} = 3\cdot\begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$$

$$X = A^{-1}\cdot B = -\frac{1}{\frac{27}{9}}\cdot 3\cdot\begin{pmatrix} 0 & 3 & -6 \\ 18 & 1 & -11 \\ -9 & -8 & 7 \end{pmatrix}\cdot\begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} = -\frac{1}{9}\cdot\begin{pmatrix} 6-6 \\ 72+2-11 \\ -36-16+7 \end{pmatrix} =$$

$$= -\frac{1}{9}\begin{pmatrix} 0 \\ 63 \\ -45 \end{pmatrix} = \begin{pmatrix} 0 \\ -7 \\ 5 \end{pmatrix}$$

$x_1 = 0$, $x_2 = -7$, $x_3 = 5$.

(6) Bir jinsli sistemaning noldan farqli yechimlari bor-yo'qligini aniqlang, agar bor bo'lsa, ularni toping.

$$\begin{cases} x_1 + x_2 + x_3 = 0 \\ 2x_1 - 3x_2 + 4x_3 = 0 \\ 4x_1 - 11x_2 + 10x_3 = 0 \end{cases}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -3 & 4 \\ 4 & -11 & 10 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 \\ 0 & -5 & 2 \\ 0 & -15 & 6 \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} 1 & 1 & 1 \\ 0 & -5 & 2 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{vmatrix} 1 & 1 \\ -5 & 2 \end{vmatrix} \neq 0$$

rang $A = 2 < 3$ demak 0 dan farqli yechimlari bor.

$$\begin{cases} x_1 + x_2 = -x_3 \\ 2x_1 - 3x_2 = -4x_3 \end{cases}$$

Kramer qoidasi bo'yicha:

$$\Delta = \begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix} = -3 - 2 = -5$$

$$\Delta x_1 = \begin{vmatrix} -x_3 & 1 \\ -4x_3 & -3 \end{vmatrix} = 3x_3 + 4x_3 = 7x_3$$

$$\Delta x_2 = \begin{vmatrix} 1 & -x_3 \\ 2 & -4x_3 \end{vmatrix} = -4x_3 + 2x_3 = -2x_3$$

$$x_1 = \frac{\Delta x_1}{\Delta} = \frac{7x_3}{-5}$$

$$x_2 = \frac{\Delta x_2}{\Delta} = \frac{+2x_3}{+5}$$

$x_3 = 5t$

$x_1 = -7t \quad x_2 = 2t \quad x_3 = 5t$

⑦ Berilgan tenglamalar sistemasining birgalikda ekanligini tekshiring, agar birgalikda bolsa, ularni a) kramer qoidasidan foydalanib, b) Matritsa usuli, c) Gauss usuli b-n yeching.

$$\begin{cases} 3x_1 + 2x_2 + x_3 = 5 \\ 2x_1 + 3x_2 + x_3 = 1 \\ 2x_1 + x_2 + 3x_2 = 11 \end{cases} \quad A = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \\ 3 & 1 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \\ 0 & -5 & -7 \end{pmatrix} \sim$$

$\sim \det \neq 0 \quad rang A = rang B = 3 \rightarrow$ sistema birgalikda

a) kramer qoidasi.

$$\Delta = \begin{vmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 2 & 1 & 3 \end{vmatrix} = 27 + 4 + 2 - (6 + 12 + 3) = 33 - 21 = 12 \neq 0$$

$$\Delta_{x_1} = \begin{vmatrix} 5 & 2 & 1 \\ 1 & 3 & 1 \\ 11 & 1 & 3 \end{vmatrix} = 45 + 22 + 1 - (33 + 6 + 5) = 68 - 44 = 24$$

$$\Delta_{x_2} = \begin{vmatrix} 3 & 5 & 1 \\ 2 & 1 & 1 \\ 2 & 11 & 3 \end{vmatrix} = 9 + 10 + 22 - (2 + 30 + 33) = 41 - 65 = -24$$

$$\Delta_{x_3} = \begin{vmatrix} 3 & 2 & 5 \\ 2 & 3 & 1 \\ 2 & 1 & 11 \end{vmatrix} = 99 + 4 + 10 - (30 + 44 + 3) = 113 - 77 = 36$$

$$x_1 = \frac{\Delta_{x_1}}{\Delta} = \frac{24}{12} = 2$$

$$x_2 = \frac{\Delta_{x_2}}{\Delta} = \frac{-24}{12} = -2$$

$$x_3 = \frac{\Delta_{x_3}}{\Delta} = \frac{36}{12} = 3$$

$x_1 = 2$
$x_2 = -2$
$x_3 = 3$

b) Matrixa usuli.

$$A=\begin{pmatrix}3 & 2 & 1\\ 2 & 3 & 1\\ 2 & 1 & 3\end{pmatrix} \quad B=\begin{pmatrix}5\\1\\11\end{pmatrix} \quad X=\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} \quad X=A^{-1}\cdot B$$

$\det A = 12 \neq 0$

$A_{11}=\begin{vmatrix}3 & 1\\ 1 & 3\end{vmatrix}=8 \quad A_{21}=-\begin{vmatrix}2 & 1\\ 1 & 3\end{vmatrix}=-5 \quad A_{31}=\begin{vmatrix}2 & 1\\ 3 & 1\end{vmatrix}=-1$

$A_{12}=-\begin{vmatrix}2 & 1\\ 2 & 3\end{vmatrix}=-4 \quad A_{22}=\begin{vmatrix}3 & 1\\ 2 & 3\end{vmatrix}=7 \quad A_{32}=-\begin{vmatrix}3 & 1\\ 2 & 1\end{vmatrix}=-1$

$A_{13}=\begin{vmatrix}2 & 3\\ 2 & 1\end{vmatrix}=-4 \quad A_{23}=-\begin{vmatrix}3 & 2\\ 2 & 1\end{vmatrix}=1 \quad A_{33}=\begin{vmatrix}3 & 2\\ 2 & 3\end{vmatrix}=5$

$$X=\frac{1}{12}\begin{pmatrix}8 & -5 & -1\\ -4 & 7 & -1\\ -4 & 1 & 5\end{pmatrix}\cdot\begin{pmatrix}5\\1\\11\end{pmatrix}=\frac{1}{12}\begin{pmatrix}40-5-11\\-20+7-11\\-20+1+55\end{pmatrix}=\frac{1}{12}\begin{pmatrix}24\\-24\\36\end{pmatrix}=\begin{pmatrix}2\\-2\\3\end{pmatrix}$$

$x_1=2, \quad x_2=-2, \quad x_3=3$

c) Gauss usuli.

$$\begin{cases}3x_1+2x_2+x_3=5 & /\cdot 2\\ 2x_1+3x_2+x_3=1 & /\cdot 3\\ 2x_1+x_2+3x_3=11 & /\cdot 3\end{cases}\Rightarrow\begin{cases}6x_1+4x_2+2x_3=10\\ 6x_1+9x_2+3x_3=3\\ 6x_1+3x_2+9x_3=33\end{cases}\Rightarrow$$

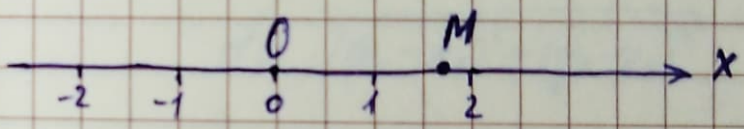
$$\Rightarrow\begin{cases}6x_1+4x_2+2x_3=10\\ 0+5x_2+x_3=-7\\ 0+x_2+7x_3=23\end{cases}\Rightarrow\begin{cases}3x_1+2x_2+x_3=5\\ 5x_2+x_3=-7\\ -5x_2+35x_3=115\end{cases}\Rightarrow$$

$$\Rightarrow\begin{cases}3x_1+2x_2+x_3=5\\ 5x_2+x_3=-7\\ 36x_3=108\end{cases}\Rightarrow x_3=3,\ x_2=-2,\ x_1=2$$

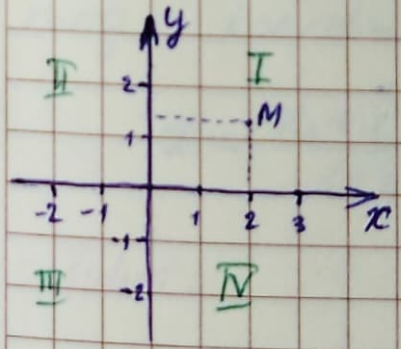
Analitik geometriya.

17-savol. Dekart koordinatalar sistemasi.

Matematikaning geometrik masalalar algebraik usul bilan yechiladigan bolimi analitik geometriya deb ataladi. Analitik geometriyaning asosi koordinatalar usuli bolib, uni XVII asrda fransuz matematigi va faylasufi Rene Dekart kiritgan.



OM kattalik M nuqtaning koordinatasi deyiladi
Ox – koordinata oqi.

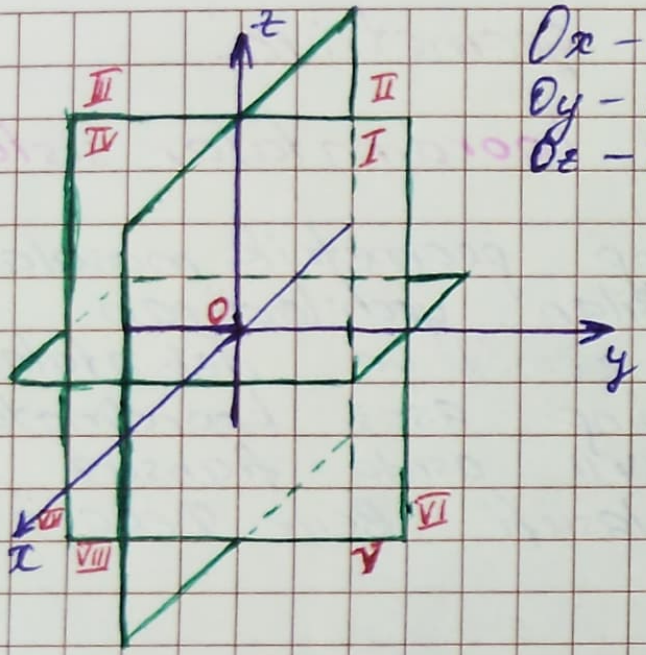


Ox – absissalar oqi (gorizontal oq)
Oy – ordinatalar oqi (vertikal oq)

Birgalikda – koordinatalar oqi.

O nuqta – koordinatalar boshi
Ox va Oy oqi joylashgan tekislik koordinatal tekisligi (Oxy bn belgilanadi)

I, II, III, IV – choraklar.



Ox - abssissalar o'qi
Oy - ordinatalar o'qi
Oz - applikatalar o'qi.

18-savol. Ikki nuqta orasidagi masofa. Kesmani berilgan nisbatda bolish.

1. Oqdagi $A(x_1)$ va $B(x_2)$ nuqtalar orasidagi masofa:

$$d = |x_2 - x_1| = \sqrt{(x_2 - x_1)^2}$$

2. Tekislikdagi $A(x_1, y_1)$ va $B(x_2, y_2)$ nuqtalar orasidagi masofa:

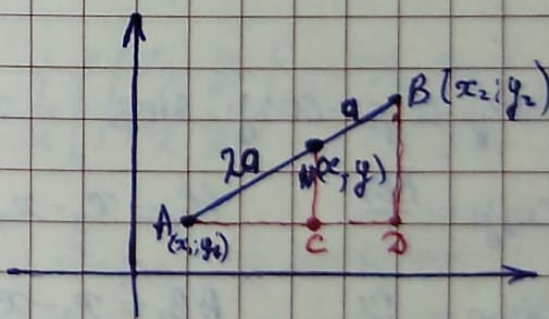
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

3. Kesmani berilgan nisbatda bolish.

$A(x_1, y_1)$ va $B(x_2, y_2)$ nuqtalar AB kesmani $AN:NB = \lambda$ nisbatda bolluvchi $N(x, y)$ nuqtaning koordinatalari:

$$x = \frac{x_1 + \lambda x_2}{1 + \lambda}, \quad y = \frac{y_1 + \lambda y_2}{1 + \lambda}.$$

Isbot:



$\triangle ABD \rightarrow \frac{AC}{AD} = \frac{\lambda q}{q(\lambda+1)} \rightarrow \frac{x - x_1}{x_2 - x_1} = \frac{\lambda}{\lambda+1}$

$$(x - x_1) \cdot (\lambda + 1) = \lambda (x_2 - x_1)$$

$$x \cdot (1 + \lambda) - x_1\lambda - x_1 = \lambda x_2 - \lambda x_1$$

$$x \cdot (1 + \lambda) = x_1 + \lambda x_2 \longrightarrow \boxed{x = \frac{x_1 + \lambda x_2}{1 + \lambda}}$$

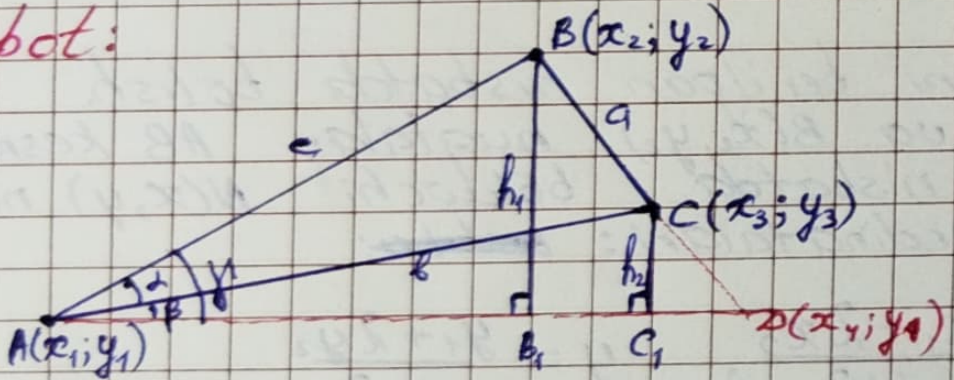
$\triangle ABD \to \frac{NC}{BD} = \frac{y-y_1}{y_2-y_1} \Rightarrow \frac{\lambda a}{(1+\lambda)a} \to \boxed{y = \frac{y_1+\lambda y_2}{1+\lambda}}$.

– $\lambda = 1$ bolganda $x = \frac{x_1+x_2}{2}$, $y = \frac{y_1+y_2}{2}$.

4. Uchlari $A(x_1;y_1)$, $B(x_2;y_2)$, $C(x_3;y_3)$ nuqtalarda bolgan ABC uchburchakning yuzi quyidagi formula b-n topiladi:

$$S = \pm\frac{1}{2}\left[\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_1 & y_1 \end{vmatrix}\right]$$

Isbot:



$S_{\triangle ABC} = \frac{1}{2} \cdot bc \cdot \sin\alpha$

1-usul: $\sin\alpha = \sin(\gamma-\beta) = \sin\gamma\cdot\cos\beta - \cos\gamma\cdot\sin\beta$

$\sin\gamma = \frac{h_1}{c}$ $\quad \cos\gamma = \frac{AB_1}{c}$ $\quad AC_1 = x_3 - x_1$ $\quad h_1 = y_2 - y_1$

$\cos\beta = \frac{AC_1}{b}$ $\quad \sin\beta = \frac{h_2}{b}$ $\quad AB_1 = x_2 - x_1$ $\quad h_2 = y_3 - y_1$

$$\sin\alpha = \frac{h_1}{c}\cdot\frac{x_3-x_1}{b} - \frac{x_2-x_1}{c}\cdot\frac{h_2}{b} =$$

$$= \frac{(y_2-y_1)(x_3-x_1) - (x_2-x_1)\cdot(y_3-y_1)}{bc}$$

$$S_{\Delta ABC} = \frac{1}{2} bc \cdot \sin\alpha = \frac{1}{2} bc \cdot \frac{(y_2-y_1)(x_3-x_1) - (x_2-x_1)(y_3-y_1)}{bc} =$$

$$\Rightarrow \left[S_{\Delta ABC} = \frac{1}{2} \cdot \left((y_2-y_1)(x_3-x_1) - (x_2-x_1)(y_3-y_1) \right) \right]$$

$$S_{\Delta ABC} = \frac{1}{2} \left(x_3(y_2-y_1) - x_1(y_2-y_1) - x_2(y_3-y_1) + x_1(y_3-y_1) \right)$$

$$S_{\Delta ABC} = \frac{1}{2} \left(x_3 \cdot (y_2-y_1) - x_2(y_3-y_1) - x_1(y_2-y_1-y_3+y_1) \right)$$

$$\left[S_{\Delta ABC} = \frac{1}{2} \left(x_1(y_3-y_2) + x_2(y_1-y_3) + x_3(y_2-y_1) \right) \right]$$

$$S_{\Delta ABC} = \frac{1}{2} \left(x_1y_3 - x_1y_2 + x_2y_1 - x_2y_3 + x_3y_2 - x_3y_1 \right)$$

$$S_{\Delta ABC} = \frac{1}{2} \left(-(x_1y_2 - x_2y_1) - (x_2y_3 - x_3y_2) - (x_3y_1 - x_1y_3) \right)$$

$$\left[S_{\Delta ABC} = \pm \frac{1}{2} \left(\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_1 & y_1 \end{vmatrix} \right) \right] \quad (\cdot)$$

2-usul: $S_{\Delta ABC} = \frac{1}{2} bc \sin\alpha$

$$\sin\alpha = \sqrt{1-\cos^2\alpha} \rightarrow \cos\alpha = \frac{\overrightarrow{AC} \cdot \overrightarrow{AB}}{|\overrightarrow{AB}| \cdot |\overrightarrow{AC}|} \rightarrow \begin{matrix} |\overrightarrow{AB}| = c \\ |\overrightarrow{AC}| = b \end{matrix}$$

$$S_\Delta = \frac{1}{2} \cdot bc \cdot \sqrt{1 - \frac{(\overrightarrow{AB} \cdot \overrightarrow{AC})^2}{(bc)^2}} = \frac{1}{2} \cdot bc \cdot \frac{\sqrt{(bc)^2 - (\overrightarrow{AB} \cdot \overrightarrow{AC})^2}}{bc}$$

$$S_\Delta = \frac{1}{2} \cdot \sqrt{(bc)^2 - (\overrightarrow{AB} \cdot \overrightarrow{AC})^2}$$

$$S_{\Delta} = \frac{1}{2} \cdot \sqrt{\left((x_3-x_1)^2+(y_3-y_1)^2\right)\cdot\left((x_2-x_1)^2+(y_2-y_1)^2\right) - \left((x_2-x_1)\cdot(x_3-x_1)+(y_2-y_1)\cdot(y_3-y_1)\right)^2}$$

$x_2-x_1=p$
$x_3-x_2=q$
$y_2-y_1=m$
$y_3-y_1=n$

$$S_{\Delta} = \frac{1}{2}\sqrt{(q^2+n^2)(p^2+m^2) - (p\cdot q+mn)^2}$$

$$S_{\Delta} = \frac{1}{2}\cdot\sqrt{(qp)^2+(np)^2+(qm)^2+(mn)^2-(pq)^2-2pqmn-(mn)^2}$$

$$S_{\Delta} = \frac{1}{2}\cdot\sqrt{(np-mq)^2}$$

$$S_{\Delta ABC} = \pm\frac{1}{2}(np-mq)$$

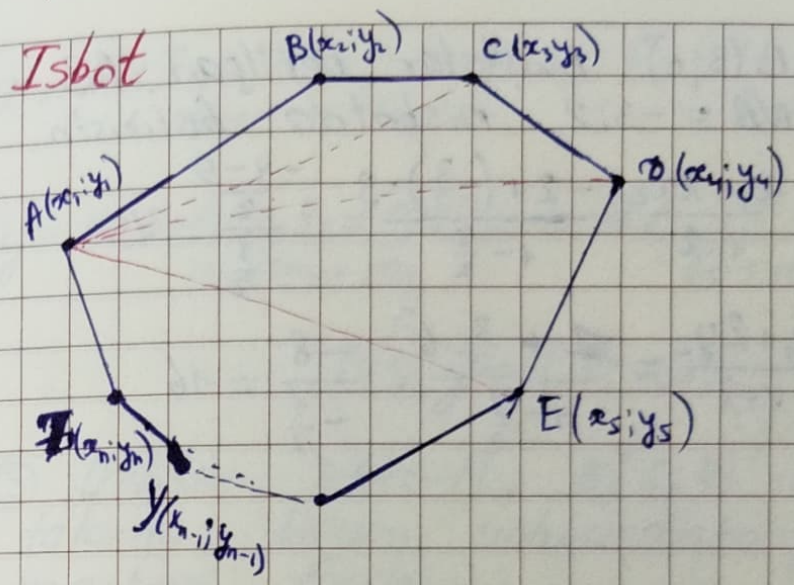
$$\left[S_{\Delta ABC} = \pm\frac{1}{2}\left((y_3-y_1)(x_2-x_1)-(y_2-y_1)(x_3-x_2)\right)\right]$$

Bu formula 1-usul bo'yicha determinantga keltiriladi (•)

5. Uchlari $A(x_1;y_1)$, $B(x_2,y_2)$, $C(x_3,y_3)$, ..., $Z(x_n,y_n)$ nuqtalarda bo'lgan ko'pburchak yuzi

$$\left[S = \pm\frac{1}{2}\left[\begin{vmatrix}x_1 & y_1\\ x_2 & y_2\end{vmatrix}+\begin{vmatrix}x_2 & y_2\\ x_3 & y_3\end{vmatrix}+\ldots+\begin{vmatrix}x_n & y_n\\ x_1 & y_1\end{vmatrix}\right]\right]$$

Isbot



$$S_{ABCD\ldots YZ} = S_{ABC} + S_{ACD} + S_{ADE} + \ldots + S_{AYZ}$$

$$S_{ABC\ldots YZ} = \pm\frac{1}{2}\left[\left(\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_1 & y_1 \end{vmatrix}\right) + \left(\begin{vmatrix} x_1 & y_1 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} + \begin{vmatrix} x_4 & y_4 \\ x_1 & y_1 \end{vmatrix}\right) +$$

$$+ \left(\begin{vmatrix} x_1 & y_1 \\ x_4 & y_4 \end{vmatrix} + \begin{vmatrix} x_4 & y_4 \\ x_5 & y_5 \end{vmatrix} + \begin{vmatrix} x_5 & y_5 \\ x_1 & y_1 \end{vmatrix}\right) + \ldots + \left(\begin{vmatrix} x_1 & y_1 \\ x_{n-1} & y_{n-1} \end{vmatrix} + \begin{vmatrix} x_{n-1} & y_{n-1} \\ x_n & y_n \end{vmatrix} + \begin{vmatrix} x_n & y_n \\ x_1 & y_1 \end{vmatrix}\right)$$

$$S_{ABC\ldots YZ} = \pm\frac{1}{2}\left[\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} + \ldots + \begin{vmatrix} x_n & y_n \\ x_1 & y_1 \end{vmatrix}\right] \quad (\cdot)$$

Misollar:

① $A(-2;1)$ va $B(3;6)$ nuqtalar yasalsin. AB kesmani $AN:NB = 3:2$ nisbatan boluvchi $N(x;y)$ nuqta topilsin.

$\lambda = \frac{3}{2}$

$$x = \frac{x_1 + \lambda x_2}{1+\lambda} = \frac{-2 + \frac{3}{2}\cdot 3}{1+\frac{3}{2}} = \frac{\frac{-4+9}{2}}{\frac{5}{2}} = 1$$

$$y = \frac{y_1 + \lambda y_2}{1+\lambda} = \frac{1 + \frac{3}{2}\cdot 6}{1+\frac{3}{2}} = \frac{10}{\frac{5}{2}} = 4 \qquad (x;y) = (1;4)$$

② $A(-2;1)$ va $B(3;6)$ nuqtalar berilgan. AB kesma $AN:NB = -3:2$ nisbatda bo'linsin.

$\lambda = -\frac{3}{2}$

$$x = \frac{x_1 + \lambda x_2}{1+\lambda} = \frac{-2 + \left(-\frac{3}{2}\right)\cdot 3}{1 - \frac{3}{2}} = \frac{-4-9}{\frac{2}{-\frac{1}{2}}} = 13$$

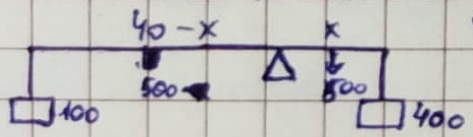
$$y = \frac{y_1 + \lambda y_2}{1+\lambda} = \frac{1 - \frac{3}{2}\cdot 6}{1 - \frac{3}{2}} = \frac{-8}{-\frac{1}{2}} = 16$$

$(x;y) = (13;16)$

6. Ox o'qning $A(x_1)$, $B(x_2)$ va $C(x_3)$ nuqtalariga mos ravishda m_1, m_2, m_3 massalar joylashtirilgan. Bu sistema massalarining markazi:

$$\boxed{x = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}}$$

Misol: ① Uzunligi 40 sm va og'irligi 500 g bo'lgan bir jinsli sterjenning uchlariga og'irliklari 100 va 400 g sharlar osilgan. Shu sistemaning og'irlik markazi aniqlansin.



$$400\cdot x + 500\cdot\frac{x}{2} = 100\cdot(40-x) + 500\cdot\frac{40-x}{2} \quad /\cdot 2$$

$$650x = 4000 - 100x + 10000 - 250x$$

$$1000x = 14000$$

$$x = 14 \qquad 40 - x = 26$$

② $A(-2;4)$, $B(3;-1)$ va $C(2;3)$ nuqtalarga mos ravishda 60, 40, 100 g massalar qo'yilgan. Shu sistemaning massalarining markazi aniqlansin.

$$x = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1+m_2+m_3} = \frac{60\cdot(-2) + 40\cdot 3 + 100\cdot 2}{60+40+100} = 1$$

$$y = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1+m_2+m_3} = \frac{60\cdot 4 + 40\cdot(-1) + 100\cdot 3}{60+40+100} = 2{,}5$$

$$M(x;y) = (1;\ 2{,}5)$$

③ Uchlari $A(1;-1)$, $B(6;4)$ va $C(2;6)$ nuqtalarda bolgan uchburchakning ogirlik markazi topilsin.

$$x = \frac{x_1+x_2+x_3}{3} = \frac{1+6+2}{3} = 3$$

$$y = \frac{y_1+y_2+y_3}{3} = \frac{-1+4+6}{3} = 3$$

$(3;3)$

④ Uchlari $A(2;0)$, $B(5;3)$ va $C(2;6)$ nuqtalarda bolgan uchburchakning yuzi hisoblansin.

$$S = \frac{1}{2}\left(\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_1 & y_1 \end{vmatrix}\right) =$$

$$= \frac{1}{2}\left(\begin{vmatrix} 2 & 0 \\ 5 & 3 \end{vmatrix} + \begin{vmatrix} 5 & 3 \\ 2 & 6 \end{vmatrix} + \begin{vmatrix} 2 & 6 \\ 2 & 0 \end{vmatrix}\right) = \frac{1}{2}\cdot(6+24-12) = 9$$

⑤ Uchlari $A(3;1)$, $B(4;6)$ va $C(6;3)$, $D(5;-2)$ nuqtalarda bolgan tortburchakning yuzi hisoblansin.

$$S = \frac{1}{2}\left(\begin{vmatrix} 3 & 1 \\ 4 & 6 \end{vmatrix} + \begin{vmatrix} 4 & 6 \\ 6 & 3 \end{vmatrix} + \begin{vmatrix} 6 & 3 \\ 5 & -2 \end{vmatrix} + \begin{vmatrix} 5 & -2 \\ 3 & 1 \end{vmatrix}\right) = \frac{1}{2}(14-24-27+11) = -13$$

19-savol. Tog'ri chiziq va uning tenglamalari.

① Tog'ri chiziqning burchak koeffitsiyentli tenglamasi:

$$[y = k \cdot x + b]$$

k - burchak koeffitsiyenti ($k = tg\alpha$)
b - boshlang'ich ordinata yoki Oy o'q ajratgan kesma kattaligi.

② Tog'ri chiziqning umumiy tenglamasi

$$[Ax + By + C = 0]$$

Xususiy hollar:

a) $C = 0 \rightarrow y = -\frac{A}{B}x \rightarrow$ to'g'ri chiziq koordinatalar boshidan o'tadi.

b) $B = 0 \rightarrow x = -\frac{C}{A} \rightarrow$ to'g'ri chiziq Ox o'qqa parallel bo'ladi.

c) $A = 0 \rightarrow y = -\frac{C}{B} \rightarrow$ to'g'ri chiziq Oy o'qqa parallel bo'ladi.

d) $B = C = 0 \rightarrow Ax = 0 \rightarrow x = 0 \rightarrow$ to'g'ri chiziq Oy o'qdan iborat.

e) $A = C = 0 \rightarrow By = 0 \rightarrow y = 0 \rightarrow$ to'g'ri chiziq Ox o'qdan iborat.

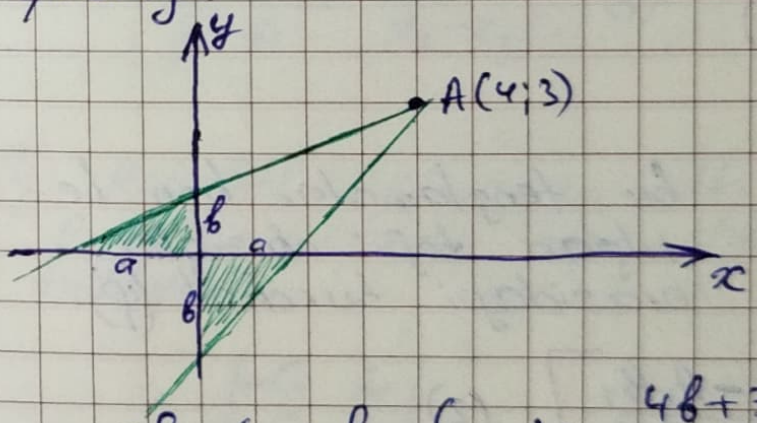
③ Tōpri chiziqning ōqlardan ajratgan kesmalar bōyicha tenglamasi.

$$\left[\frac{x}{a}+\frac{y}{b}=1\right]$$

Bu yerda:
a va b – tōpri chiziqning ōqlardan kespan kesmalari kattaliklari.

Misollar.

① A(4;3) nuqtadan ōtuvchi va koordinatalar burchagidan yuzi 3ta birlikka teng uchburchak kesuvchi tōpri chiziq tenglamasini yozing.



$$\begin{cases}\frac{ab}{2}=3\\ \frac{4}{a}+\frac{3}{b}=1\end{cases}$$

$ab=6 \to b=\frac{6}{a}$ ~~→~~ $\frac{4b+3a}{ab}=1$

$4b+3a=6$

$4\cdot\frac{6}{a}+3a=6 \quad /\cdot\frac{a}{3}$

$8+a^2=2a$

$a^2-2a+8=0$

$(a-4)(a+2)=0$

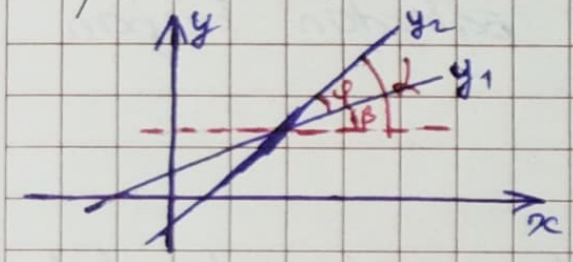
$a_1=4 \quad a_2=-2$

$b_1=\frac{3}{2} \quad b_2=-3$

J: $\frac{x}{4}+\frac{2y}{3}=1$ va $\frac{x}{-2}+\frac{y}{-3}=1$

20-savol: Ikki to'g'ri chiziq orasidagi burchak. Berilgan ikki to'g'ri chiziqning kesishish nuqtasi.

① $y=k_1x+b_1$ to'g'ri chiziqdan $y=k_2x+b_2$ to'g'ri chiziqgacha soat strelkasiga qarshi yo'nalishda hisoblanuvchi φ burchak:



$$\boxed{tg\varphi=\frac{k_2-k_1}{1+k_1k_2}} \quad (1)$$

Isbot: $tg\varphi=tg(\alpha-\beta)=\frac{tg\alpha-tg\beta}{1+tg\alpha\cdot tg\beta}$

$$\boxed{tg\varphi=\frac{k_2-k_1}{1+k_1k_2}} \quad (\cdot)$$

– $A_1x+B_1y+C_1=0$, $A_2x+B_2y+C_2=0$ bu tenglamalar b-n berilgan to'g'ri chiziqlar orasidagi burchak (φ):

$$\boxed{tg\varphi=\frac{A_1B_2-A_2B_1}{A_1A_2+B_1B_2}} \quad (2)$$

Isbot: $Ax+By+C=0 \rightarrow y=-\frac{A}{B}x-\frac{C}{B}$

$$k=-\frac{A}{B},\quad b=-\frac{C}{B}$$

$(1)\rightarrow tg\varphi=\frac{k_2-k_1}{1+k_1k_2}$

$$tg\varphi=\frac{-\frac{A_1}{B_1}+\frac{A_2}{B_2}}{1+\frac{A_1A_2}{B_1B_2}}=\frac{\frac{A_2B_2-A_1B_2}{B_1B_2}}{\frac{A_1A_2+B_1B_2}{B_1B_2}}=\frac{A_2B_1-A_1B_2}{A_1A_2+B_1B_2} \quad (\cdot)$$

Parallellik sharti: $k_1 = k_2$ yoki $\frac{A_1}{A_2} = \frac{B_1}{B_2}$ $(A_2B_1 - A_1B_2 = 0)$

Perpendikularlik sharti: $k_2 = -\frac{1}{k_1}$ yoki $A_1A_2 + B_1B_2 = 0$

② Berilgan $A(x_1; y_1)$ nuqtadan o'tuvchi to'g'ri chiziqlar to'plamining tenglamasi:

$$\boxed{y - y_1 = k(x - x_1)} \quad (3)$$

③ Berilgan ikki $A(x_1; y_1)$ va $B(x_2; y_2)$ nuqtalardan o'tuvchi to'g'ri chiziq tenglamasi:

$$\boxed{\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}} \quad (4)$$

④ Parallel bo'lmagan ikki $A_1x + B_1y + C_1 = 0$ va $A_2x + B_2y + C_2 = 0$ to'g'ri chiziqning kesishish nuqtasi:

$$x = \frac{\begin{vmatrix} -C_1 & B_1 \\ -C_2 & B_2 \end{vmatrix}}{\begin{vmatrix} A_1 & B_1 \\ A_2 & B_2 \end{vmatrix}}, \qquad y = \frac{\begin{vmatrix} A_1 & -C_1 \\ A_2 & -C_2 \end{vmatrix}}{\begin{vmatrix} A_1 & B_1 \\ A_2 & B_2 \end{vmatrix}}$$

Isbot:

$$\begin{cases} A_1x + B_1y + C_1 = 0 \\ A_2x + B_2y + C_2 = 0 \end{cases} \rightarrow \begin{cases} A_1x + B_1y = -C_1 \\ A_2x + B_2y = -C_2 \end{cases}$$

Kramer qoidasiga ko'ra: $x = \frac{\Delta_x}{\Delta}$; $y = \frac{\Delta_y}{\Delta}$ (•)

Misollar:

① Quyidagi to'g'ri chiziqlar orasidagi burchak aniqlansin:

a) $\begin{cases} y = 2x - 3 \\ y = \frac{1}{2}x + 1 \end{cases}$

$$tg\varphi = \frac{k_1 - k_2}{1 + k_1 k_2} = \frac{2 - \frac{1}{2}}{1 + 2 \cdot \frac{1}{2}} = \frac{\frac{3}{2}}{2} = \frac{3}{4}$$

$$\varphi = arctg\frac{3}{4}$$

b) $\begin{cases} 5x - y + 7 = 0 \\ 2x - 3y + 1 = 0 \end{cases}$

$$tg\varphi = \frac{A_1B_2 - A_2B_1}{A_1A_2 + B_1B_2} = \frac{5 \cdot (-3) - 2 \cdot (-1)}{5 \cdot 2 + (-1) \cdot (-3)} = \frac{-13}{13} = -1$$

$\varphi = 135°$ yoki $45°$

c) $\begin{cases} \frac{x}{a} + \frac{y}{b} = 1 \ /ab \\ \frac{x}{b} + \frac{y}{a} = 1 \ /ab \end{cases} \Rightarrow \begin{cases} bx + ay = ab \\ ax + by = ab \end{cases}$

$$tg\varphi = \frac{A_1B_2 - A_2B_1}{A_1A_2 + B_1B_2} = \frac{b^2 - a^2}{2ab} \rightarrow \varphi = arctg\frac{b^2 - a^2}{2ab}$$

② Uchburchak tomonlari $x + 2y = 0$, $x + 4y - 6 = 0$, $x - 4y - 6 = 0$ tenglamalar bilan berilgan. Uning ichki burchaklari topilsin.

$$tg\varphi_{12} = \frac{1 \cdot 4 - 1 \cdot 2}{1 \cdot 1 + 2 \cdot 4} = \frac{2}{9} \qquad tg\varphi_{23} = \frac{1 \cdot (-4) - 1 \cdot 4}{1 \cdot 1 - 4 \cdot 4} = \frac{-8}{-15} = \frac{8}{15}$$

$$tg\varphi_{31} = \frac{1 \cdot 2 - 1 \cdot (-4)}{1 \cdot 1 + 2 \cdot (-4)} = \frac{6}{-7} = -\frac{6}{7}$$

③ $A(-1;1)$ nuqtadan o'tib, $2x+3y=6$ to'g'ri chiziq bilan $45°$ burchak tashkil etuvchi to'g'ri chiziq tenglamasi yozilsin.

$y-y_1=k(x-x_1)$
$y-1=k(x+1)$
$y=kx+(k+1)$

$3y=-2x+6$
$y=-\frac{2}{3}x+2$

$k_1=k \quad k_2=-\frac{2}{3}$

$tg\varphi=tg45°=1=\frac{-\frac{2}{3}-k}{1-\frac{2}{3}k}$

$1-\frac{2}{3}k=-\frac{2}{3}-k \;/\cdot 3$

$3-2k=-2-3k$

$k=-5 \rightarrow y=-5x-4$

$tg45°=1=\frac{k+\frac{2}{3}}{1-\frac{2}{3}k} \rightarrow k+\frac{2}{3}=1-\frac{2}{3}k \;/\cdot 3$

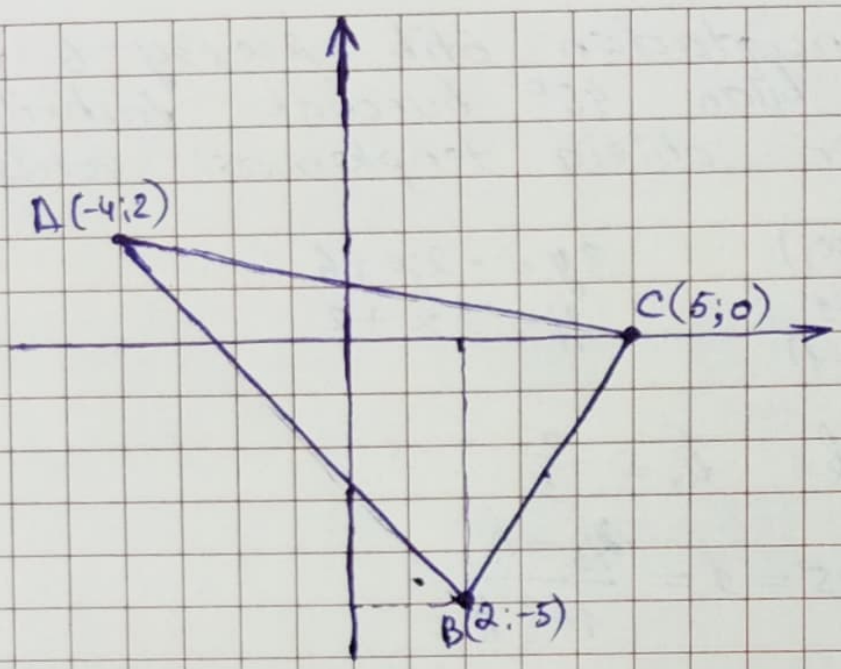
$3k+2=3-2k$
$5k=1 \rightarrow k=\frac{1}{5}$

$y=\frac{1}{5}x+\frac{6}{5} \rightarrow x-5y+6=0$

④ Uchlari $A(-4;2)$, $B(2;-5)$, $C(5;0)$ nuqtalarda bo'lgan uchburchakning mediana va ~~bissektrisa~~ balandliklari kesishish nuqtalarini toping.

Medianalar kesishish nuqtasi:

$x=\frac{x_1+x_2+x_3}{3}=\frac{-4+2+5}{3}=1 \qquad y=\frac{y_1+y_2+y_3}{3}=\frac{2-5+0}{3}=-1$


A(-4;2)
C(5;0)
B(2;-5)

21-savol: Tōpri chiziqning normal tenglamasi. Nuqtadan tōpri chiziqgacha bolgan masofa.

1. Tōpri chiziqning normal tenglamasi.

$$x\cdot\cos\beta + y\cdot\sin\beta - p = 0$$

p - koordinata boshidan tōpri chiziqqa tushirilgan perpendikulyar (normal) uzunligi.

β - shu perpendikulyarning Ox ōqiga ōgish burchagi.

2. $(x_0; y_0)$ nuqtadan $x\cdot\cos\beta + y\cdot\sin\beta - p = 0$ tōpri chiziqgacha bolgan masofa:

$$d = |x_0\cos\beta + y_0\sin\beta - p|$$

yoki.

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2+B^2}}$$

Tōpri chiziqning umumiy tenglamasi $Ax+By+C=0$ ni normal korinishga keltirish u-n uni $M = \pm\frac{1}{\sqrt{A^2+B^2}}$ (normallashtirish koepaytuvchisi)ga kopaytirish k-k. Uning ishorasi C ga teskari qilib olinadi.

3. $Ax+By+C=0$ va $A_1x+B_1y+C_1=0$ tōpri chiziqlar orasidagi burchak bissektrisasi tenglamasi.

$$\frac{Ax+By+C}{\sqrt{A^2+B^2}} = \pm\frac{A_1x+B_1y+C_1}{\sqrt{A_1^2+B_1^2}}$$

① Tǒpri chiziqlarni normal kǒrinishga keltiring.

a) $3x-4y-20=0 \quad / \cdot \frac{1}{\sqrt{3^2+4^2}}=\frac{1}{5}$

$\frac{3}{5}x-\frac{4}{5}y-4=0$

b) $x+y+3=0 \quad \cdot / -\frac{1}{\sqrt{1+1}}=-\frac{1}{\sqrt{2}}$

$-\frac{\sqrt{2}}{2}x-\frac{\sqrt{2}}{2}y-\frac{3}{\sqrt{2}}=0$

c) $y=kx+b \rightarrow kx-y+b=0 \quad / \cdot -\frac{1}{\sqrt{k^2+1}}$

$-\frac{k}{\sqrt{1+k^2}}x+\frac{1}{\sqrt{1+k^2}}y-\frac{b}{\sqrt{k^2+1}}=0$

② $A(4;3)$, $B(2;1)$ va $C(1;0)$ nuqtadan $3x+4y-10=0$ tǒpri chiziqqacha bǒlgan masofani toping.

$3x+4y-10=0 \quad / \cdot \frac{1}{\sqrt{3^2+4^2}}$

$\frac{3}{5}x+\frac{4}{5}y-2=0$

$A(4;3) \rightarrow \frac{3}{5}\cdot 4+\frac{4}{5}\cdot 3-2=\frac{14}{5}=2\frac{4}{5}$

③ $2x-3y=6$ va $4x-6y=25$ tǒpri chiziqlar ǒzaro parallel ekanligi kǒrsatilsin va ular orasidagi masofa aniqlansin.

$y=\frac{2}{3}x-2 \qquad y=\frac{2}{3}x-\frac{25}{6} \qquad k_1=k_2$ demak $\parallel$

$x=0 \rightarrow y=-2.$

$(0;-2)$ nuqtadan $6y-4x+25=0$ tǒpri chiziqqacha bǒlgan masofani topamiz.

$$d=\frac{|Ax_0+By_0+C|}{\sqrt{A^2+B^2}}=\frac{|-4\cdot 0+6\cdot(-2)+25|}{\sqrt{16+36}}=\frac{13}{2\sqrt{13}}=\frac{\sqrt{13}}{2}$$

④ $y=kx+5$ to'g'ri chiziq koordinatalar boshidan $d=\sqrt{5}$ masofa uzoqlikda bo'lsa, k topilsin.

$$kx-y+5=0$$

$$d=\frac{|Ax_0+By_0+C|}{\sqrt{A^2+B^2}}=\sqrt{5}$$

$$\frac{|k\cdot 0-1\cdot 0+5|}{\sqrt{k^2+1}}=\sqrt{5}\rightarrow\frac{5}{\sqrt{5}}=\sqrt{k^2+1}$$

$$5=k^2+1\rightarrow \boxed{k=\pm 2}$$

⑤ $2x+3y=12$ va $3x+2y=12$ to'g'ri chiziqlar orasidagi burchaklar bissektrisalarining tenglamalari yozilsin.

$$\frac{Ax+By+C_1}{\sqrt{A_1^2+B_1^2}}=\pm\frac{Ax_2+By_2+C}{\sqrt{A_2^2+B_2^2}}$$

$$\frac{2x+3y-12}{\sqrt{4+9}}=\pm\frac{3x+2y-12}{\sqrt{9+4}}$$

$$2x+3y-12=3x+2y-12 \qquad 2x+3y-12=-3x-2y+12$$
$$x-y=0 \qquad 5x+5y-24=0$$

⑥ $A(2;4)$ nuqtadan o'tuvchi va koordinatalar boshidan $d=2$ uzoqlikda bo'lgan to'g'ri chiziq tenglamasi yozilsin

$x_0 \cos\varphi + y_0 \sin\varphi - 2 = 0$

$2\cos\varphi + 4\sin\varphi - 2 = 0$

$\cos\varphi + 2\sin\varphi - 1 = 0$

$\cos^2\frac{\varphi}{2} - \sin^2\frac{\varphi}{2} + 4\sin\frac{\varphi}{2}\cdot\cos\frac{\varphi}{2} - \cos^2\frac{\varphi}{2} - \sin^2\frac{\varphi}{2} = 0$

$4\sin\varphi/2 \cdot \cos\varphi/2 - 2\sin^2\varphi/2 = 0$

$\sin\frac{\varphi}{2}\left(2\cos\frac{\varphi}{2} - \sin\frac{\varphi}{2}\right) = 0$

$\sin\frac{\varphi}{2} = 0$ $\qquad$ $\operatorname{tg}\varphi/2 = 2$

$\frac{\varphi}{2} = 0 \to \varphi = 0$ $\qquad$ $\frac{\varphi}{2} = \operatorname{arctg}2 \to \varphi = 2\operatorname{arctg}2$

$x - 2 = 0 \to x = 2$

$\cos\varphi = \cos(2\operatorname{arctg}2) = \cos^2(\operatorname{artg}2) - \sin^2(\operatorname{arctg}2) =$

$= \frac{1}{5} - \frac{4}{5} = -\frac{3}{5}$

$\sin\varphi = \sin(2\operatorname{arctg}2) = 2\sin(\operatorname{arctg}2)\cdot\cos(\operatorname{arctg}2) =$

$= 2\cdot\frac{2}{\sqrt{5}}\cdot\frac{1}{\sqrt{5}} = \frac{4}{5}$

$x\cdot\left(-\frac{3}{5}\right) + y\cdot\frac{4}{5} - 2 = 0 \quad / \cdot(-5)$

$3x - 4y + 10 = 0$

22-savol. Aylana. Ellips.

Aylana

Markazi $C(a;b)$ nuqtada va radiusi R bolgan aylana tenglamasi:

$$\boxed{(x-a)^2+(y-b)^2=R^2}$$

Misollar.

① $x^2+y^2+5x=0$ aylana va $x+y=0$ to'g'ri chiziq yasalsin va ularning kesishgan nuqtalari topilsin.

$x+y=0 \to x=-y$

$(-y)^2+y^2+5(-y)=0$

$2y^2-5y=0$

$y(2y-5)=0$

$y=0 \quad y=2,5$

$x=0 \quad x=-2,5$

$(0;0),\ (-2,5;2,5)$

② $A(1;2)$ nuqtadan o'tuvchi va koordinata o'qlariga urinuvchi aylana tenglamasi yozilsin.

Yechish: Koordinata o'qlariga urinuvchi aylana-ning markazi ordinatasi va abssissasi teng bo'ladi.

$O(x;y) \to x=y=R$

$(x-a)^2+(y-b)^2=R^2$

$(x-1)^2+(x-2)^2=R^2$

$2x^2-6x+5=x^2$

$x^2-6x+5=0$

$x=1 \quad x=5$

$O_1(1;1)$ $(x-1)^2+(y-1)^2=1$

$O_2(5;5)$ $(x-5)^2+(y-5)^2=25$

(5) A(-1;3), B(0;2), C(1;-1) nuqtalardan o'tuvchi aylana tenglamasini yozing.

$$x^2+y^2+mx+ny+p=0$$

$$\begin{cases} A \to 1+9-m+3n+p=0 \\ B \to 4+2n+p=0 \\ C \to 1+1+m-n+p=0 \end{cases} \Rightarrow \begin{cases} m-3n-p=10 \\ 2n+p=-4 \quad /(-1) \\ m-n+p=-2 \end{cases}$$

$$\begin{cases} m-3n-p=10 \\ m-3n=2 \end{cases} \Rightarrow 2-p=10$$

$p=-8$ $n=2$ $m=8$

$$x^2+y^2+8x+2y-8=0$$

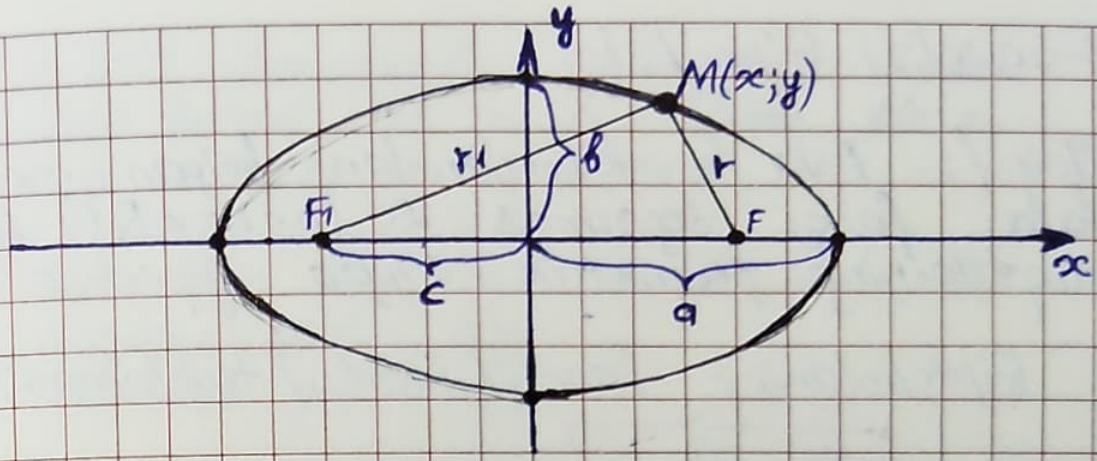
Ellips

Tarif: Fokuslar deb atalgan F va F_1 nuqtalargacha bo'lgan masofalar yig'indisi o'zgarmas $2a$ $(2a > FF_1)$ miqdorga teng bo'lgan nuqtalarning geometrik yig'indisi **ellips** deb ataladi.

Ellipsning kanonik tenglamasi:

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

Bu yerda:
a, b – yarim o'qlar.
Agar $a>b$ bo'lsa, u holda F va F_1 fokuslar koordinata boshidan $c=\sqrt{a^2-b^2}$ masofada yotadi.
$\frac{c}{a}=\varepsilon<1$ → ekssentrisitet



$M(x;y)$ nuqtadan ellipsning F va F_1 fokuslari-gacha bolgan masofalar fokal radius-vektorlar deyiladi va quyidagi formulalar b-n topiladi:

$$\boxed{r = a - \varepsilon x} \qquad \boxed{r_1 = a + \varepsilon x}$$

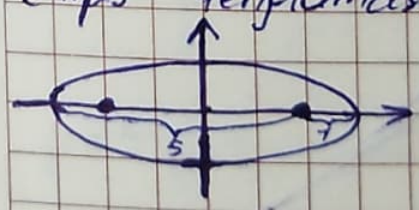
Misollar:

① $x^2 + 4y^2 = 16$ ellipsni yasang, uning fokusi va ekstentrisitetini toping.

$$x^2 + 4y^2 = 16 \;/:16$$
$$\frac{x^2}{4^2} + \frac{y^2}{2^2} = 1 \rightarrow \begin{matrix} a = 4 \\ b = 2 \end{matrix}$$

$$c = \sqrt{a^2 - b^2} = \sqrt{16 - 4} = \boxed{2\sqrt{3}} \qquad \varepsilon = \frac{c}{a} = \frac{2\sqrt{3}}{4} = \boxed{\frac{\sqrt{3}}{2}}$$

② Fokuslaridan biridan to katta o'qining uchlarigacha bolgan masofalar 5 va 1 bolgan ellips tenglamasini tuzing.



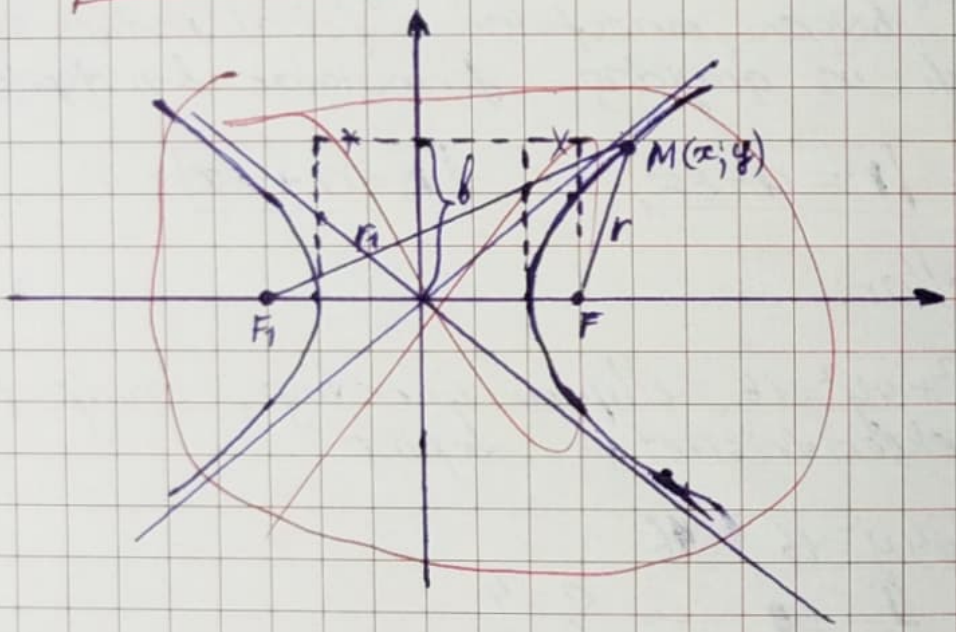
$$c = \frac{5-1}{2} = 2 \qquad a = 3 \qquad b = \sqrt{a^2 - c^2} = \sqrt{9-4} = \sqrt{5}$$
$$\frac{x^2}{9} + \frac{y^2}{5} = 1 \;/\cdot 45$$
$$\boxed{5x^2 + 9y^2 = 45}$$

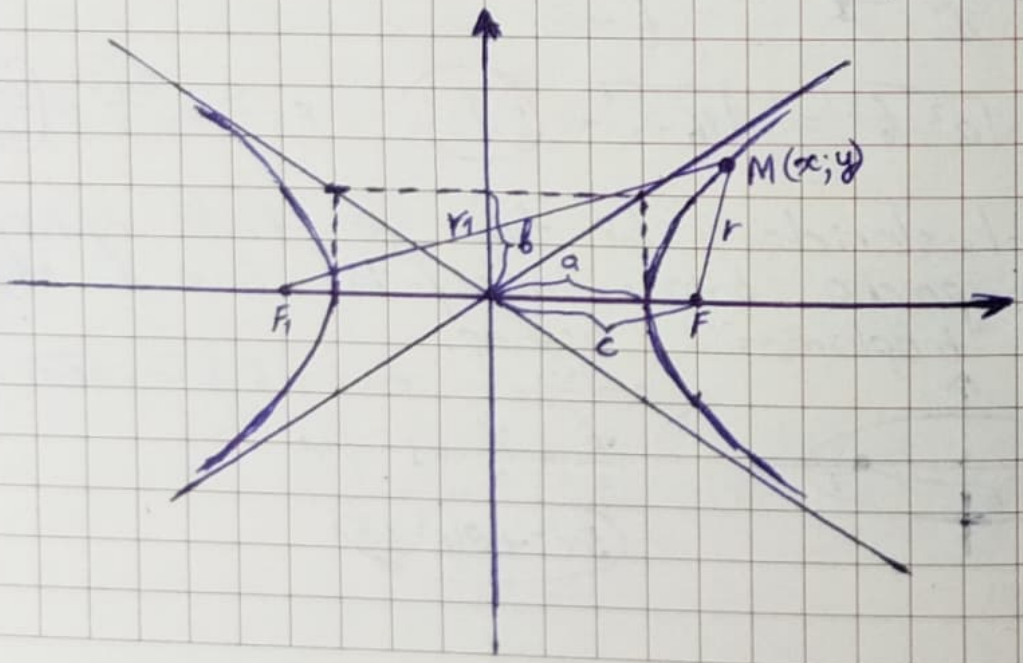
23-saəd: Giperbola

Tə'rif: F və F_1 fokuslarından böyük məsafələr fərqi özgarmaz 2a ($0 < 2a < F_1F$) böyük nuqtaların həndəsi yerinə giperbola deyilir.

Giperbolanın kanonik (adi) tənlaməsi:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$





Giperbola koordinata o'qlariga nisbatan simmetrik bo'ladi. U ox o'qi bilan $A(a;0)$, $A_1(-a;0)$ nuqtalarda kesib o'tadi, bu nuqtalar giperbolaning uchlari deyiladi.

Giperbola Oy o'qini kesib o'tmaydi.

a - haqiqiy yarim o'q

b - mavhum yarim o'q

Koordinatalar boshidan uning fokusigacha bo'lgan masofa:

$$c = \sqrt{a^2 + b^2}$$

Eksentrisitet:

$$\frac{c}{a} = \varepsilon > 1$$

Asimptota:

$$y = \pm \frac{b}{a} x$$

Fokal radius-vektorlar:

$$r = |\varepsilon x - a|, \qquad r_1 = |\varepsilon x + a|$$

Misollar. $x^2 - 4y^2 = 16$ giperbolani yasang, uning asimptotasini toping. Fokuslar, eksentrisitet va asimptotalarini ~~toping~~. orasidagi $<$ ni toping.

$$x^2 - 4y^2 = 16 \quad / :16$$

$$\frac{x^2}{4^2} - \frac{y^2}{2^2} = 1$$

$a = 4 \quad b = 2$

$$c = \sqrt{a^2 + b^2} = \sqrt{16 + 4} = 2\sqrt{5}$$

$$y = \pm \frac{2}{4} x = \pm \frac{1}{2} x \rightarrow \text{asimptota}$$

$$\varepsilon = \frac{c}{a} = \frac{2\sqrt{5}}{4} = \frac{\sqrt{5}}{2} \rightarrow \text{eksentrisitet}$$

$k_1 = \frac{1}{2} \quad k_2 = -\frac{1}{2} \quad tg\varphi = \frac{k_1 - k_2}{1 + k_1 k_2} = \frac{\frac{1}{2} + \frac{1}{2}}{1 - \frac{1}{4}} = \frac{1}{\frac{3}{4}} = \frac{4}{3}$

$\varphi = arctg\frac{4}{3}$

24-savol. Parabola.

Ta'rif: Fokus va direktrisadan deb ataluvchi nuqta va tekislikdagi chiziqdan bir xil masofada joylashgan nuqtalarning geometrik o'rniga parabola deyiladi.

Parabolaning 2 xil ko'rinishidagi kanonik tenglamasi mavjud:

1) $y^2=2px$ – parabola Ox o'qiga nisbatan simmetrik.

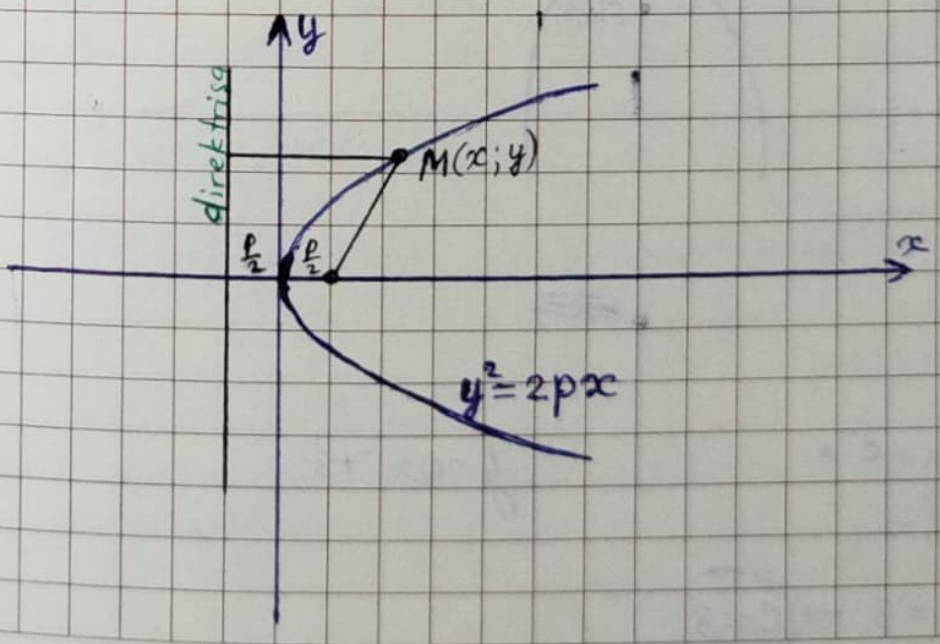
2) $x^2=2py$ – parabola Oy o'qiga nisbatan simmetrik.

Ikkala holatda ham uning uchi koordinatalar boshida bo'ladi.

① $y^2=2px$ fokusi $F(\frac{p}{2};0)$ nuqtada bo'ladi.

direktrisasi $x=-\frac{p}{2}$

fokal radius-vektori $r=x+\frac{p}{2}$



② $x^2 = 2py$ ning fokusi $F(0; \frac{p}{2})$ nuqtada yotadi.

direktrisasi: $y = -\frac{p}{2}$

fokal radius-vektori: $r = y + \frac{p}{2}$

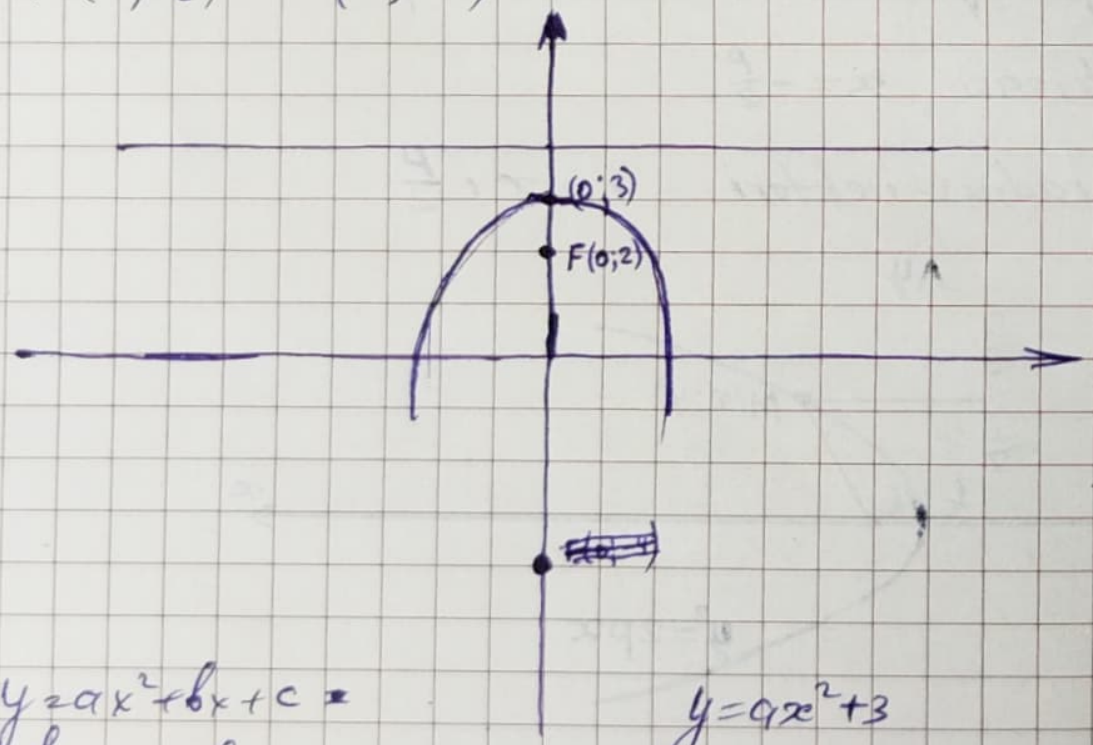
Misollar:

① $F(0;2)$ nuqtadan va $y=4$ to'g'ri chiziqdan bir xil uzoqlashgan nuqtalar geometrik o'rnining tenglamasi tuzilsin. Bu egri chiziqning koordinata o'qlari bilan kesishgan nuqtalari topilsin va u yasalsin.

$y = 4 \rightarrow$ direktrisa $\quad x^2 = 2py$

$y = -\frac{p}{2} = 4 \rightarrow p = -8 \rightarrow x^2 = -16y$

~~$F(0; \frac{p}{2}) = F(0; -4)$~~



$y = ax^2 + bx + c$

$y = ax^2 + 3$

$x_0 = -\frac{b}{2a} = 0 \rightarrow b = 0$

$y_0 = ax_0^2 + c = 3 \rightarrow \boxed{c = 3}$

$x_0 = 0$

$x^2 = 2py \rightarrow (0;3)$ nuqtasi parabolaning uchi

demak direktrisa $y = 4 - 3 = 1 = -\frac{p}{2} \rightarrow p = -2$

$x^2 = -4y \rightarrow \boxed{y = -\frac{1}{4}x^2}$

$\boxed{y = -\frac{x^2}{4} + 3}$ $\qquad y = 0 \rightarrow x^2 = 12$

$\boxed{x = \pm 2\sqrt{3}}$

② koordinatalar boshidan va $x = -4$ to'g'ri chiziqdan bir xil uzoqlikda bo'lgan nuqtalar geometrik o'rnining tenglamasi tuzilsin. Bu egri chizig'ning koordinata o'qlari bilan kesishgan nuqtalari topilsin va u yasalsin.

Parabolaning uchi: $(-2;0)$ nuqtada

demak direktrisa: $x = -2 = -\frac{p}{2} \rightarrow p = 4$.

$y^2 = 2px = 8x$

$\boxed{y^2 = 8(x+2)}$

③ $y^2 = 4x$ tenglama b-n berilgan parabola fokusi, direktrisasi tenglamasi tuzilsin.

$y^2 = 2px = 4x \rightarrow p = 2$

$F\left(\frac{p}{2};0\right) = \boxed{F(1;0)}$ $\qquad \boxed{y = -\frac{p}{2} = -1}$

25-savol: Ikkinchi tartibli egri chiziqning direktrisasi, diametri, markazi va ularga o'tkazilgan urinmalar

1. Oy ⊥ga parallel va undan $\frac{a}{\varepsilon}$ masofada joylashgan to'g'ri chiziqlar $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $(a>b)$ ellipsning va $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ giperbolaning direktrisalari d-di, bunda ε - egri chiziqning ekssentrisiteti.

Direktrisalarning tenglamalari:

$$\boxed{x=\pm\frac{a}{\varepsilon}}$$

Direktrisalarning (~~xossalari~~:) xossasi: Egri chiziq ixtiyoriy nuqtasining fokuspacha va mos direktrisagacha masofalarning nisbati ekssentrisitetga teng: $\boxed{\frac{r}{d}=\varepsilon}$

2. Ikkinchi tartibli egri chiziqning diametri deb, parallel vatarlar o'rtalarining geometrik o'rniga aytiladi. Ellips~~[illegible]~~ bilan giperbolaning diametrlari ularning markazlaridan o'tuvchi to'g'ri chiziqlar tesmalaridan va nurlaridan iboratdir.

$\frac{x^2}{a^2}\pm\frac{y^2}{b^2}=1$ egri chiziqlar uchun o'qmaliklari $k=\operatorname{tg}\alpha$ bo'lgan vatarlarni teng bo'luvchi diametrning tenglamasi $\boxed{y=\pm\frac{b^2}{a^2k}x}$ bo'lsa,

$y^2 = 2px$ parabola u-n:

$$y = \frac{p}{k}$$

boladi.

Ellips va giperbolada bir diametr 2-diametrga parallel bolgan vatarlarni teng 2 ga bolsa, bunday 2 diametr ozaro qoshma deyiladi. Qoshma digmetrlarning burchak koeffitsiyentlari k va k_1 bolsa:

$$kk_1 = -\frac{b^2}{a^2} \text{ (ellips uchun)}$$

$$kk_1 = \frac{b^2}{a^2} \text{ (giperbola uchun)}$$

tengliklar bilan boglanganadir.

3. $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1\right)$ ellipsga otkazilgan urinmaning tenglamasi $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$

$\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\right)$ giperbolaga otkazilgan urinmaning tenglamasi:

$$\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$$

$(y^2 = 2px)$ parabolaga otkazilgan urinmaning tenglamasi

$$y \cdot y_0 = p(x + x_0)$$

x_0, y_0 – urinish nuqtasi

Misollar:

① $\frac{x^2}{25}+\frac{y^2}{9}=1$ ellips va uning direktrisalari yasalsin. Ellipsning $x=-3$ abssissasidan ong fokusigacha va ong direktrisasigacha bolgan masofalar topilsin.

$a=5,\ b=3 \qquad c=\sqrt{a^2-b^2}=4$

$\varepsilon=\frac{c}{a}=\frac{4}{5}$

$x=\pm\frac{a}{\varepsilon}=\pm\frac{5}{\frac{4}{5}}=\pm\frac{25}{4}$

$d_1=4-(-3)=7.$

$d_2=\frac{25}{4}-(-3)=\frac{37}{4}$

② $\frac{x^2}{16}-\frac{y^2}{9}=1$ giperbola va uning direktrisalari yasalsin, giperbolaning $x=5$ abssissasidan chap fokusigacha va chap direktrisasigacha bolgan masofalar topilsin.

$a=4 \quad b=3 \qquad c=\sqrt{a^2+b^2}=5. \qquad \varepsilon=\frac{c}{a}=\frac{5}{4}$

$x=\pm\frac{a}{\varepsilon}=\pm\frac{4}{\frac{5}{4}}=\pm\frac{16}{5}$

$d_1=5-(-5)=10 \qquad d_2=5-\left(-\frac{16}{5}\right)=\frac{41}{5}$

③ katta yarim ogi 2 ga teng, direktrisalari $x=\pm\frac{4}{\sqrt{3}}$ toping chizgilardan iborat ellipsning kanonik tenglamasi yozilsin.

$x = \pm\frac{a}{\varepsilon} = \pm\frac{4}{\sqrt{3}}$

$\frac{a^2}{c} = \frac{4}{\sqrt{3}}$ $\quad$ $\frac{a^2}{\sqrt{a^2-b^2}} = \frac{4}{\sqrt{3}}$

$\frac{4}{\sqrt{4-b^2}} = \frac{4}{\sqrt{3}} \rightarrow 4-b^2=3$ $\quad b=1$.

$\frac{x^2}{4} + \frac{y^2}{1} = 1 \rightarrow x^2+4y^2=4$

④ Asimptotalari $y=\pm x$, direktrisalari esa $x=\pm\sqrt{6}$ bolgan giperbolaning tenglamasi yozilsin.

$y = \pm\frac{b}{a}x = \pm x \rightarrow a=b$.

$x = \pm\frac{a}{\varepsilon} = \pm\frac{a^2}{c} = \pm\sqrt{6}$.

$\frac{a^2}{\sqrt{a^2+b^2}} = \sqrt{6} \rightarrow \frac{a^2}{\sqrt{2}a} = \sqrt{6} \rightarrow a=2\sqrt{3}=b$

$x^2-y^2=12$

⑤ $x^2+2y^2=8$ ellipsga $(0;6)$ nuqtadan o'tkazilgan urinmalarning tenglamalari yozilsin.

$\frac{x^2}{8} + \frac{y^2}{4} = 1$

$\frac{x\cdot 0}{8} + \frac{y\cdot 6}{4} = 1 \rightarrow y=\frac{2}{3}$

$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$ $\quad$ $\frac{x^2}{8} + \frac{y^2}{4} = 1$

6) $\rightarrow \frac{0\cdot x_0}{8} + \frac{6\cdot y_0}{4} = 1 \rightarrow y_0=\frac{2}{3}$ $\quad$ $\frac{x_0^2}{8} + \frac{\left(\frac{2}{3}\right)^2}{4} = 1$

$\frac{x_0^2}{8} = \frac{8}{9} \rightarrow x_0 = \pm\frac{8}{3}$

$$\frac{x \cdot x_0}{8} + \frac{y \cdot y_0}{4} = 1$$

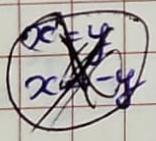
$$\frac{x \cdot \left(\pm\frac{8}{3}\right)}{8} + \frac{y \cdot \left(\frac{2}{3}\right)}{4} = 1$$

$$\pm\frac{x}{3} + \frac{y}{6} = 1 \quad / \cdot 6$$

$$\pm 2x + y = 6 \rightarrow \quad y = \pm 2x + 6$$

⑥ $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ellipsning koordinata o'qlaridan teng kesmalar ajratuvchi urinmasining tenglamasi yozilsin.

$$\frac{x \cdot x_0}{a^2} + \frac{y \cdot y_0}{b^2} = 1$$



26-savol. Vektorlar va ular ustida amallar.

Ta'rif: Yo'naltirilgan $\overline{AB}$ kesma vektor d-di.
A – vektorning boshi
B – vektorning oxiri.
$|AB|$ yoki $|a|$ – vektorning moduli (uzunligi)

Bir to'g'ri chiziqqa parallel bo'lgan vektorlar **kollinear vektorlar** d-di.

Bir tekislikka parallel bo'lgan vektorlar **komplanar vektorlar** d-di.

Agar 2 a va b vektorlar:
1) teng modulga ega
2) o'zaro kollinear
3) bir tomonga yo'nalgan bo'lsa,
ular **o'zaro teng** d-di.

Vektorning skalyarga ko'paytmasi.
a vektorning biror m songa (skalyarga) ko'paytmasi deb, uzunligi $a|m|$ ga teng bo'lgan va yo'nalishi esa berilgan vektor yo'nalishidagi $(m>0)$ yoki unga qarama-qarshi $(m<0)$ bo'lgan yangi vektorga aytiladi.

Misollar:
① Tekislikda $A(0;-2)$, $B(4;2)$ va $C(4;-2)$ nuqtalar berilgan. Koordinatalar boshidan $\overline{OA}$, $\overline{OB}$ va $\overline{OC}$ kuchlar qo'yilgan. Ularning teng ta'sir etuvchisi $\overline{OM}$ yasalsin va uning o'qlardagi proyeksiyalari hamda uzunligi topilsin. $\overline{OA}$, $\overline{OB}$, $\overline{OC}$ va $\overline{OM}$ kuchlar

i və j birlik vektorlar ərgəli ifadələnsin.

$\overline{OA} = (0; -2)$ $\overline{OB} = (4; 2)$ $\overline{OC}\ (4; -2)$

$\overline{OM} = \overline{OA} + \overline{OB} + \overline{OC} = (0+4+4; -2+2-2) = (8; -2)$

$pr_x \overline{OM} = 8$ $pr_y \overline{OM} = -2$

$|\overline{OM}| = \sqrt{64+4} = 2\sqrt{17}$

28-savol. Kollinearlik, komplanarlik. Bazis.

1-ta'rif. Yonalgan kesma vektor d-di.

2-ta'rif: Boshi va oxiri ustma-ust tushgan vektor nol vektor d-di.

3-ta'rif. Bitta to'g'ri chiziqda yoki parallel to'g'ri chiziqlarda yotuvchi $\vec{a}$ va $\vec{b}$ vektorlar kollinear vektorlar d-di.

4-ta'rif. $\vec{a}$ va $\vec{b}$ vektorlar kollinear, bir xil yonalgan va uzunliklari teng bolsa, ular teng vektorlar d-di.

$$\vec{a} = \vec{b}$$

5-ta'rif. Bitta tekislikda yoki parallel tekisliklarda yotuvchi vektorlar komplanar vektorlar deb ataladi.

n ta $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ vektor va shuncha $\alpha_1, \alpha_2, \dots, \alpha_n$ sonni qaraymiz. Bu sonlarning mos vektorlarga kopaytmalari yigindisi, $\alpha_1\vec{a}_1 + \alpha_2\vec{a}_2 + \dots + \alpha_n\vec{a}_n$ vektorlarning chiziqli kombinatsiyasi deb ataladi.

Ta'rif. $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ vektorlar sistemasi uchun kamida bittasi noldan farqli shunday n ta $\alpha_1, \alpha_2, \dots, \alpha_n$ sonlar mavjud bolsaki, vektorlarning chiziqli kombinatsiyasi nolga teng, ya'ni

$$\alpha_1\vec{a}_1 + \alpha_2\vec{a}_2 + \dots + \alpha_n\vec{a}_n = 0 \quad (1)$$

bolsa, u sistema chiziqli bog'liq sistema d-di

Aks holda $\bar{a}_1, \bar{a}_2, \dots, \bar{a}_n$ vektorlar chiziqli erkli deb ataladi, ular uchun (1) tenglik faqat $\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$ bo'lganda o'rinli bo'ladi.

Ta'rif. Istalgan $\bar{a}$ vektorni n-ta chiziqli erkli $\bar{e}_1, \bar{e}_2, \dots, \bar{e}_n$ vektorlarning chiziqli kombinatsiyasi orqali ifodalash mumkin bo'lsa, u holda bu vektorlar fazoning bazisi deb ataladi.

Bazis hosil qiladigan vektorlar soni fazoning o'lchami d-di.

Bazisga kiruvchi vektorlar, bazis vektorlar d-di.

29-savol. Skalyar ko'paytma.

Ta'rif. Ikkita $\bar{a}$ va $\bar{b}$ vektorlarning skalyar ko'paytmasi deb bu vektorlar uzunliklarini ular orasidagi burchak kosinusiga ko'paytmasiga teng bo'lgan skalyarga (songa) aytiladi.

$$\boxed{\bar{a}\cdot\bar{b} = |\bar{a}|\cdot|\bar{b}|\cdot\cos\varphi}$$

Agar vektorlar $a(a_x, a_y, a_z)$ va $b(b_x, b_y, b_z)$ koordinatalar orqali berilgan bo'lsa,

$$\boxed{\bar{a}\bar{b} = a_x b_x + a_y\cdot b_y + a_z\cdot b_z}$$

Ikki vektor orasidagi burchak.

$$\boxed{\cos\varphi = \frac{a\cdot b}{|a||b|} = \frac{a_x b_x + a_y b_y + a_z b_z}{\sqrt{a_x^2+a_y^2+a_z^2}\cdot\sqrt{b_x^2+b_y^2+b_z^2}}}$$

Parallellik sharti: $b = m\cdot a$ yoki $\frac{b_x}{a_x} = \frac{b_y}{a_y} = \frac{b_z}{a_z} = m$

Perpendikulyarlik sharti: $ab = 0$ yoki $a_x b_x + a_y b_y + a_z b_z = 0$

Misollar.

(1) $a = -i + j$ va $b = i - 2j + 2k$ vektorlar orasidagi burchak aniqlansin.

Yechish: $a(-1; 1; 0)$ va $b(1; -2; 2)$

$$\cos\varphi = \frac{\bar{a}\cdot\bar{b}}{|\bar{a}|\cdot|\bar{b}|} = \frac{-1\cdot 1 + 1\cdot(-2) + 0\cdot 2}{\sqrt{1^2+1^2}\cdot\sqrt{1^2+2^2+2^2}} = \frac{-3}{\sqrt{2}\cdot 3} = -\frac{\sqrt{2}}{2}$$

$$\varphi = 135^\circ$$

② Uchlari $A(2;-1;3)$, $B(1;1;1)$ va $C(0;0;5)$ nuqtalarda bolgan $\triangle ABC$ ning burchaklari aniqlansin.

$$\cos\alpha = \frac{\overline{AB}\cdot\overline{AC}}{|\overline{AB}|\cdot|\overline{AC}|} = \frac{(-1;2;-2)\cdot(-2;1;2)}{\sqrt{1^2+2^2+2^2}\cdot\sqrt{2^2+1^2+2^2}} =$$

$$= \frac{2+2-4}{} = 0 \qquad \alpha = 90^\circ$$

$$\cos\beta = \frac{\overline{BA}\cdot\overline{BC}}{|\overline{BA}|\cdot|\overline{BC}|} = \frac{(1;-2;2)\cdot(-1;-1;4)}{\sqrt{1^2+2^2+2^2}\cdot\sqrt{1^2+1^2+4^2}} =$$

$$= \frac{-1+2+8}{3\cdot 3\sqrt{2}} = \frac{1}{\sqrt{2}} \rightarrow \beta = 45^\circ \rightarrow \gamma = 45^\circ$$

③ $(2i-j)\cdot j + (j-2k)\cdot k + (i-2k)^2$ ifodadagi qavslar ochilsin.

$i\cdot i = j\cdot j = k\cdot k = 1$

$i\cdot j = j\cdot k = i\cdot k = 0$

$$2ij - j^2 + jk - 2k^2 + i^2 - 4ik + 4k^2 =$$

$$= 2\cdot 0 - 1 + 0 - 2\cdot 1 + 1 - 4\cdot 0 + 4\cdot 1 =$$

$$= 0 - 1 + 0 - 2 + 1 - 0 + 4 = 2$$

30-savol. Vektor ko'paytma. Aralash ko'paytma.

Ta'rif. a va b vektorlarning vektor ko'paytmasi deb shunday uchinchi c vektorga aytiladiki:

1) u son qiymati bo'yicha berilgan a va b vektorlarda yasalgan parallelogramm yuziga teng modulga teng.

2) u parallelogramm tekisligiga perpendikular.

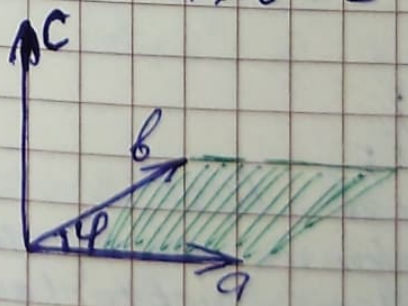
3) u shunday tomonga yo'naltirilganki, uning uchidan qaraganda a vektordan b vektorga qarab eng kichik burilish soat strelkasiga qarama-qarshi boladi.

a, b va c vektorlarning bu xildagi joylanishiga o'ng bog'lam deyiladi.

Ikki vektorning vektor ko'paytmasi $a \times b$ kommishda belgilanadi, shunday qilib, agar

1) $|c| = |a \times b| = ab\sin\varphi$
2) $c \perp a$ va $c \perp b$,
3) a, b, c lar o'ng bog'lam hosil qilsa

$a \times b = c$ boladi.



Vektor ko'paytmaning xossalari:

I) $a \times b = -b \times a$

II) $a \times (b+c) = a \times b + a \times c$

III) Agar $a \parallel b$ bo'lsa, $a \times b = 0$, xususiy holda $a \times a = 0$

Ortlarning vektor ko'paytmalari:

$i \times j = k$, $j \times k = i$, $k \times i = j$.

$ijkij \rightarrow j \cdot i = -k$, $k \cdot j = -i$, $i \cdot k = -j$

Vektor ko'paytmani ko'paytuvchilar koordinatalari $a(a_x, a_y, a_z)$ va $b(b_x, b_y, b_z)$ orqali ifodalash:

$$a \times b = \begin{vmatrix} i & j & k \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

a va b vektorlarda yasalgan parallelogram yuzi;

$$S_{\square} = |a \times b|$$

Shu vektorlarda yasalgan uchburchak yuzi:

$$S_{\triangle} = \frac{1}{2} |a \times b|$$

Misollar:

① Agar 1) $a = 5i$; $b = 2k$;
2) $a = i + j$; $b = i - j$;
3) $a = 2i + 3j$; $b = 3j + 2k$ bo'lsa,
$c = a \times b$ vektor aniqlansin va yasalsin. Har bir hol uchun berilgan vektorlarda yasalgan parallelogramm yuzi hisoblansin

1) $a = 3i$, $b = 2k$.
$c = a \times b = 3i \cdot 2k = 6 \cdot ik = -6j$ $\quad S_{\square} = 6$

2) $a = i + j$, $b = i - j \rightarrow \bar{a} = (1;1;0)$, $\bar{b} = (1;-1;0)$
$c = a \times b = i^2 - j^2 = ?$ $\quad (a-b)(a+b) = 2b \times a \rightarrow -2k$

$$c = a \times b = \begin{vmatrix} i & j & k \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} = 0 + 0 - k - (k + 0 + 0) = -2k$$

$S_{\square} = 2$

3) $a = 2i + 3j$, $b = 3j + 2k \rightarrow \bar{a} = (2;3;0)$, $\bar{b} = (0;3;2)$

$$c = a \times b = \begin{vmatrix} i & j & k \\ 2 & 3 & 0 \\ 0 & 3 & 2 \end{vmatrix} = 6i + 0 + 6k - (0 + 4j + 0) = 6i - 4j + 6k$$

$$S_{\square} = \sqrt{36 + 16 + 36} = \sqrt{88} = 2\sqrt{22}$$

(2) Uchlari $A(7;3;4)$; $B(1;0;6)$; va $C(4;5;-2)$ nuqtalarda bolgan uchburchakning yuzi hisoblansin.

$\overline{AB} = (-6;-3;2)$; $\quad \overline{AC} = (-3;2;-6)$

$$S_{\Delta} = \frac{1}{2}|AB \times AC| = \frac{1}{2} \cdot \begin{vmatrix} i & j & k \\ -6 & -3 & 2 \\ -3 & 2 & -6 \end{vmatrix} = \frac{1}{2}(18i + (-6j) + (-12k) - (9k + 36j + 4i) = \frac{1}{2}(14i - 42j - 21k)$$

$$S_{\Delta} = \frac{1}{2}\sqrt{2^2 + 6^2 + 3^2} \cdot 7 = \frac{7}{2} \cdot \sqrt{49} = 24{,}5$$

(3) $a = 2j + k$ va $b = i + 2k$ vektorlarda parallelogramm yasalsin hamda uning yuzi va balandligi aniqlansin.

$\bar{a}=(0;2;1)$, $\bar{b}=(1;0;2)$

$$S_{\square}=|a\times b|=\begin{vmatrix} i & j & k \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{vmatrix}=4i+j+0-2k-0-0=$$
$$=4i+j-2k$$

$$S_{\square}=\sqrt{16+1+4}=\sqrt{21}$$

$$h_1=\frac{S_{\square}}{|a|}=\frac{\sqrt{21}}{\sqrt{5}} \qquad h_2=\frac{S_{\square}}{|b|}=\frac{\sqrt{21}}{\sqrt{5}}$$

④ $a=k-j$, $b=i+j+k$ vektorlarda yasalgan parallelogrammning yuzi hisoblansin.

$$S_{\square}=|a\times b|=\begin{Vmatrix} i & j & k \\ 0 & -1 & 1 \\ 1 & 1 & 1 \end{Vmatrix}=|-i+j+0-(-k+i)|=$$
$$=|-2i+j+k|=\sqrt{4+1+1}=\sqrt{6}$$

⑤ $a=i+j$, $b=i-j$ $\quad S_{\square}=?$ $\quad c=?$

$$c=a\times b=(i+j)\cdot(i-j)=i\times i+j\times i-i\times j-j\times j=$$
$$=0-k-k-0=-2k$$

$$S_{\square}=2$$

⑥ $(a-b)\times(a+b)=2a\times b$ isbotlansin.

$$a\times a-b\times a+a\times b-b\times b=a\times b+a\times b=2a\times b$$
$$\quad\;\; \| \qquad\quad \downarrow \qquad\qquad \|$$
$$\quad\;\; 0 \qquad -a\times b \qquad\quad 0$$

⑦ $(a+b)\times(a-b)=2b\times a$ isbotlansin.

$$a\times a+b\times a-a\times b-b\times b=b\times a+b\times a=2b\times a$$
$$\quad\;\; \| \qquad\qquad\quad \| \qquad\quad \|$$
$$\quad\;\; 0 \qquad\qquad -b\times a \qquad\quad 0$$

⑧ $(2a+b)\times(a+2b) = 3a\times b$ isbotlansin.

$$2a\times a + b\times a + 4a\times b + 2b\times b = -a\times b + 4a\times b = 3a\times b$$

($a\times a = 0$, $b\times b = 0$, $b\times a = -a\times b$)

⑨ m va n o'zaro 30° burchak tashkil etuvchi birlik vektorlar bolsa, $a = m+2n$ va $b = 2m+n$ vektorlarda yasalgan parallelogrammning yuzi topilsin.

$$a\times b = (m+2n)\times(2m+n) = 2m\times m + 4n\times m + m\times n + 2n\times n$$

($m\times m = 0$, $n\times m = -m\times n$, $n\times n = 0$)

$$a\times b = 3n\times m = 3\cdot n\cdot m\cdot \sin 30^\circ = 3\cdot\frac{1}{2} = 1{,}5$$

Uch vektorning aralash ko'paytmasi.

Ta'rif. a, b va c vektorlarning <u>aralash ko'paytmasi</u> deb $(a\times b)\cdot c$ ko'rinishdagi ifodaga aytiladi.

Agar a, b, c vektorlar o'zlarining koordinatalari bilan berilsa, u holda

$$(a\times b)\cdot c = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}$$

Aralash ko'paytmaning xossalari

I. Aralash ko'paytmaning istalgan ikkita ko'paytuvchisining o'rinlari almashtirilsa, ko'paytmaning ishorasi o'zgaradi:

$$(a\times b)\cdot c = -(a\times c)\cdot b = -(c\times b)\cdot a$$

II. Agar berilgan uchta vektordan ikkitasi o'zaro teng yoki parallel bo'lsa, aralash ko'paytma 0 (nol) ga teng bo'ladi.

III. "Nuqta" bilan ko'rsatilgan va krest "x" bilan ko'rsatilgan amallarning o'rinlarini almashtirish m-n:

$$[(a \times b) \cdot c = a \cdot (b \times c)]$$

shuning uchun ham aralash ko'paytma abc ko'rinishda, ya'ni qavslarni va amallar belgilarini ko'rsatmasdan yozish qabul qilingan.

a, b, c vektorlarda yasalgan parallelepipedning hajmi:

$$V = \pm abc \begin{cases} + \text{ vektorlar o'ng bog'lam tashkil etsa,} \\ - \text{ vektorlar chap bog'lam tashkil etsa.} \end{cases}$$

a, b, c vektorlarda yasalgan piramida hajmi:

$$\left[V_{pir} = \pm \frac{1}{6} abc\right]$$

Komplanarlik sharti. Agar a, b va c vektorlar o'zaro komplanar bo'lsa, $abc = 0$, va aksincha so'nggi tenglik bajarilsa, berilgan uch vektor o'zaro komplanar bo'ladi. Shuning bilan birga a, b, c lar orasida $c = m \cdot a + n \cdot b$ ko'rinishdagi chiziqli bog'lanish mavjud bo'ladi.

Misollar:

① $a=3i+4j$, $b=-3j+k$, $c=2j+5k$ vektorlarda parallelepiped yasalsin hamda uning hajmi hisoblansin. Berilgan (a, b, c) vektorlar qaysi boshlamni tashkil etadi?

$a(3;4;0)$ $b(0;-3;1)$ $c(0;2;5)$

$$V=\begin{vmatrix}3&4&0\\0&-3&1\\0&2&5\end{vmatrix}=-45+0+0-(0+0+6)=-51$$

chap boshlam. $V=51$

② Uchlari $O(0;0;0)$, $A(5;2;0)$, $B(2;5;0)$ va $C(1;2;4)$ nuqtalarda bolgan piramida yasalsin hamda uning hajmi, ABC yoqining yuzi va shu yoqqa tushirilgan balandligi hisoblansin.

$\overline{OA}(5;2;0)$, $\overline{OB}(2;5;0)$ $\overline{OC}(1;2;4)$

$$V=\frac{1}{6}\begin{vmatrix}5&2&0\\2&5&0\\1&2&4\end{vmatrix}=\frac{1}{6}\cdot(100+0+0-(0+16+0))=\frac{1}{6}\cdot 84=14$$

$\overline{AB}=(-3;3;0)$ $\overline{AC}=(-4;0;4)$

$$S_{\triangle ABC}=\frac{1}{2}|\overline{AB}\times\overline{AC}|=\frac{1}{2}\cdot\begin{vmatrix}i&j&k\\-3&3&0\\-4&0&4\end{vmatrix}=\frac{1}{2}\cdot|12i-12k+12j|=$$

$$=\frac{1}{2}\cdot 12\sqrt{3}=6\sqrt{3}$$

$$h=\frac{3\cdot V}{S_{\triangle ABC}}=\frac{3\cdot 14}{6\sqrt{3}}=\frac{7\sqrt{3}}{3}$$

③ $A(2;-1;-2)$, $B(1;2;1)$, $C(2;3;0)$, $D(5;0;-6)$ nuqtalarning bir tekislikda yotishi kor-satilsin.

$\overrightarrow{AB}(-1;3;3)$, $\overrightarrow{AC}(0;4;2)$, $\overrightarrow{AD}(3;1;-4)$

$$(\overrightarrow{AB}\times\overrightarrow{AC})\cdot\overrightarrow{AD}=\begin{vmatrix}-1 & 3 & 3\\ 0 & 4 & 2\\ 3 & 1 & -4\end{vmatrix}=16+18+0-(36+0-2)=$$

$$=0$$

31-savol: Tekislik va uning tenglamasi.
32-savol. Tekislikning normal tenglamasi.

Tekislikning umumiy tenglamasi

$$Ax+By+Cz+D=0 \quad (1)$$

$N\{A; B; C\}$ vektor (1) tekislikka normal vektor deyiladi.

$Ax+By+Cz+D=0$ tenglamaning maxsus hollari.

I. $D=0$ bolganda, $Ax+By+Cz=0$ – tekislik koordinatalar boshidan otadi.

II. $C=0$ bolganda $Ax+By+D=0$ – tekislik Oz ogga parallel.

III. $C=D=0$ bolganda, $Ax+By=0$ – tekislik Oz ogdan otadi.

IV. $B=C=0$ bolganda, $Ax+D=0$ – tekislik yOz tekislikka parallel.

V. koordinata tekisliklarining tenglamalari:
$x=0$, $y=0$, $z=0$.

Tekislikning koordinata oqlaridan ajratgan kesmalar boyicha tenglamasi:

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

Berilgan $M_0(x_0; y_0; z_0)$ nuqtadan otuvchi va berilgan $\vec{n}=\{A, B, C\}$ normal vektorga ega tekislik tenglamasi: $A(x-x_0)+B(y-y_0)+C(z-z_0)=0$

Berilgan 3 ta $M_1(x_1; y_1; z_1)$, $M_2(x_2; y_2; z_2)$ va $M_3(x_3; y_3; z_3)$ nuqtadan o'tuvchi tekislik tenglamasi:

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 1$$

Tekisliklar orasidagi burchak:

$$\cos\varphi = \frac{\bar{N}_1 \cdot \bar{N}_2}{|\bar{N}_1| \cdot |\bar{N}_2|} = \frac{A_1A_2 + B_1B_2 + C_1C_2}{\sqrt{A_1^2+B_1^2+C_1^2} \cdot \sqrt{A_2^2+B_2^2+C_2^2}}$$

Tekislikning perpendikulyarlik sharti normal vektorlarning perpendikulyarlik shartiga teng kuchli:

$\bar{N}_1 \cdot \bar{N}_2 = 0$ yoki $A_1A_2 + B_1B_2 + C_1C_2 = 0$

Tekisliklarning parallellik sharti $\bar{N}_1$ va $\bar{N}_2$ vektorlarning kollinearlik shartiga teng kuchli:

$\bar{N}_1 = \lambda \bar{N}_2$ yoki $\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2} = \lambda$

Misollar.

① $2x + 3y + 6z - 12 = 0$ tekislik yasalsin va unga normal vektorning koordinata o'qlari bilan tashkil etgan burchaklari topilsin.

$2x + 3y + 6z = 12 \quad / :12$

$\frac{x}{6} + \frac{y}{4} + \frac{z}{2} = 1$

Koordinata o'qlarini (6; 4; 2) nuqtalarda kesib o'tadigan tekislik.

Normal vektor: $\bar{N}(2;3;6)$

Ox o'qi bilan tashkil qilgan burchak:

$$\cos\varphi_1 = \frac{(2;3;6)\cdot(x;0;0)}{\sqrt{2^2+3^2+6^2}\cdot\sqrt{x^2+0+0}} = \frac{2x}{7x} = \frac{2}{7}$$

Oy: $\cos\varphi_2 = \frac{3}{7}$; Oz: $\cos\varphi_3 = \frac{6}{7}$.

② $M_1(0;-1;3)$ va $M_2(1;3;5)$ nuqtalar berilgan. M_1 nuqtadan o'tuvchi va $N=\bar{M_1M_2}$ vektorga perpendikulyar tekislik tenglamasi yozilsin.

$$N = (1;4;2) = (A;B;C)$$

$$x+4y+2z+D=0$$

$$M_1 \to 0-4+6+D=0 \to D=-2$$

$$\boxed{x+4y+2z-2=0}$$

③ $A(a;-\frac{a}{2};a)$ va $B(0;\frac{a}{2};0)$ nuqtalardan teng uzoqlikda bo'lgan nuqtalar geometrik o'rnining tenglamasi yozilsin.

$$\frac{x\cdot a - y\cdot\frac{a}{2}+z\cdot a}{\sqrt{a^2+\frac{a^2}{4}+a^2}} = \frac{y\cdot\frac{a}{2}}{\frac{a}{2}}$$

$$\frac{x\cdot a - y\frac{a}{2}+z\cdot a}{\frac{3}{2}a} = y\cdot\frac{\frac{a}{2}}{\frac{a}{4}}$$

$$2x-y+2z=3y \to 2x$$

$$d = \frac{Ax_0 + By_0 + Cz_0 + D}{\sqrt{A^2+B^2+C^2}} = \frac{Ax_{02} + By_{02} + Cz_{02} + D}{\sqrt{A^2+B^2+C^2}}$$

$$A\cdot a + B\cdot\left(-\frac{a}{2}\right) + C\cdot a = A\cdot 0 + B\cdot\frac{a}{2} + C\cdot 0$$

$$Aa - \frac{Ba}{2} + Ca = 0$$

$$(x-a)^2 + \left(y+\frac{a}{2}\right)^2 + (z-a)^2 = \left(y-\frac{a}{2}\right)^2 + x^2 + y^2$$

$$-2ax + a^2 + ay + \frac{a^2}{4} - 2az + a^2 = -ay + \frac{a^2}{4}$$

$$2ax - 2ay + 2az - 2a^2 = 0 \quad /:2a$$

$$\boxed{x - y + z = a}$$

(4) Oz ogdan va $M_1(2; -4; 3)$ nuqtadan o'tuvchi tekislik tenglamasini yozing.

$C=0$ $D=0$ $\quad Ax + By = 0$

$2A - 4B = 0 \rightarrow A = 2B$

$$A(x-x_0) + B(y-y_0) + C(z-z_0) = 0$$
$$A(x-2) + B(y+4) + C(z-3) = 0$$
$$Ax + By + Cz - 2A + 4B - 3C = 0$$

$C=0, D=0 \rightarrow -(2A - 4B + 3C) = 0 \rightarrow A = 2B$

$$2Bx + By = 0$$
$$\boxed{2x + y = 0}$$

(5) Oy ogga parallel, Ox va Oz ogʻlardan a va c kesmalar ajratuvchi tekislik tenglamasi yozilsin.

$B=0$. $\quad Ax + Cz + D = 0$

$A \cdot a + C \cdot c + D = 0 \quad D = -Aa - Cc$

$Ax + Cz - Aa - Cc = 0 \;/\; : AC$

$\frac{x}{a} + \frac{z}{c} = 1$

⑥ $M(2;-1;+3)$ nuqtadan o'tuvchi va koordinata o'qlaridan teng kesmalar ajratuvchi tekislik tenglamasini yozing.

$\frac{x}{a} + \frac{y}{a} + \frac{z}{a} = 1$

$x + y + z = a$

$2 - 1 + 3 = a \rightarrow a = 4.$

$x + y + z = 4$

⑦ $M_1(-4;0;4)$ nuqtadan o'tuvchi va Ox va Oy o'qlaridan $a=4$ va $b=3$ kesmalar ajratuvchi tekislikning tenglamasi yozilsin.

$\frac{x}{4} + \frac{y}{3} + \frac{z}{c} = 1$

$\frac{-4}{4} + \frac{0}{3} + \frac{4}{c} = 1$

$\frac{4}{c} = 2 \rightarrow c = 2$

$\frac{x}{4} + \frac{y}{3} + \frac{z}{2} = 1$

Matematik analiz

33-savol. Sonlar ketma-ketligi limiti. Yaqinlashuvchi ketma-ketliklarning xossalari.

Tarif. Natural sonlar to'plamida aniqlangan funksiya **sonli ketma-ketlik** deyiladi va $\{x_n\}$ ko'rinishda belgilanadi.

Agar har qanday natural n u-n

$$x_{n+1} > x_n$$

tengsizlik bajarilsa, $\{x_n\}$ **o'suvchi**, $x_n > x_{n+1}$ tengsizlik bajarilsa, $\{x_n\}$ **kamayuvchi ketma-ketlik** d-di.

Faqat o'suvchi yoki kamayuvchi ketma-ketlik **monoton ketma-ketlik** d-di.

Agar ixtiyoriy $\varepsilon > 0$ son uchun shunday $N = N(\varepsilon) > 0$ son mavjud bo'lsaki, barcha $n \geq N$ lar uchun

$$|x_n - a| < \varepsilon$$

tengsizlik bajarilsa, o'zgarmas a son $\{x_n\}$ ketma-ketlikning **limiti** d-di va quyidagicha yoziladi:

$$\boxed{\lim_{n \to \infty} x_n = a}$$

Agar $\{x_n\}$ ketma-ketlik limitga ega bo'lsa, u **yaqinlashuvchi**, aks holda **uzoqlashuvchi ketma-ketlik** d-di.

Har qanday chegaralangan va monoton ketma-ketlik limitga ega.

Misol. $\lim\limits_{n\to\infty}\frac{2n+3}{2n+1}=1$ ekanligini isbot qiling va $N(\varepsilon)$ ni toping.

Yechish: Agar ixtiyoriy $\varepsilon>0$ uchun shunday $N(\varepsilon)$ soni mavjud bolsaki, barcha $n\geq N(\varepsilon)$ lar uchun

$$|x_n-a|=\left|\frac{2n-3}{2n-1}-1\right|<\varepsilon$$

tengsizlik bajarilsa, limitning ta'rifiga kora qoyilgan masala hal boladi. Yuqoridagi tengsizlik quyidagiga teng kuchli:

$$\frac{2}{2n+1}<\varepsilon$$

bundan: $2n+1>\frac{2}{\varepsilon}$ yoki $n>\frac{2-\varepsilon}{2\varepsilon}$

tengsizlikka ega bolamiz.

Demak: $N=N(\varepsilon)=\frac{2-\varepsilon}{2\varepsilon}$, $\lim\limits_{n\to\infty}\frac{2n+3}{2n+1}=1$

Limitlar haqida teoremalar:

a) $\lim\limits_{x\to a}C=C$ $\quad(C=\text{const})$

b) Agar $\lim\limits_{x\to a}f(x)$ va $\lim\limits_{x\to a}\varphi(x)$ mavjud bolsa,

$$\lim_{x\to a}(f(x)+\varphi(x))=\lim_{x\to a}f(x)+\lim_{x\to a}\varphi(x)$$

e) $\lim\limits_{x\to a} f(x)\cdot\varphi(x) = \lim\limits_{x\to a} f(x)\cdot\lim\limits_{x\to a}\varphi(x)$.

d) $\lim\limits_{x\to a}\varphi(x)\neq 0 \to \lim\limits_{x\to a}\dfrac{f(x)}{\varphi(x)} = \dfrac{\lim\limits_{x\to a} f(x)}{\lim\limits_{x\to a}\varphi(x)}$

Misollar.

① Limitni hisoblang.

$$\lim_{n\to\infty}\frac{3n^2+5n-1}{n^2+3} = \lim_{n\to\infty}\frac{3+\frac{5}{n}-\frac{1}{n^2}}{1+\frac{3}{n^2}} = \frac{3}{1} = 3$$

② $$\lim_{n\to\infty}\frac{(n+2)!+(n+1)!}{(n+3)!} = \frac{(n+1)!\cdot(n+3)}{(n+1)!\cdot(n+2)\cdot(n+3)} = \frac{1}{n+2} = 0$$

③ $$\lim_{n\to\infty}\frac{(n+2)^3+(n-2)^3}{n^4+2n^2-1} = \frac{\frac{2n^3+24n}{n^4}}{1+\frac{2}{n^2}-\frac{1}{n^4}} =$$

$$= \lim_{n\to\infty}\frac{\frac{2}{n}+\frac{24}{n^3}}{1+\frac{2}{n^2}-\frac{1}{n^4}} = \frac{0}{1} = 0$$

④ $$\lim_{n\to\infty}\frac{(2^n+7^n):7^n}{(2^n-7^{n-1}):7^n} = \frac{\left(\frac{2}{7}\right)^n+1}{\left(\frac{2}{7}\right)^n-\frac{1}{7}} = \frac{1}{-\frac{1}{7}} = -7$$

⑤ $$\lim_{n\to\infty}\sqrt{n}\left(\sqrt{n+2}-\sqrt{n-3}\right) = \lim_{n\to\infty}\left(\sqrt{n^2+2n}-\sqrt{n^2-3n}\right)$$

$$= \lim_{n\to\infty}\frac{\sqrt{n}\cdot\left(\sqrt{n+2}-\sqrt{n-3}\right)\cdot\left(\sqrt{n+2}+\sqrt{n-3}\right)}{\sqrt{n+2}+\sqrt{n-3}} =$$

$$= \lim_{n\to\infty}\frac{\sqrt{n}\cdot(n+2-n+3)}{\sqrt{n+2}+\sqrt{n-3}} = \lim_{n\to\infty}\frac{5\cdot\sqrt{n}}{\sqrt{n+2}+\sqrt{n-3}}\Big|:\sqrt{n} =$$

$$= \lim_{n\to\infty} \frac{5}{\sqrt{1+\frac{2}{n}} + \sqrt{1+\frac{3}{n}}} = \frac{5}{2}$$

⑥ $$\lim_{n\to\infty} \frac{(n+1)^3-(n+1)^2}{(n-1)^3-(n+1)^3} = \lim_{n\to\infty} \frac{(n+1)^2\cdot(n+1-1)}{-6n^2-2} =$$

$$= \lim_{n\to\infty} \frac{(n^3+2n^2+n):n^2}{(-6n^2-2):n^2} = \frac{n+2+\frac{1}{n^2}}{-6-\frac{2}{n^2}} = \frac{n+2}{-6} = -\infty$$

⑦ $$\lim_{n\to\infty} \frac{(3-n)^3}{(n+1)^2-(n+1)^3}$$

$$\frac{(3-n)^3}{(n+1)^2(1-n-1)} = \frac{(3-n)^3}{(n+1)^2\cdot(-n)} = \frac{3-27n+9n^2-n^3}{-n^3-2n^2-n}$$

$$\lim_{n\to\infty} \frac{\frac{3}{n^3}-\frac{27}{n^2}+\frac{9}{n}-1}{-1-\frac{2}{n}-\frac{1}{n^2}} = \frac{-1}{-1} = 1$$

⑧ $$\lim_{n\to\infty} \sqrt{n^3+8}\cdot\left(\sqrt{n^3+2}-\sqrt{n^3-1}\right)$$

$$= \lim_{n\to\infty} \frac{\sqrt{n^3+8}\left(\sqrt{n^3+2}-\sqrt{n^3-1}\right)\cdot\left(\sqrt{n^3+2}+\sqrt{n^3-1}\right)}{\sqrt{n^3+2}+\sqrt{n^3-1}} =$$

$$= \lim_{n\to\infty} \frac{\sqrt{n^3+8}\cdot(n^3+2-n^3+1)}{\sqrt{n^3+2}+\sqrt{n^3-1}} = \frac{3\cdot\sqrt{1+\frac{8}{n^3}}}{\sqrt{1+\frac{2}{n^3}}+\sqrt{1-\frac{1}{n^3}}} = \frac{3}{2}$$

Funksiyaning limitini hisoblash.

① $$\lim_{x\to 2} \frac{3x-2}{x^2+1} = \frac{3\cdot 2-2}{2^2+1} = \frac{4}{5}$$

② $\lim\limits_{x\to 1} \dfrac{x^3-x^2-x+1}{x^3+x^2-x-1}$

$x=1$ da $\frac{0}{0}$ aniqmaslik mavjud.

$$\lim_{x\to 1} \frac{x^2(x-1)-(x-1)}{x^2(x+1)-(x+1)} = \frac{(x-1)(x^2-1)}{(x+1)(x^2-1)} = \frac{x-1}{x+1} = \frac{0}{2} = 0$$

③ $\lim\limits_{x\to 2}\left(\dfrac{4}{x^2-4} - \dfrac{1}{x-2}\right)$

$x=2$ $\infty-\infty$ aniqmaslik mavjud.

$$\lim_{x\to 2} \frac{4-(x+2)}{x^2-4} = \lim_{x\to 2} \frac{-(x-2)}{(x-2)(x+2)} = -\frac{1}{2+2} = -\frac{1}{4}$$

④ $\lim\limits_{x\to 0} \dfrac{\sqrt{2+x}-\sqrt{2}}{x^2+2x}$

$x=0$ da $\frac{0}{0}$ aniqmaslik mavjud.

$$\lim_{x\to 0} \frac{(\sqrt{2+x}-\sqrt{2})(\sqrt{2+x}+\sqrt{2})}{(x^2+2x)(\sqrt{2+x}+\sqrt{2})} = \lim_{x\to 0} \frac{x}{(x^2+2x)\sim x} =$$

$$= \lim_{x\to 0} \frac{1}{(x+2)(\sqrt{2+x}+\sqrt{2})} = \frac{1}{2\cdot 2\sqrt{2}} = \frac{\sqrt{2}}{8}.$$

34-savol. Cheksiz katta va cheksiz kichik miqdor, ular orasidagi bog'lanish.

Cheksiz kichik o'zgaruvchi. Agar har qanday musbat ε son uchun o'zgaruvchining shunday a_0 qiymati mavjud bo'lsaki, a ning undan so'nggi har bir qiymatining absolyut miqdori ε dan kichik bo'lsa, a o'zgaruvchi cheksiz kichik d-di.

Agar a cheksiz kichik bo'lsa, u nolga intiladi d-di va $a \to 0$ ko'rinishda yoziladi.

Cheksiz katta o'zgaruvchi. Agar har qanday musbat c son uchun o'zgaruvchining shunday qiymati mavjud bo'lsaki, x ning undan so'nggi har bir qiymatining absolyut miqdori c dan katta bo'lsa, x o'zgaruvchi cheksiz katta d-di. Bu $x \to \infty$ ko'rinishda yoziladi.

Cheksiz katta o'zgaruvchiga teskari miqdor cheksiz kichik miqdor va aksincha bo'ladi.

O'zgaruvchining limiti. Agar o'zgarmas a va o'zgaruvchi x orasidagi ayirma cheksiz kichik miqdor, ~~yoki~~ bo'lsa, o'zgarmas a o'zgaruvchi x ning limiti deyiladi ($\lim x = a$)

35-savol. Sonli ketma-ketliklarning yaqinlashish alomatlari.

51-savol. Yaqinlashuvchi sonli qatorlar va ular haqida teoremalar

Ta'rif. Ushbu

$$u_1+u_2+\dots+u_n+\dots \quad (1)$$

ifoda cheksiz qator yoki to'g'ridan-to'g'ri qator deyiladi.

$u_1, u_2, \dots, u_n, \dots$ – qatorning hadlari.

u_n – umumiy hadi.

Agar qatorning hadlari sonlardan iborat bo'lsa sonli qatorlar, funksiyalardan iborat bo'lsa funksional qatorlar deyiladi.

Ta'rif. (1) qatorning dastlabki n ta hadining yig'indisi

$$S_n=u_1+u_2+\dots+u_n$$

shu qatorning n-xususiy yig'indisi deyiladi.

Xususiy yig'indilar:

$$S_1=u_1$$
$$S_2=u_1+u_2$$
$$\dots$$
$$S_n=u_1+u_2+\dots+u_n$$

Ta'rif. Agar (1) qatorning xususiy yig'indilaridan iborat ketma-ketlik $S_1, S_2, \dots, S_n, \dots$ chekli limitga ega bo'lsa, bu qator yaqinlashuvchi qator deyiladi.

Qatorning yig'indisi: $S=\lim\limits_{n\to\infty} S_n$

$$S=u_1+u_2+\dots+u_n+\dots=\sum_{n=1}^{\infty} u_n$$

Tarif. Agar (1) qatorning xususiy yig'indilari ketma-ketligi chekli limitga ega bo'lmasa, bu qator uzoqlashuvchi qator deyiladi.

Sonlar nazariyasining mazmuni qatorning yaqinlashuvchi yoki uzoqlashuvchi ekanligini aniqlash va yaqinlashuvchi qatorlar yig'indisini hisoblashdan iborat.

Cheksiz geometrik progressiya $|q|<1$ da yaqinlashuvchi va $|q|\geq 1$ da uzoqlashuvchi qator.

Qator yaqinlashishining zaruriy sharti

Teorema. Agar qator yaqinlashuvchi bo'lsa, u holda qatorning umumiy hadi n cheksiz o'sganda nolga intiladi.

Demak, agar qatorning umumiy hadi $n\to\infty$ da nolga intilmasa, u holda qator uzoqlashadi.

Masalan: $\frac{1}{3}+\frac{2}{5}+\frac{3}{7}+\ldots+\frac{n}{2n+1}+\ldots$

qator uzoqlashuvchi, chunki

$$\lim_{n\to\infty} u_n = \lim_{n\to\infty}\frac{n}{2n+1}=\frac{1}{2}\neq 0$$

Garmonik qator deb ataluvchi

$$1+\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{n}+\ldots$$

qator uchun $\lim\limits_{n\to\infty} u_n=\lim\limits_{n\to\infty}\frac{1}{n}=0$ bo'lishiga qaramay u yaqinlashuvchi emasligi aniqlangan

Isbot.

$$1+\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{n}+\ldots = 1+\frac{1}{2}+\left(\frac{1}{3}+\frac{1}{4}\right)+\left(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}\right)+$$

$$+\ldots \Rightarrow 1+\frac{1}{2}+\left(\frac{1}{4}+\frac{1}{4}\right)+\left(\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}\right)+\ldots$$

Har qaysi qavs ichidagi qo'shiluvchilar yig'indisi kichiklashdi va 1/2 ga teng bo'ldi. Oxirgi qator cheksiz ko'p qavslarga ega bo'lganligi sababli ularning yig'indisi cheksizlikka intiladi.

Qatorlar ustida sodda amallar bajarish.

1-teorema (qatorni songa ko'paytirish haqida). Agar $u_1+u_2+\ldots+u_n+\ldots$ qator yaqinlashuvchi bo'lib, uning yig'indisi S ga teng bo'lsa, u holda

$$\lambda u_1+\lambda u_2+\ldots+\lambda u_n+\ldots$$

qator ham yaqinlashuvchi bo'ladi.

2-teorema: (qatorlarni qo'shish haqida). Agar

$$u_1+u_2+\ldots+u_n+\ldots,$$
$$v_1+v_2+\ldots+v_n+\ldots$$

qatorlar yaqinlashuvchi va ularning yig'indilari mos ravish s va S bo'lsa, u holda

$$(u_1+v_1)+(u_2+v_2)+\ldots+(u_n+v_n)+\ldots$$

qator ham yaqinlashuvchi bo'ladi va u/= yig'indisi $s+S$ bo'ladi.

3-teorema: $(u_1-v_1)+(u_2-v_2)+\ldots+(u_n-v_n)+\ldots$ ham yaqinlashuvchi.

4-teorema. Agar qator yaqinlashuvchi bolsa u holda berilgan qatorga chekli sondagi hadlarni qoshish yoki unda chekli sondagi hadlarni tashlab yuborishdan hosil bolgan qator ham yaqinlashuvchi boladi.

Yaqinlashish alomatlari:

1) Hadlari musbat hamda kamayuvchi bolgan qatorning yaqinlashishi uchun integral alomat:

Agar $u_n = f(n)$ deb olinsa, bunda $f(n)$ kamayuvchi funksiya va

$$\int_1^{\infty} f(x)\,dx = \begin{cases} A \text{ bolsa qator yaqinlashadi.} \\ \infty \text{ bolsa qator uzoqlashadi.} \end{cases}$$

2) Musbat hadli qatorning yaqinlashishi uchun Dalamber alomati:

Agar

$$d = \lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \begin{cases} <1 \text{ bolsa, yaqinlashuvchi} \\ >1 \text{ bolsa, uzoqlashuvchi} \\ =1 \text{ bolsa, masala yechilmay qoladi.} \end{cases}$$

yani yaqin. ham uzoq. ham bolishi m-n

Misol: $\sum_{n=0}^{\infty} \frac{2^n \cdot n!}{n^n}$ qator yaqinlashishini tekshiring.

$$u_n = \frac{2^n \cdot n!}{n^n} \qquad u_{n+1} = \frac{2^{n+1}(n+1)!}{(n+1)^{n+1}}$$

$$d = \lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{2^{n+1}(n+1)!\,n^n}{(n+1)^{n+1} \cdot 2^n \cdot n!} = \lim_{n\to\infty} \frac{2n^n}{(n+1)^n} =$$

$$= 2\lim_{n\to\infty} \frac{1}{\frac{(n+1)^n}{n^n}} = 2\lim_{n\to\infty} \frac{1}{\left(1+\frac{1}{n}\right)^n} = \frac{2}{e} < 1 \quad \text{yaqinlashuvchi}$$

Qator yaqinlashishining yetarlilik sharti:

Teorema.

Ikkita musbat hadli

$$a_1+a_2+\dots+a_n+\dots \quad (1)$$
$$b_1+b_2+\dots+b_n+\dots \quad (2)$$

qatorlarni qaraymiz.

1-teorema: Agar (1) qatorning barcha hadlari (2) qatorning mos hadlaridan katta bolmasa, yani $a_n \le b_n$ $(n=1,2,3,\dots)$ bolsa va (2) qator yaqinlashuvchi va yigindisi B ga teng bolsa, (1) qator ham yaqinlashadi va uning A yigindisi B dan katta bolmaydi.

Misol. Qator yaqinlashishini tekshiring

$$1+\frac{1}{2!}+\frac{1}{3!}+\dots+\frac{1}{n!}+\dots$$

Berilgan qatorni maxraji $q=\frac{1}{2}<1$ bolgan yaqinlashuvchi

$$1+\frac{1}{2}+\frac{1}{2^2}+\dots+\frac{1}{2^{n-1}}+\dots$$

cheksiz kamayuvchi geometrik progressiya bn taqqoslaymiz. Har bir $n>2$ uchun

$$\frac{1}{n!}=\frac{1}{2\cdot3\cdot\dots\cdot n}<\frac{1}{2\cdot2\cdot2\cdot\dots\cdot2}=\frac{1}{2^{n-1}}$$

tengsizlik orinli. 1-teoremaga kora berilgan qator ham yaqinlashuvchi.

2-teorema. Agar (1) qatorning barcha hadlari (2) qatorning mos hadlaridan kichik bolmasa, yani $a_n \ge b_n$ $(n=1,2,3,\dots)$ bolsa va (2) qator uzoqlashuvchi bolsa, (1) qator ham uzoqlashuvchi boladi.

Misol. Qator yaqinlashuvchi ekanligini tekshiring:

$$\frac{1}{\ln 2}+\frac{1}{\ln 3}+\dots+\frac{1}{\ln n}+\dots$$

Ixtiyoriy $n\geq 2$ uchun $\ln n<n$ tengsizlik o'rinli va bundan esa

$$\frac{1}{n}<\frac{1}{\ln n}$$

tengsizlikka ega bolamiz.

$$1+\frac{1}{2}+\frac{1}{3}+\dots+\frac{1}{n}+\dots$$

garmonik qatorning uzoqlashuvchi ekanligini kördik, shuning uchun 2-teoremaga asosan berilgan qator uzoqlashuvchi.

Taqqoslashning limit alomati

3-teorema. Agar (1) va (2) qatorlar umumiy hadlari uchun noldan farqli chekli

$\lim\limits_{n\to\infty}\frac{a_n}{b_n}=p$ $(0<p<\infty)$ limit mavjud bolsa, (1) va (2) qatorlar bir vaqtda yoki yaqinlashadi yoki uzoqlashadi.

Misol.

① $\operatorname{tg}1+\operatorname{tg}\frac{1}{2}+\dots+\operatorname{tg}\frac{1}{n}+\dots$ yaqinlashishga tekshiring.

Ushbu qatorni

$$1+\frac{1}{2}+\dots+\frac{1}{n}+\dots$$

garmonik qator bilan taqqoslaymiz. $\frac{u_n}{v_n}$

nisbatni tuzamiz va uning limitini topamiz.

$$\lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\frac{\operatorname{tg}\frac{1}{n}}{\frac{1}{n}}=1>0$$

chunki $n\to\infty$ da $\operatorname{tg}\frac{1}{n}\simeq\frac{1}{n}$

Shunday qilib berilgan qator uzoqlashuvchi, chunki garmonik qator uzoqlashuvchi

② Ushbu

$$\sin\frac{1}{2}+\sin\frac{1}{2^2}+\ldots+\sin\frac{1}{2^n}+\ldots$$

qatorlarni

$$\frac{1}{2}+\frac{1}{2^2}+\ldots+\frac{1}{2^n}+\ldots+\frac{1}{2^n}+\ldots$$

qator bilan taqqoslaymiz, oxirgi qator yaqinlashuvchi, chunki uning hadlari maxraji $q=1/2$ bo'lgan geometrik progressiya tashkil qiladi.

$\frac{u_n}{v_n}$ nisbatni tuzamiz va uning limitini topamiz: $n\to\infty$ da $\sin\frac{1}{2^n}\sim\frac{1}{2^n}$ bo'lgani uchun: $\lim\limits_{n\to\infty}\frac{u_n}{v_n}=\lim\limits_{n\to\infty}\frac{\sin 2^{-n}}{2^{-n}}=1>0$. Shunday qilib, berilgan qator yaqinlashuvchi.

Koshining radikal alomati.

Teorema. (1) qatorning hamma hadlari musbat va $l=\lim\limits_{n\to\infty}\sqrt[n]{a_n}$ chekli yoki cheksiz limit mavjud bo'lsin, u holda $l<1$ bo'lganda (1) qator yaqinlashuvchi, $l>1$ bo'lganda esa uzoqlashuvchi bo'ladi.

Bu yerda ham Dalamber alomatidagi singari $l=1$ holda qator yaqinlashishi haqida masala ochiq qoladi, ya'ni bu holda qator yaqinlashishi ham uzoqlashishi ham mumkin.

Misol. $\sum\limits_{n=0}^{\infty}\frac{1}{3^n}\left(\frac{n}{n+1}\right)^{n^2}$ qator yaqinlashishini tekshiring.

Koshining radikal alomati bo'yicha tekshiramiz:

$$l=\lim_{n\to\infty}\sqrt[n]{a_n}=\lim_{n\to\infty}\sqrt[n]{\frac{1}{3^n}\left(\frac{n}{n+1}\right)^{n^2}}=\frac{1}{3}\lim_{n\to\infty}\frac{1}{\left(1+\frac{1}{n}\right)^n}=\frac{1}{3e}<1$$

demak berilgan qator yaqinlashuvchi.

Koshining integral alomati.

Teorema. Musbat hadli $\sum_{n=0}^{\infty} a_n$ qatorning hadlari $[1;+\infty)$ oraliqda uzluksiz monoton kamayuvchi $f(x)$ funksiyaning qiymatlari sifatida, ya'ni $a_1=f(1)$, $a_2=f(2)$, ..., $a_n=f(n)$, ... ko'rinishda berilgan bo'lsin. U holda

1) Agar $\int_1^{+\infty} f(x)dx$ xosmas integral yaqinlashsa, $\sum_{n=0}^{\infty} a_n$ qator ham yaqinlashadi.

2) Agar $\int_1^{+\infty} f(x)dx$ xosmas integral uzoqlashsa $\sum_{n=0}^{\infty} a_n$ qator ham uzoqlashadi.

Misol. $\sum_{n=0}^{\infty} \frac{1}{n\cdot \ln n}$ qator yaqinlashishini tekshiring.

Koshining integral alomatidan foydalanamiz. $f(x)=\frac{1}{x\ln x}$ funksiya teoremaning barcha shartlarini qanoatlantiradi. Uning integralini hisoblaymiz:

$$\int_1^{+\infty} f(x)dx = \int_1^{+\infty} \frac{1}{x\ln x}dx = \ln|\ln x|\Big|_2^{+\infty} = \infty$$

Demak, berilgan $\sum_{n=0}^{\infty} \frac{1}{n\cdot \ln n}$ qator uzoqlashuvchi bo'lar ekan.

Leybnis alomati.

Agar

1) Qator hadlarining absolyut qiymatlari ketma-ketligi monoton kamaysa, ya'ni $a_1 > a_2 > a_3 > \dots > a_n > \dots$;

2) Umumiy had nolga intilsa: $\lim\limits_{n\to\infty} a_n = 0$,

u holda (1) qator yaqinlashadi va bunda qatorning S yig'indisi

$$0 < S < a_1$$

tengsizliklarni qanoatlantiradi.

Misollar:

① Quyidagi qatorlar uchun yaqinlashishning zaruriy sharti bajariladimi?

a) $\frac{1}{2} + \frac{3}{4} + \frac{5}{6} + \frac{7}{8} + \dots$

$u_n = \frac{2n-1}{2n}$ $\lim\limits_{n\to\infty} \frac{2n-1}{2n} = \lim\limits_{n\to\infty} \frac{2-\frac{1}{n}}{2} = 1 \neq 0$

bajarilmadi.

b) $\frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \dots$ $u_n = \frac{1}{2n-1}$

$\lim\limits_{n\to\infty} \frac{1}{2n-1} = \lim\limits_{n\to\infty} \frac{\frac{1}{n}}{2-\frac{1}{n}} = \frac{0}{2} = 0$ bajarildi.

c) $\frac{2}{3} + \frac{4}{9} + \frac{6}{27} + \frac{8}{81} + \dots$ $u_n = \frac{2n}{3^n}$

$\lim\limits_{n\to\infty} \frac{2n}{3^n}$ = Dalamber alomatidan foydalanamiz

$\lim\limits_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim\limits_{n\to\infty} \frac{\frac{2(n+1)}{3^{n+1}}}{\frac{2n}{3^n}} = \lim\limits_{n\to\infty} \frac{2n+2}{3\cdot 2n} = \lim\limits_{n\to\infty} \frac{2+\frac{2}{n}}{6} = \frac{2}{6} = \frac{1}{3} < 1$

Qator yaqinlashuvchan.

Integral alomatiga doir misollar.

① $1+\frac{1}{3}+\frac{1}{5}+\dots$ $\quad u_n=\frac{1}{2n-1}$ (kamayuvchi)

$$\int_1^{\infty}\frac{1}{2x-1}dx=\frac{1}{2}\ln(2x-1)\Big|_1^{\infty}=\infty$$ uzoqlashadi

② $1+\frac{1}{\sqrt{4}}+\frac{1}{\sqrt{7}}+\frac{1}{\sqrt{10}}+\dots$ $\quad u_n=\frac{1}{\sqrt{3n-2}}$ (kamayuvchi)

$$\int_1^{\infty}\frac{1}{\sqrt{3x-2}}dx=-\frac{2}{9}\cdot(3x-2)^{-\frac{3}{2}}\Big|_1^{\infty}=-\frac{2}{9}\cdot\frac{1}{\sqrt{(3x-2)^3}}\Big|_1^{\infty}=$$

$$=0+\frac{2}{9}=\frac{2}{9}$$ Yaqinlashadi.

③ $\frac{1}{2^3}+\frac{2}{3^3}+\frac{3}{4^3}+\dots$ $\quad u_n=\frac{n}{(n+1)^3}$ kamayuvchi

$$\int_1^{\infty}\frac{x}{(x+1)^3}dx=\int_1^{\infty}\frac{x+1-1}{(x+1)^3}dx=\int_1^{\infty}\frac{1}{(x+1)^2}dx-\int_1^{\infty}\frac{1}{(x+1)^3}dx=$$

$$=-\frac{1}{3}\cdot\frac{1}{(x+1)^3}\Big|_1^{\infty}+\frac{1}{4}\cdot\frac{1}{(x+1)^4}\Big|_1^{\infty}=$$

$$=0+\frac{1}{24}+0-\frac{1}{64}=\text{A son}$$

Dalamber alomatiga oid misollar:

① $\frac{2}{3}+\frac{4}{9}+\frac{6}{27}+\frac{8}{81}+\dots$ $\quad u_n=$ ~~u~~

② $1+\frac{2}{2!}+\frac{4}{3!}+\frac{8}{4!}+\dots$ $\quad u_n=\frac{2^{n-1}}{n!}$ $\quad u_{n+1}=\frac{2^n}{(n+1)!}$

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{\frac{2^n}{(n+1)!}}{\frac{2^{n-1}}{n!}}=\lim_{n\to\infty}\frac{2}{n+1}=0<1$$ yaqinlashuvchi

③ $1+\frac{1\cdot2}{1\cdot3}+\frac{1\cdot2\cdot3}{1\cdot3\cdot5}+\dots$

$u_n=\frac{n!}{n!!}$ $u_{n+1}=\frac{(n+1)!}{(n+1)!!}$

$u_n=\frac{n!}{(2n-1)!!}$ $u_{n+1}=\frac{(n+1)!}{(2n+1)!!}$

$$\lim_{n\to\infty}\frac{\frac{(n+1)!}{(2n+1)!!}}{\frac{n!}{(2n-1)!!}}=\lim_{n\to\infty}\frac{n+1}{2n+1}=\lim_{n\to\infty}\frac{1+\frac{1}{n}}{2+\frac{1}{n}}=\frac{1}{2}<1$$

yaqinlashadi.

Garmonik qator yoki kamayuvchi progressiya bilan taqqoslab quyidagi qatorlarning yaqinlashishi tekshirilsin.

① $a=1+\frac{1}{\sqrt2}+\frac{1}{\sqrt3}+\frac{1}{\sqrt4}+\dots$

$b=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\dots$

$b_n<a_n$
b-uzoqlashuvchi
demak a ham uzoqlashuvchi

② $a=1+\frac{1}{2\cdot5}+\frac{1}{3\cdot5^2}+\frac{1}{4\cdot5^3}+\dots$

$a_n<b_n$

$b=1+\frac{1}{2^2}+\frac{1}{2^3}+\dots+\frac{1}{2^n}+\dots$

$b_n=\frac{1}{2^n}$ $b_{n+1}=\frac{1}{2^{n+1}}$

Dalamber alomatiga ko'ra:

$$\lim_{n\to\infty}\frac{b_{n+1}}{b_n}=\lim_{n\to\infty}\frac{\frac{1}{2^{n+1}}}{\frac{1}{2^n}}=\lim_{n\to\infty}\frac{1}{2}=\frac{1}{2}<1$$ yaqinlashuvchi

demak a yaqinlashuvchi.

Teylor va Makloren qatorlari.

1) $f(x)=f(0)+\frac{f'(0)}{1!}x+\frac{f''(0)}{2!}x^2+\dots+R_n(x)$ (1)

ko'rinishdagi formula Makloren formulasi deyiladi, bunda

$$R_n(x)=\frac{x^n}{n!}(f^{(n)})(\theta x);\quad 0\le\theta<1.$$

2) $f(x)=f(a)+\frac{f'(a)}{1!}(x-a)+\frac{f''(a)}{2!}(x-a)^2+\dots+R_n(x)$ (2)

ko'rinishdagi formula Teylor formulasi. d-di, bunda

$$R_n(x)=\frac{(x-a)^n}{n!}f^{(n)}[a+\theta(x-a)]$$

3) Teylor va Makloren qatorlari. (1) va (2) formulalarda n cheksizlikka intilganda ($n\to\infty$) R_n nolga intilsa ($R_n(x)\to 0$) u holda bu formulalardan x ning $\lim\limits_{n\to\infty}R_n(x)=0$ bo'lgandagi qiymatlari uchun $f(x)$ ga yaqinlashuvchi quyidagi

$$f(x)=f(0)+\frac{f'(0)}{1!}x+\frac{f''(0)}{2!}x^2+\dots \quad (3)$$

$$f(x)=f(a)+\frac{f'(a)}{1!}(x-a)+\frac{f''(a)}{2!}(x-a)^2+\dots \quad (4)$$

cheksiz qatorlar hosil bo'ladi.

Elementar funksiyalarning qatorlarga yoyilmalari.

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

Bu qatorlar x ning har qanday qiymatlarida mos (ko'rsatilgan) funksiyalarga yaqinlashadi.

$\left[(1+x)^m = 1 + \frac{m}{1}x + \frac{m(m-1)}{1\cdot 2}\cdot x^2 + \dots\right]$ – binomial qator

bo'lib, $|x|<1$ bo'lganda $(1+x)^m$ binomga yaqinlashadi.

$\left[\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} \dots\right]$ qator $-1 \le x \le 1$ bo'lganda $\ln(1+x)$ ga yaqinlashadi

$\left[\operatorname{arctg} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots\right]$ qator $|x| \le 1$ bo'lganda $\operatorname{arctg} x$ ga yaqinlashadi.

Misollar.

① Quyidagi funksiyalar x ning darajalari bo'yicha qatorga yoyilsin

a) $\cos(x-\alpha) = \cos x \cdot \cos\alpha + \sin x \sin\alpha =$

$= \cos\alpha \cdot \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) + \sin\alpha \cdot \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)$

b) $\sin^2 x = \frac{1}{2}(1 - \cos 2x) = \frac{1}{2}\cdot\left(1 - \left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \dots\right)\right) =$

$= \frac{2x^2}{2!} - \frac{2^3x^4}{4!} + \frac{2^5x^6}{6!} - \dots$

c) $x \cdot e^x = x\left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots\right) = x + \frac{x^2}{1!} + \frac{x^3}{2!} + \dots$

d) $\sin\left(mx + \frac{\pi}{3}\right) = \sin mx \cdot \cos\frac{\pi}{3} + \cos mx \cdot \sin\frac{\pi}{3} =$

$= \frac{1}{2}\left(mx - \frac{(mx)^3}{3!} + \frac{(mx)^5}{5!} - \dots\right) + \frac{\sqrt{3}}{2}\left(1 - \frac{(mx)^2}{2!} + \frac{(mx)^4}{4!} - \dots\right)$

② $f(x) = \ln(1 + e^{kx})$ funksiyaning qatorga yoyilmasidagi birinchi uchta hadi yozilsin.

$\ln(1+e^{kx}) \Rightarrow f(x) = f(0) + \frac{f'(0)}{1!} \cdot x + \frac{f''(0)}{2!} \cdot x^2 + \dots$

$\ln(1+e^{kx}) = \ln 2 + \frac{kx}{2} + \frac{k^2 \cdot x^2}{2! \cdot 2^2} + \dots$

$f'(0) = \left(\frac{1}{1+e^{kx}} \cdot e^{kx} \cdot k\right)(0) = \frac{1}{2}k$

$f''(x) = k \cdot \left(\frac{e^{kx}}{1+e^{kx}}\right)' = k \cdot \left(\frac{k \cdot e^{kx} \cdot (1+e^{kx}) - e^{kx} \cdot k \cdot e^{kx}}{(1+e^{kx})^2}\right)$

$f''(0) = k \cdot \left(\frac{2k - k}{4}\right) = \frac{k^2}{4}$

③ $y = x^4 - 3x^2 + 2x + 2$ funksiyani $(x-1)$ ikkihad darajalari bo'yicha yoying.

Yechish: Teylor formulasini yozamiz:

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \dots + \frac{f^n(x_0)}{n!}(x - x_0)^n + \dots$$

$x_0 = 1 \to f(1) = 2$

$f'(1) = (4x^3 - 6x + 2)(1) = 0$

$f''(1) = 12 \cdot 1^2 - 6 = 6.$

$f'''(1) = 24 \cdot 1 = 24$

$f^{IV}(1) = 24$

$f^{V}(1) = 0$ va h.k.

demak:

$f(x)=x^4-3x^2+2x+2=2+\frac{6}{2!}(x-1)^2+\frac{24}{3!}(x-1)^3+\frac{24}{4!}(x-1)^4$

$f(x)=2+3(x-1)^2+4(x-1)^3+(x-1)^4$.

④ $y=\frac{1}{x}$ funksiya uchun $x_0=1$ nuqtada n-darajali Teylor kophadini yozing.

$y(1)=1$

$y'(1)=-\frac{1}{x^2}\Big|_{x=1}=-1$

$y''(1)=+\frac{1\cdot 2}{x^3}\Big|_{x=1}=2!$

$y'''(1)=-\frac{1\cdot 2\cdot 3}{x^4}\Big|_{x=1}=-3!$

$y^{IV}(1)=\frac{1\cdot 2\cdot 3\cdot 4}{x^5}\Big|_{x=1}=4!$ ...

$y^n(1)=(-1)^n\cdot n!$

demak:

$y=1-\frac{1!}{1!}(x-1)+\frac{2!}{2!}\cdot(x-1)^2-\frac{3!}{3!}\cdot(x-1)^3+\ldots \Rightarrow$

$y=1-(x-1)+(x-1)^2-(x-1)^3+(x-1)^4-(x-1)^5+\ldots+(-1)^n(x-1)^n+\ldots$

⑤ $y=2^x$ funksiyani Makloren qatoriga yoying.

$y(0)=1$

$y'(0)=2^x\cdot\ln 2\,|_{x=0}=\ln 2$

$y''(0)=2^x\cdot\ln 2\cdot\ln 2\,|_{x=0}=\ln^2 2$

$\ldots\ldots$

$y^n(0)=\ln^n 2$

$y = 2^x = 1 + \ln 2 x + \frac{\ln^2 2 \cdot x^2}{2!} + \frac{\ln^3 2}{3!} \cdot x^3 + \dots + \frac{\ln^n 2}{n!} x^n + \dots$

⑥ ~~$f(x) = x^4 - 4x + 2x^3 +$~~

$f(x) = x^4 - 5x^3 + x^2 - 3x + 4$ kophadni $(x-4)$ ikkihadning darajalari bo'yicha yoying.

$x = 4$.

$f(4) = 256 - 5 \cdot 64 + 16 - 12 + 4 = -56$

$f'(4) = 4x^3 - 15x^2 + 2x - 3 \,|_{x=4} = 256 - 15 \cdot 16 + 8 - 3 = 21.$

$f''(4) = 12x^2 - 30x + 2 \,|_{x=4} = 192 - 120 + 2 = 74$

$f'''(4) = 24x - 30 \,|_{x=4} = 66.$

$f^{IV}(4) = 24.$

~~$f(x) = \dots + 21$~~.

$f(x) = -56 + 21x + \frac{74x^2}{2!} + \frac{66}{3!} \cdot x^3 + \frac{24}{4!} \cdot x^4$

⑦ $f(x) = x^2 \cdot e^x$ funksiyani Makloren qatoriga yoying va yaqinlashish sohasini toping.

$f(0) = 0$

$f'(0) = 2x \cdot e^x + x^2 \cdot e^x \,|_{x=0} = 0$

~~$f''(0) = \dots$~~

~~$f'''(0) = 2 \cdot e^x + 2x \cdot e^x$~~

$f''(0) = 2 \cdot e^x + 2x \cdot e^x + 2x \cdot e^x + x^2 e^x \,|_{x=0} = 2$

$f'''(0) = 2 \cdot e^x + 2 \cdot e^x + 2 \cdot x \cdot e^x + f''(0) \,|_{x=0} = 6$

$f^{IV}(0) = 6e^x + 4e^x + 4x \cdot e^x + f''(0) \,|_{x=0} = 12$

$\dots$

$f^n(0) = (n-1) \cdot n$

$f(x) = x^2 \cdot e^x = \frac{1 \cdot 2}{2!} \cdot x^2 + \frac{2 \cdot 3}{3!} \cdot x^3 + \frac{3 \cdot 4}{4!} x^4 + \dots + \frac{(n-1)n}{n!} \cdot x^n$

$$f(x)=\frac{x^2}{0!}+\frac{x^3}{1!}+\frac{x^4}{2!}+\dots+\frac{x^n}{(n-2)!}$$

Darajali qatorlarning tatbiqi.

Funksiya qiymatini taqribiy hisoblash.

① e sonini 0,00001 gacha aniqlik bilan toping.

Yechish: $x=1$ da e^x ning qatorga yoyilmasidan foydalanamiz:

$$e=1+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\dots+\frac{1}{n!}+\dots$$

n sonni shunday aniqlaymizki,

$$e\approx 2+\frac{1}{2!}+\frac{1}{3!}+\dots+\frac{1}{n!}$$

taqribiy tenglikning xatoligi 0,00001 dan oshmasin. Qoldiqni baholaymiz:

$$R_n=\frac{1}{(n+1)!}+\frac{1}{(n+2)!}+\dots=\frac{1}{n!}\cdot\left[\frac{1}{n+1}+\frac{1}{(n+1)(n+2)}+\dots\right]<$$

$$<\frac{1}{n!}\cdot\left[\frac{1}{(n+1)}+\frac{1}{(n+1)^2}+\dots\right]=\frac{1}{n!}\cdot\frac{\frac{1}{(n+1)}}{1-\frac{1}{n+1}}=\frac{1}{n!\cdot n}$$

Endi $R<\frac{1}{n!\cdot n}<0{,}00001$

tengsizlikni yechib, $n\geq 8$ ni topamiz. Demak

$$e\approx 2+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\frac{1}{5!}+\frac{1}{6!}+\frac{1}{7!}+\frac{1}{8!}$$

Buni hisoblab, talab qilingan aniqlikdagi javobni olamiz:

$$e\approx 2{,}71828$$

52-savol. Fure qatori.

Agar $y=f(x)$ funksiya (a, b) oraliqda chegaralangan (ya'ni $a<x<b$ da $|f(x)|<M$, bu yerda M - o'zgarmas) va bo'lakli - monoton (ya'ni (a, b) oraliqni har birida bu funksiya monoton bo'lgan chekli sondagi oraliqlarga ajratish mumkin) bo'lsa, u holda bu funksiya (a, b) oraliqda Dirixle shartlarini qanoatlantiradi deyiladi.

Agar $y=f(x)$ funksiya uzunligi 2π ga teng $(-\pi; \pi)$ oraliqda Dirixle shartlarini qanoatlantirsa, u holda bu oraliqning $f(x)$ uzluksiz bo'lgan har qanday x nuqtasida funksiyani Fure trigonometrik qatoriga yoyish mumkin, ya'ni

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx)$$

bu yerda a_n, b_n - Fure koeffitsiyentlari bo'lib, ular quyidagi formulalar bo'yicha hisoblanadi:

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos nx\,dx \quad (n=0, 1, 2, \dots),$$

$$b_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\sin nx\,dx \quad (n=1, 2, 3, \dots)$$

Agar $x\in(-\pi; \pi)$ nuqta $f(x)$ funksiyaning uzilish nuqtasi bo'lsa, Fure qatori yig'indisi $S(x)$ funksiyaning chap va o'ng limitlarining o'rta arifmetigiga teng:

$$S(x)=\frac{1}{2}[f(x-0)+f(x+0)]$$

Oraliq oxirlari $x=\pi$ va $x=-\pi$ nuqtalarda;

$$S(\pi)=S(-\pi)=\frac{1}{2}[f(-\pi+0)+f(\pi+0)]$$

Agar $f(x)$ - juft (ya'ni $f(-x)=f(x)$) bolsa, u holda Fure qatorida faqat kosinuslar qatnashadi, chunki barcha $b_n=0$ bolib, $a_n=\frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$ boladi. Agar $f(x)$ funksiya toq (ya'ni $f(-x)=-f(x)$) bolsa, Fure qatorida faqat sinuslar qatnashadi, chunki barcha $a_n=0$ bolib, $b_n=\frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$ boladi

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$(0;\pi)$ oraliqda berilgan $f(x)$ funksiya $(-\pi;0)$ oraliqda yo juft, yo toq funksiya kabi davom ettirilishi mumkin. Demak, uni zarur bolsa, $(0;\pi)$ oraliqda kosinuslar yoki sinuslar boyicha toliq bolmagan Fure qatoriga yoyish mumkin.

Davri 2π bolgan har qanday davriy $f(x)$ funksiya va istalgan $a\in R$ uchun

$$\int_{-\pi}^{\pi} f(x)dx=\int_{a-\pi}^{a+\pi} f(x)\,dx$$

bolgani uchun Fure koeffitsiyentlarini quyidagi formulalar boyicha hisoblash mumkin:

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx, \qquad b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx$$

bu yerda $n = 0, 1, 2, \ldots$

Misollar:

① Davri 2π bolgan ushbu

$$f(x) = \begin{cases} 0, & \text{agar } -\pi < x < 0 \text{ bolsa} \\ x, & \text{agar } 0 \le x \le \pi \text{ bolsa} \end{cases}$$

funksiyani Fure qatoriga yaying.

Yechish:

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\int_0^{\pi} x\,dx = \frac{1}{\pi}\cdot\frac{x^2}{2}\Big|_0^{\pi} = \frac{\pi}{2}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_0^{\pi} x\cos nx\,dx = \left\{\begin{matrix} u = x, & du = dx \\ dv = \cos nx\,dx, & v = \frac{1}{n}\sin nx \end{matrix}\right\} =$$

$\{uv - \int v\,du\}$

$$= \frac{1}{\pi}\left(\frac{x}{n}\sin nx\Big|_0^{\pi} - \frac{1}{n}\int_0^{\pi}\sin nx\,dx\right) = \frac{1}{\pi n^2}\cos nx\Big|_0^{\pi} = \frac{1}{\pi n^2}\left((-1)^n - 1\right)$$

n - juft bolganda, $a_n = 0$; n - toq bolsa $a_n = -\frac{2}{\pi n^2}$,

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_0^{\pi} x\sin nx\,dx = \left\{\begin{matrix} u = x, & du = dx, \\ dv = \sin nx\,dx, & \\ v = -\frac{1}{n}\cos nx & \end{matrix}\right\} =$$

$$= \frac{1}{\pi}\left(-\frac{x}{n}\cos nx\Big|_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos nx\,dx\right) = \frac{1}{\pi}\left(-\frac{\pi}{n}\cos\pi n + \frac{1}{n^2}\sin nx\Big|_0^{\pi}\right) =$$

$$= \frac{1}{n}(-1)^{n+1}$$

Topilgan koeffitsiyentlardan foydalanib Fure qatorini tuzamiz:

$$f(x_0)=\frac{\pi}{4}+\sum_{n=1}^{\infty}\left(-\frac{2}{\pi(2n-1)^2}\cos(2n-1)x+\frac{(-1)^{n+1}}{n}\cdot\sin nx\right)$$

Bu qator berilgan funksiyaga barcha $x\neq(2n-1)\pi$ larda yaqinlashadi. $x=(2n-1)\pi$ nuqtalarda qator yig'indisi

$$S((2n-1)\pi)=\frac{f(-\pi+0)+f(\pi-0)}{2}=\frac{0+\pi}{2}=\frac{\pi}{2}$$

formula bo'yicha hisoblanadi.

② $-\pi\le x\le\pi$ oraliqda $f(x)=x$ funksiyani Fure qatoriga yoying.

$$a_0=\frac{1}{\pi}\cdot\int_{-\pi}^{\pi}x\,dx=\frac{1}{\pi}\cdot\frac{x^2}{2}\cdot\Big|_{-\pi}^{\pi}=0.$$

$f(x)$ toq funksiya $(x=-(-x))$. shuning uchun faqat sinuslar qatnashadi.

$$b_n=\frac{2}{\pi}\int_0^{\pi}f(x)\sin nx\,dx=\frac{2}{\pi}\cdot\int_0^{\pi}x\cdot\sin nx\,dx=$$

$$=\left\{\begin{matrix}u=x & du=dx\\ dv=\sin nx\,dx & v=-\frac{1}{n}\cos nx\end{matrix}\right\}=\frac{2}{\pi}\cdot\left(-\frac{1}{n}\cdot x\cdot\cos nx+\frac{1}{n}\int\cos nx\,dx\right)\Big|_0^{\pi}=$$

$$=\frac{2}{\pi}\cdot\frac{1}{n}\cdot\left(\frac{1}{n}\cdot\sin nx-x\cos nx\right)\Big|_0^{\pi}=\frac{2}{\pi n}\cdot\left(\frac{1}{n}\cdot\sin\pi n-\pi\cdot(-1)^n-\right.$$

$$\left.-0+0\right)=\frac{2}{\pi n}\cdot\pi\cdot(-1)^{n+1}=\frac{2}{n}\cdot(-1)^{n+1}$$

$$f(x)=\sum_{n=1}^{\infty} b_n \sin nx = 2\cdot\sum_{n=1}^{\infty}(-1)^{n+1}\cdot\frac{\sin nx}{n}$$

3) Ushbu funksiyani Fure qatoriga yoying:

$$f(x)=\begin{cases}-2x, & \text{agar } -\pi\le x<0 \text{ bolsa},\\ 3x, & \text{agar } 0\le x\le\pi \text{ bolsa},\end{cases}$$

$$a_0=\frac{1}{\pi}\cdot\int_{-\pi}^{\pi} f(x)dx=\frac{1}{\pi}\left(\int_{-\pi}^{0}-2x\,dx+\int_{0}^{\pi}3x\,dx\right)=$$

$$=\frac{1}{\pi}\left(-x^2\Big|_{-\pi}^{0}+\frac{3}{2}x^2\Big|_{0}^{\pi}\right)=\frac{1}{\pi}\cdot\left(+\pi^2+\frac{3}{2}\pi^2\right)=\frac{5}{2}\pi.$$

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cdot\cos nx\,dx=\frac{1}{\pi}\int_{0}^{\pi}3x\cos nx\,dx=\begin{Bmatrix}u=x & du=dx\\ dv=\cos nx\,dx & v=\frac{1}{n}\sin nx\end{Bmatrix}=$$

$$=\frac{1}{\pi}\left(uv-\int v\,du\right)\Big|_{0}^{\pi}=\frac{1}{\pi}\left(\frac{x}{n}\cdot\sin nx-\frac{1}{n}\cdot\left(-\frac{1}{n}\right)\cos nx\right)\Big|_{0}^{\pi}=$$

$$=\frac{1}{\pi}\cdot\frac{1}{n^2}\cdot\left((-1)^n-1\right)$$

$$b_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx=$$

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx=\frac{1}{\pi}\cdot\left(\int_{0}^{\pi}3x\cos nx+\int_{-\pi}^{0}-2x\cos nx\,dx\right)=$$

$$=\frac{1}{\pi}\cdot\left(3\cdot\left(\frac{x}{n}\cdot\sin nx+\frac{1}{n^2}\cdot\cos nx\right)\Big|_{0}^{\pi}-2\left(\frac{x}{n}\cdot\sin nx+\frac{1}{n^2}\cos nx\right)\Big|_{-\pi}^{0}\right)=$$

$$=\frac{1}{\pi}\cdot\frac{3}{n^2}\cdot\left((-1)^n-1\right)-\frac{2}{n^2}\left((-1)^n-(-1)^n\right)=\frac{1}{\pi}\cdot\left(\frac{5}{n^2}\cdot(-1)^n-\frac{5}{n^2}\right)$$

$$b_n = \frac{1}{\pi} \cdot \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{1}{\pi} \cdot \left(\int_0^{\pi} 3x \sin nx \, dx + \int_{-\pi}^{0} -2x \sin nx \, dx \right) =$$

$$= \frac{1}{\pi} \left(\left(-\frac{3x}{n} \cdot \cos nx - \frac{3}{n^2} \cdot \sin nx \right) \Big|_0^{\pi} + \left(\frac{2x}{n} \cdot \cos nx + \frac{2}{n^2} \sin nx \right) \Big|_{-\pi}^{0} \right) =$$

$$= \frac{1}{\pi} \cdot \left(\left(-\frac{3\pi}{n} \right) \cdot (-1)^n - \frac{2\pi}{n} \cdot (-1)^n \right) = \frac{1}{\pi} \cdot \left(-\frac{5\pi}{n} \right) \cdot (-1)^n$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) =$$

$$= \frac{5\pi}{4} + \sum_{n=1}^{\infty} \left(\frac{5}{\pi n^2} ((-1)^n - 1) \cdot \cos nx + \left(-\frac{5}{n} \right) \cdot (-1)^n \sin nx \right)$$

38-savol. Cheksiz katta va cheksiz kichik funksiyalar va ularni taqqoslash.

Birinchi va ikkinchi ajoyib limitlar.

$\left[\lim\limits_{\alpha\to 0}\frac{\sin\alpha}{\alpha}=1\right]$ – birinchi ajoyib limit.

$\left[\lim\limits_{x\to\infty}\left(1+\frac{1}{x}\right)^x=\lim\limits_{\alpha\to 0}(1+\alpha)^{\frac{1}{\alpha}}=e\right]$ – ikkinchi ajoyib limit.

Misollar yechganda quyidagi tengliklarni nazarda tutish foydali:

$\left[\lim\limits_{\alpha\to 0}(1+k\alpha)^{\frac{1}{\alpha}}=\lim\limits_{x\to\infty}\left(1+\frac{k}{x}\right)^x=e^k\right]$

$\left[\lim\limits_{x\to 0}\frac{\ln(1+x)}{x}=1\right]$

$\left[\lim\limits_{x\to 0}\frac{e^x-1}{x}=1\right]$ $\left[\lim\limits_{x\to 0}\frac{a^x-1}{x}=\ln a\ (a>0)\right]$

Misollar:

① $\lim\limits_{x\to 0}\frac{\sin 3x}{x}=\lim\limits_{x\to 0}\frac{\sin 3x}{3x}\cdot 3=3$

② $\lim\limits_{x\to\frac{\pi}{2}}\frac{\cos x}{\pi-2x}=\lim\limits_{x\to\frac{\pi}{2}}\frac{\sin\left(\frac{\pi}{2}-x\right)}{2\cdot\left(\frac{\pi}{2}-x\right)}\to$ $\frac{\pi}{2}-x=z$, $x\to\frac{\pi}{2}\to z\to 0$

$\lim\limits_{z\to 0}\frac{\sin z}{2z}=\frac{1}{2}$

③ $\lim\limits_{x\to+\infty}\left(\frac{2x+1}{2x-3}\right)^{4x-1}=\lim\limits_{x\to\infty}\left(\frac{2x-3+4}{2x-3}\right)^{4x-1}=$

$=\lim\limits_{x\to\infty}\left(1+\frac{1}{\frac{2x-3}{4}}\right)^{\frac{2x-3}{4}\cdot\frac{4(4x-1)}{2x-3}}=\lim\limits_{x\to\infty}e^{\frac{4(4x-1)}{2x-3}}=$

$=\lim\limits_{x\to\infty}e^{\frac{4(4-\frac{1}{x})}{2-\frac{3}{x}}}=e^{8}$

④ $\lim\limits_{x\to0}\frac{1-\cos6x}{x\cdot\sin5x}=\lim\limits_{x\to0}\frac{2\sin^2 3x}{3x\cdot\sin5x}\cdot3=$

$=6\cdot\lim\limits_{x\to0}\frac{\sin3x}{3x}\cdot\lim\limits_{x\to0}\frac{\sin3x}{\sin5x}=6\cdot\lim\limits_{x\to0}\frac{\sin3x}{3x}\cdot\frac{5x}{\sin5x}\cdot\frac{3}{5}=$

(first limit $=1$, second limit $=1$)

$=6\cdot\frac{3}{5}=\frac{18}{5}$

⑤ $\lim\limits_{x\to\pi}\frac{\sin3x}{\sin2x}=\lim\limits_{x\to\pi}\frac{-\sin(3\pi-3x)}{+\sin(2\pi-2x)}\Rightarrow$ $\pi-x=z$, $x\to\pi\Rightarrow z\to0$

$\lim\limits_{z\to0}\frac{-\sin3z}{\sin2z}=\lim\limits_{z\to0}\frac{-\sin3z}{3z}\cdot\frac{2z}{\sin2z}\cdot\frac{3}{2}=-\frac{3}{2}$

⑥ $\lim\limits_{x\to0}\frac{e^{3x}-e^{2x}}{3x}=\lim\limits_{x\to0}\frac{e^{2x}(e^{x}-1)}{3x}=\frac{1}{3}$ ($e^{2x}\to1$)

⑦ $\lim\limits_{x\to1}(3-2x)^{\frac{4}{x-1}}=\lim\limits_{x\to1}(1+2(1-x))^{\frac{4}{x-1}}=$

$=\lim\limits_{x\to1}\left(1+\frac{1}{\frac{1}{2(1-x)}}\right)^{\frac{1}{2(1-x)}\cdot(-8)}\Rightarrow$ $1-x=z$, $x\to1\Rightarrow z\to0$

e^{-8}

(8) $\lim\limits_{x\to 0} \frac{\sin 3x}{\operatorname{tg} 5x} = \lim\limits_{x\to 0} \frac{\sin 3x}{\sin 5x} \cdot \cos 5x =$

$= \underbrace{\lim\limits_{x\to 0} \frac{\sin 3x}{\sin 5x}}_{3/5} \cdot \underbrace{\lim\limits_{x\to 0} \cos 5x}_{1} = \frac{3}{5}$

Ekvivalent cheksiz kichik funksiyalar va ular yordamida limitlarni hisoblash

Agar $\alpha(x)$ va $\beta(x)$ $x \to x_0$ da cheksiz kichik funksiyalar bolib,

$$\lim_{x\to x_0} \frac{\alpha(x)}{\beta(x)} = 1$$

bolsa, u holda ekvivalent d-di va $x \to x_0$ da $\alpha(x) \sim \beta(x)$ kabi belgilanadi. Masalan $\lim\limits_{x\to 0} \frac{\sin x}{x} = 1$, shu sababli $x \to 0$ da $\sin x \sim x$

Shunga oxshash $x \to 0$ da quyidagi cheksiz kichik funksiyalar ekvivalentdir:

$\operatorname{arctg} x \sim x$,

$1 - \cos x \sim \frac{x^2}{2}$

$\operatorname{tg} x \sim x$

$\operatorname{arctg} x \sim x$

$e^x - 1 \sim x$

$a^x - 1 \sim x \ln a$

$\ln(1+x) \sim x$

$(1+x)^m - 1 \sim mx$ va h.k.

Ikkita cheksiz kichik funksiyalar nisbatining limiti ularga ekvivalent cheksiz kichik funksiyalar nisbatining limitiga teng, ya'ni $x \to x_0$ da $\alpha(x) \sim \alpha_1(x)$ va $\beta(x) \sim \beta_1(x)$ bolsa, u holda

$$\left[\lim_{x \to x_0} \frac{\alpha(x)}{\beta(x)} = \lim_{x \to x_0} \frac{\alpha_1(x)}{\beta_1(x)}\right]$$

Misollar.

① Limitni hisoblang.

$$\lim_{x \to 0} \frac{1-\cos 4x}{tg^2 3x}$$

$$1-\cos 4x \sim \frac{16x^2}{2} = 8x^2$$

$$tg^2 3x \sim (3x)^2 = 9x^2$$

$$\lim_{x \to 0} \frac{8x^2}{9x^2} = \frac{8}{9}$$

② Limitni hisoblang.

$$\lim_{x \to 0} \frac{\sqrt[3]{8+3x} - 2}{\sqrt[4]{16+5x} - 2}$$

Yechish: kasrning surat va maxrajini 2 ga bolib, songra ularni ekvivalent cheksiz kichik funksiyalar b-n almashtiramiz:

$$\lim_{x \to 0} \frac{\sqrt[3]{1+\frac{3}{8}x} - 1}{\sqrt[4]{1+\frac{5}{16}x} - 1} = \lim_{x \to 0} \frac{\left(1+\frac{3}{8}x\right)^{\frac{1}{3}} - 1}{\left(1+\frac{5}{16}x\right)^{\frac{1}{4}} - 1} = \lim_{x \to 0} \frac{\frac{1}{3} \cdot \frac{3}{8}x}{\frac{1}{4} \cdot \frac{5}{16}x} = \frac{8}{5}$$

Cheksiz kichik funksiyalarni taqqoslash.

$x \to x_0$ da $\alpha(x)$ va $\beta(x)$ cheksiz kichik funksiyalar bolsin. Bu funksiyalarni taqqoslash uchun ular nisbatining $x \to x_0$ dagi limiti hisoblanadi:

a) agar $\lim\limits_{x \to x_0} \frac{\alpha(x)}{\beta(x)} = 0$ bolsa, uholda $\alpha(x)$ funksiya $\beta(x)$ ga nisbatan yuqori tartibli cheksiz kichik funksiya deyiladi va $\alpha = o(\beta)$ kabi belgilanadi.

b) agar $\lim\limits_{x \to x_0} \frac{\alpha(x)}{\beta(x)} = \infty$ bolsa, u holda $\alpha(x)$ funksiya $\beta(x)$ ga nisbatan quyi tartibli cheksiz kichik funksiya deyiladi. Ravshanki bu holda $\lim\limits_{x \to 0} \frac{\beta(x)}{\alpha(x)} = 0$ yoki $\beta = o(\alpha)$

c) agar $\lim\limits_{x \to 0} \frac{\alpha(x)}{\beta(x)} = A \neq 0$ va A chekli son bolsa, u holda $\alpha(x)$ va $\beta(x)$ bir xil tartibli cheksiz kichik funksiyalar deyiladi.

Xususan, agar $A = 1$ bolsa, u holda ekvivalent cheksiz kichik funksiyalarga ega bolamiz.

d) Agar $\alpha(x^k)$ va $\beta(x)$ bir xil tartibli cheksiz kichik funksiyalar bolib $k > 0$ bolsa, u holda $\beta(x)$ cheksiz kichik funksiya $\alpha(x)$ ga nisbatan k-tartibga ega deyiladi.

Misollar.

① $x \to 0$ da $\alpha = x\sin^2 x$ va $\beta = 2x\sin x$ cheksiz kichik funksiyalarni taqqoslang.

$$\lim_{x\to 0} \frac{\alpha(x)}{\beta(x)} = \lim_{x\to 0} \frac{x\sin^2 x}{2x\sin x} = \lim_{x\to 0} \frac{\sin x}{2} = 0 \to \alpha = o(\beta)$$

② $x \to 0$ $\alpha = x \cdot \ln(1+x)$, $\beta = x\sin x$.

$$\lim_{x\to 0} \frac{\alpha}{\beta} = \lim_{x\to 0} \frac{x\cdot\ln(1+x)}{x\sin x}$$

$\ln(1+x) \sim x$

$$\lim_{x\to 0} \frac{x}{\sin x} = 1 \to \alpha \sim \beta$$

③ $x = 0 \to \alpha = x^2\sin^2 x$ $\beta = x \cdot \operatorname{tg} x$.

$$\lim_{x\to 0} \frac{\alpha}{\beta} = \lim_{x\to 0} \frac{x^2\sin^2 x}{x\operatorname{tg} x}$$

$\operatorname{tg} x \sim x$

$$\lim_{x\to 0} \frac{x^2\sin^2 x}{x^2} = \lim_{x\to 0} \sin^2 x = 0 \to \alpha = o(\beta)$$

④ $x \to 0$ $\alpha = a^x - 1 \sim x\ln a$ $b = x\ln a$

$\to (a \sim b)$

39-savol. Funksiya uzluksizligi. Uzluksiz funksiyalar ustida arifmetik amallar.

40-savol. Funksiya uzilishi, uzilish turlari.

Agar x_0 va uning atrofida aniqlangan $y=f(x)$ funksiya shu nuqtada chekli limitga ega bolib, bu limit funksiyaning x_0 nuqtadagi qiymatiga teng, ya'ni

$$\lim_{x\to x_0} f(x) = f(x_0)$$

bolsa, u holda bu funksiya x_0 nuqtada **uzluksiz** d-di.

Funksiyaning uzluksizligi haqidagi quyidagi ta'rif yuqoridagi ta'rifga teng kuchlidir.

Agar $y=f(x)$ funksiya x_0 nuqtada va uning atrofida aniqlangan bolib, argumentning cheksiz kichik orttirmasiga funksiyaning cheksiz kichik orttirmasi mos kelsa, ya'ni

$$\lim_{\Delta x\to 0} \Delta y = 0$$

bolsa, u holda funksiya x_0 nuqtada **uzluksiz** d-di. Bu yerda:

$\Delta x = x - x_0$ - argument orttirmasi

$\Delta y = f(x_0+\Delta x) - f(x_0)$ - funksiya orttirmasi

$f(x)$ funksiyaning x_0 nuqtada uzluksiz bo'lishi uchun uzluksizlikning quyidagi shartlari zarur va yetarlidir:

a) funksiya x_0 nuqta va uning atrofida aniqlangan.

b) funksiyaning $x=x_0$ nuqtadagi chap va o'ng limitlari teng.

$$f(x_0-0) = f(x_0+0)$$

c) $x=x_0$ nuqtadagi bir tomonli limitlar $f(x_0)$ ga teng, ya'ni $f(x_0-0) = f(x_0+0) = f(x_0)$

$f(x)$ funksiya x_0 nuqtaning atrofida aniqlangan, ammo bu nuqtaning o'zida uzluksizlik shartlaridan aqalli bittasi bajarilmasa, bu funksiya x_0 nuqtada **uzilishga ega** deyiladi.

Agar $f(x)$ funksiya uchun chekli bir tomonli $f(x_0-0)$ va $f(x_0+0)$ limitlar mavjud bo'lsa va, shu bilan birga, $f(x_0)$, $f(x_0-0)$, $f(x_0+0)$ sonlar o'zaro teng bo'lmasa, u holda x_0 nuqta **1-tur uzilish nuqtasi** d-di.

Xususan agar $f(x_0-0) = f(x_0+0) \neq f(x_0)$ bo'lsa, u holda x_0 **bartaraf qilinadigan uzilish nuqtasi** d-di.

Agar $f(x_0-0)$ yoki $f(x_0+0)$ bir tomonli limitlardan aqalli bittasi ∞ ga teng bo'lsa, x_0 nuqta **2-tur uzilish nuqtasi** d-di.

Misollar

① $f(x)=\frac{x-2}{|x-2|}$ funksiyaning uzilish nuqtasi va turi aniqlansin.

Yechish: funksiya $x_0=2$ nuqtada aniqlanmagan.

$f(x)=\frac{x-2}{-(x-2)}=-1$ $\quad f(x)=\frac{x-2}{x-2}=1$

demak $\lim\limits_{x\to 2^-} f(x)=-1$, $\lim\limits_{x\to 2^+} f(x)=1$.

1-tur uzilish nuqtasi.

② $y=\frac{x}{x-4}$ funksiyaning uzilish nuqtasini toping va uzilish turini aniqlang.

Yechish: $x_0=4$ da aniqlanmagan.

$\lim\limits_{x_0\to 4^-} \frac{x}{x-4}=\lim\limits_{x_0\to 4^+} \frac{x}{x-4}=\infty$ 2-tur uzilish nuqtasi

③ $f(x)=\frac{|x|}{x}$

Yechish: $f(0^+)=\lim\limits_{x\to 0^+}\frac{|x|}{x}=\lim\limits_{x\to 0^+}\frac{+x}{x}=1$

$f(-0)=\lim\limits_{x\to 0^-}\frac{|x|}{x}=\lim\limits_{x\to 0^-}\frac{-x}{x}=-1$

x_0 - 1-tur uzilish nuqtasi

④ $f(x)=2^{\frac{1}{x-1}}$ $\quad x_0=1$.

$f(1^-)=\lim\limits_{x\to 1^-} 2^{\frac{1}{x-1}}=2^{-\infty}=0$ $\quad x_0$ - 2-tur uzilish nuqtasi.

$f(1^+)=\lim\limits_{x\to 1^+} 2^{\frac{1}{x-1}}=2^{\infty}=\infty$

41-savol. Funksiya hosilasi. Teskari funksiya hosilasi. Murakkab funksiyaning hosilasi.

Ta'rif: Funksiya orttirmasi Δy ning argument orttirmasi Δx ga nisbatining $\Delta x \to 0$ dagi limiti y funksiyaning hosilasi deb ataladi. (y', $f'(x_0)$, $\frac{dy}{dx}$, $\frac{df}{dx}$).

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x_0+\Delta x) - f(x_0)}{\Delta x}$$

Teskari funksiya hosilasi

Teskari funksiyaning hosilasi funksiya hosilasiga teskari miqdorga tengdir.

$$x'_y = \frac{1}{y'_x}$$

Teskari funksiya: Agar $f(x_1)=y_1$ va $f(x_2)=y_2$ da $x_1=y_2$, $x_2=y_1$ bolsa demak y_1 va y_2 funksiyalar o'zaro teskari funksiyalardir.

Misollar.

① $f(x) = x^3+1$ $\quad (f^{-1}(9))' - ?$

a) $f(x_1)=y_1$ $\quad f^{-1}(x_2)=y_2$

$x_1=y_2$ $\quad x_2=y_1=9$.

$(x_1; y_1) \leftrightarrow (x_2; y_2)$

$(x_1; 9) \leftrightarrow (9; y_2)$

$x_1^3+1=9 \to x_1=2$

$(2;9) \leftrightarrow (9;2)$

$$(f^{-1}(9))' = \frac{1}{f'(2)} = \frac{1}{3x^2}\Big|_{x=2} = \frac{1}{12}$$

в) $f(x) = x^3 + 1 \rightarrow \quad f^{-1}(x) = \sqrt[3]{x-1}$

$$(f^{-1}(9))' = \frac{1}{3\sqrt[3]{(-1)^2}} = \frac{1}{3 \cdot 2^2} = \frac{1}{12}$$

② ~~$y_1 = x$~~ $\quad f(x) = x^5 - 26 \qquad (f^{-1}(6))' - ?$

$x_2 = y_1 = 6.$

$x_1^5 - 26 = 6 \rightarrow x_1 = 2$

$(2;6) \leftrightarrow (6;2)$

$$(f^{-1}(6))' = \frac{1}{f'(2)} = \frac{1}{5x^4}\Big|_{x=2} = \frac{1}{80}$$

43-savol. Funksiya differensiali, yuqori tartibli hosila va differensial.

$$f''(x_0) = \frac{d^2 f(x_0)}{dx^2}$$

$$f^{(n+1)}(x) = \left(f^{(n)}(x)\right)'$$

Misollar:

① $(a^x)^{(n)} - ?$

$$(a^x)' = a^x \ln a$$
$$(a^x)'' = a^x (\ln a)^2$$

② $(x^a)' = a x^{a-1}$

$(x^a)'' = a(a-1) \cdot x^{a-2}$

$\dots$

$(x^a)^{(n)} = a(a-1)(a-2)\dots(a-n+1) \cdot x^{a-n}$.

③ $f(x) = \frac{1}{x} \quad (x > 0)$

$$\left(\frac{1}{x}\right)^{(n)} = \frac{(-1)^n \cdot n!}{x^{n+1}}$$

44-savol. Differensial hisobning asosiy teoremalari. Teylor formulasi.

Roll teoremasi: Agar $y=f(x)$ funksiya $[a, b]$ kesmada uzluksiz, (a,b) oraliqda differensiallanuvchi va $f(a)=f(b)$ bolsa, u holda agalli bitta shunday $x=c$ $(a<c<b)$ nuqta mavjudki, unda $f'(c)=0$ boladi.

Bu teorema hosilaning nollari yoki ildizlari haqidagi teorema ham d-di.

Lagranj teoremasi. Agar $y=f(x)$ funksiya $[a;b]$ kesmada uzluksiz, (a, b) oraliqda differensiallanuvchi bolsa, u holda agalli bitta shunday $x=c$ $(a<c<b)$ nuqta mavjudki,

$$f(b)-f(a)=f'(c)(b-a)$$

boladi.

Bu teorema chekli ayirmalar haqidagi teorema ham deyiladi.

Koshi teoremasi. Agar $y=f(x)$ va $y=\varphi(x)$ funksiyalar $[a, b]$ kesmada uzluksiz, (a, b) oraliqda differensiallanuvchi, shu bilan birga bu oraliqda $\varphi'(x)\neq 0$ bolsa, u holda agalli bitta shunday $x=c$ $(a<c<b)$ nuqta mavjudki,

$$\frac{f(b)-f(a)}{\varphi(b)-\varphi(a)}=\frac{f'(c)}{\varphi'(c)}$$

boladi, bunda $\varphi(b)\neq\varphi(a)$

Misollar.

① $[1,5]$ kesmada $f(x)=x^2-6x+100$ funksiya uchun Roll teoremasi o'rinlimi? x ning qanday qiymatlarida $f'(x)=0$ boladi?

Yechish: $f(1)=f(5)=95$ demak Roll teoremasi shartlari bajariladi.

$f'(x_0)=0$ boladigan x_0 ni topamiz.

$f'(x_0)=2x_0-6=0$

$x_0=3$

② $f(x)=\sqrt[3]{(x-8)^2}$ funksiya uchun $[0;10]$ kesmada Lagranj teoremasi o'rinlimi?

Yechish: $f(x)$ funksiya x ning barcha qiymatlarida uzluksiz, ammo uning $f'(x)=\frac{2}{3\sqrt[3]{x-8}}$ hosilasi $(0;10)$ oraliqning $x=8$ nuqtasida mavjud emas shunga ko'ra Lagranj teoremasi o'rinli emas.

46-savol:

Aniqmasliklarni ochishning Lopital qoidasi.

($\frac{0}{0}$ yoki $\frac{\infty}{\infty}$ ko'rinishdagi aniqmasliklarni ochish.)

$f(x)$ va $\varphi(x)$ funksiyalar x_0 nuqtaning biror atrofida (x_0 nuqtaning o'zidan tashqari) differensiallanuvchi va $\varphi'(x)\neq 0$ bo'lsin. Agar $\lim\limits_{x\to x_0} f(x)=\lim\limits_{x\to x_0}\varphi(x)=0$ yoki $\lim\limits_{x\to x_0} f(x)=\lim\limits_{x\to x_0}\varphi(x)=\infty$ bolib, $\lim\limits_{x\to x_0}\frac{f'(x)}{\varphi'(x)}$ mavjud bo'lsa, u holda $\left[\lim\limits_{x\to x_0}\frac{f(x)}{\varphi(x)}=\lim\limits_{x\to x_0}\frac{f'(x)}{\varphi'(x)}\right]$ boladi.

$x\to\infty$ da ham Lopital qoidasi o'rinli.

$0\cdot\infty$ yoki $\infty-\infty$ shaklidagi aniqmasliklar algebraik almashtirishlar orqali $0/0$ yoki ∞/∞ ko'rinishdagi aniqmasliklarga keltirilib, so'ngra Lopital qoidasidan foydalaniladi.

0^0, ∞^0 yoki 1^∞ ko'rinishdagi aniqmasliklar logarifmlash orqali $0/0$ yoki ∞/∞ ko'rinishdagi aniqmasliklarga keltiriladi.

Misollar.

① $\lim\limits_{x\to 0}\dfrac{x-\sin x}{x^3}$ ni toping.

Yechish: Ifodaning surati va maxraji $x\to 0$ da nolga intiladi, shu sababli $0/0$ shaklidagi aniqmaslikka egamiz. Lopital qoidasidan foydalansak,

$$\lim_{x\to 0}\frac{x-\sin x}{x^3}=\lim_{x\to 0}\frac{1-\cos x}{3x^2}=\lim_{x\to 0}\frac{\sin x}{6x}=\frac{1}{6}$$

Bu yerda Lopital qoidasi 2 marta qo'llanildi.

② $\lim\limits_{x\to 0}x^2\cdot\ln x=?$

$$\lim_{x\to 0}x^2\ln x=\lim_{x\to 0}\frac{\ln x}{\frac{1}{x^2}}=\lim_{x\to 0}\frac{\frac{1}{x}}{-\frac{2}{x^3}}=-\frac{1}{2}\lim_{x\to 0}x^2=0$$

③ $\lim\limits_{x\to 0}(\sin x)^x-?$ 0^0 shaklidagi aniqmaslik.

$$y=\sin x^x$$
$$\ln y=x\ln\sin x=\frac{\ln\sin x}{\frac{1}{x}}$$

$$\lim_{x\to 0}\ln y=\lim_{x\to 0}\frac{\ln\sin x}{\frac{1}{x}}=\lim_{x\to 0}\frac{\frac{\cos x}{\sin x}}{-\frac{1}{x^2}}=-\lim_{x\to 0}\frac{x^2\cdot\cos x}{\sin x}=$$

$= -\lim\limits_{x\to 0}\left(x\cdot\frac{x}{\sin x}\cdot\cos x\right) = 0$

$\lim\limits_{x\to 0} y = e^0 = 1$

④ $[-1;0]$ va $[0;1]$ kesmalarda $f(x)=x-x^3$ funksiya uchun Roll teoremasi o'rinlimi? Agar o'rinli bo'lsa, u holda x ning tegishli qiymatlarini toping.

- $[-1;0]$, $[0;1]$ kesmalarda uzluksiz.
- $f'(x)=1-3x^2 \to (a;b)$ kesmada differensiallanuvchan.
- $f(-1)=0$, $f(0)=0$, $f(1)=0$ $f(-1)=f(0)=f(1)=0$ demak Roll teoremasi o'rinli.

$f'(x)=0$

$1-3x^2=0 \to \boxed{x=\pm\frac{1}{\sqrt{3}}}$

⑤ $[-1;2]$ kesmada $\frac{4}{x}$ va $1-\sqrt[3]{x^2}$ funksiyalarga Lagranj teoremasini qo'llab bo'ladimi?

$y=\frac{4}{x}$ funksiya $x=0$ da aniqlanmagan.

$y=1-\sqrt[3]{x^2} \to y'=-\frac{2}{3\sqrt[3]{x}} \to x=0$ da aniqlanmagan ya'ni $[a,b]$ oraliqda differensiallanuvchan emas.

demak, ikkala funksiya uchun ham Lagranj teoremasi o'rinsiz.

Lopital qoidasiga doir misollar:

① $\lim\limits_{x\to 1}\frac{x^3-3x^2+4x+2}{x^3-5x+4}$ $\Rightarrow$ ($x=1$ da nolga teng) $\to \frac{0}{0}$ $\Rightarrow$ ($x=1$ da nolga teng)

$$\lim_{x\to 1}\frac{3x^2-4x+4}{3x^2-5}=\frac{-7}{-2}=\frac{7}{2}.$$

② $$\lim_{x\to 0} tgx\cdot \ln x=\lim_{x\to 0}\frac{\ln x}{ctgx}=\lim_{x\to 0}\frac{\frac{1}{x}}{\frac{1}{\sin^2 x}}=$$

$$=\lim_{x\to 0}\frac{-\sin^2 x}{x}=\lim_{x\to 0}\frac{-\sin 2x}{1}=0$$

③ $$\lim_{x\to 0}(tgx)^{\sin x}$$

$$y=(tgx)^{\sin x}$$

$$\ln y=\sin x\cdot \ln(tgx)$$

$$\lim_{x\to 0}\ln y=\lim_{x\to 0}\frac{\ln tgx}{\frac{1}{\sin x}}=\lim_{x\to 0}\frac{\frac{1}{tgx}\cdot\frac{1}{\cos^2 x}}{-\frac{1}{\sin^2 x}}=\lim_{x\to 0} tgx=0$$

$$\ln y=0\to y=e^0=1.$$

Teylor formulasi.

Agar $y=f(x)$ funksiya x_0 nuqtaning biror atrofida $(n+1)$-tartibgacha hosilalarga ega bolsa, $((n+1)$-tartibli hosila ham tiradi), u holda bu atrofning har qanday x nuqtasi uchun n-tartibli Teylor formulasi orinli:

$$f(x)=f(x_0)+\frac{f'(x_0)}{1!}(x-x_0)+\frac{f''(x_0)}{2!}(x-x_0)^2+\dots+$$

$$+\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n+R_n(x)$$

bunda: $R_n(x)=\frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}$ Teylor formulasining Lagranj shaklidagi qoldiq hadi deyiladi. ξ nuqta x va x_0 nuqtalar orasida yotadi, yani $\xi=x_0+\theta(x-x_0)$ va $0<\theta<1$

Misollar. ① $f(x)=x^3-2x^2+3x+5$ ko'phadni $(x-2)$ ikkihadning butun musbat darajalari bo'yicha yoying.

Yechish: $x_0=2$

$f'(x)=3x^2-4x+3$

$f''(x)=6x-4$

$f'''(x)=6$ $\quad n\geq 4$ uchun $f^{(n)}(x)=0$

$f(2)=11,\ f'(2)=7,\ f''(2)=8\ \ f'''(2)=6$

demak,

$$f(x)=x^3-2x^2+3x+5=11+7(x-2)+4(x-2)^2+(x-2)^3$$

② $x_0=-1$ da $f(x)=e^x$ funksiya uchun 3-tartibli Teylor formulasini yozing.

Yechish: $f^{(n)}(x)=e^x$ $\quad f^{(n)}(-1)=\frac{1}{e}$

$$e^x=\frac{1}{e}+\frac{1}{e}\cdot\frac{x+1}{1!}+\frac{1}{e}\cdot\frac{(x+1)^2}{2!}+\frac{1}{e}\cdot\frac{(x+1)^3}{3!}+R_3(x)$$

45-savol. Funksiyaning ekstremum qiymatlari. Funksiyaning qavariq va botiqligi.

Misollar: ① $y = x^4 - 2x^2 + 3$ funksiyaning $[-3; 2]$ kesmadagi ekstremumlarini toping.

$y' = 4(x^3 - x)$ $x_1 = -1$, $x_2 = 0$, $x_3 = 1$.

$y(0) = 3$ $y(1) = 2$, $y(-1) = 2$, $y(-3) = 66$, $y(2) = 11$.

$y_{max} = 66$ $y_{min} = 2$.

② Funksiyaning monotenlik oralig'ini toping.

$y = 2 - 3x + x^3$

$y' = -3 + 3x^2 = 3(x^2 - 1)$

$y' > 0 \rightarrow 3(x-1)(x+1) > 0$

$(-\infty; -1) \cup (1; \infty)$ - o'sadi

$(-1; 1)$ - kamayadi

$y = f(x)$ funksiyaning grafigi (a, b) oralig'ining istalgan nuqtasida o'tkazilgan urinmadan pastda yotsa, u holda funksiya grafigi qavariq, yuqorida yotsa botiq deyiladi.

Funksiya grafigining qavariq qismini botiq qismidan ajratuvchi $M_0(x_0, f(x_0))$ nuqta grafikning egilish nuqtasi deyiladi.

Funksiya grafigining qavariq yoki botiq bo'lishining yetarlilik shartlari.

Agar (a, b) oraliqda differensiallanuvchi $y = f(x)$ funksiyaning 2-tartibli hosilasi manfiy, ya'ni $f''(x) < 0$ bo'lsa, u holda bu oraliqda funksiya grafigi qavariq, $f''(x) > 0$ bo'lsa botiq bo'ladi.

Misol:
$y=xe^x$ funksiyaning qavariqlik, botiqlik oraliqlarini va egilish nuqtalarini toping.

Yechish: Aniqlanish sohasi: $x\in(-\infty;\infty)$

$y'=e^x(1+x)$, $y''=e^x(2+x)$

$x_0=-2$

$(-\infty;-2)\to$ botiq

$(-2;\infty)\to$ qavariq

Agar $y=f(x)$ funksiya grafigidagi nuqta shu grafik boylab cheksiz uzoqlashda undan biror to'g'ri chiziqgacha bo'lgan masofa nolga intilsa, bu to'g'ri chiziq $y=f(x)$ funksiya grafigining **asimptotasi** deb ataladi.

Agar $\lim\limits_{x\to\infty} f(x)=\infty$ bo'lsa, $x=a$ to'g'ri chiziq $y=f(x)$ funksiya grafigining **vertikal asimptotasi** deyiladi.

Agar $k=\lim\limits_{x\to+\infty}\frac{f(x)}{x}$ va $b=\lim\limits_{x\to+\infty}(f(x)-kx)$

$k=\lim\limits_{x\to-\infty}\frac{f(x)}{x}$ va $b=\lim\limits_{x\to-\infty}(f(x)-kx)$

limitlar mavjud bo'lsa, u holda $y=kx+b$ to'g'ri chiziq $y=f(x)$ funksiyaning **og'ma asimptotasi** deyiladi.

Misol: $y=\frac{x^2-2x+3}{x+2}$ funksiyaning asimptotalarini toping.

Yechish: $\lim\limits_{x\to -2} y = \infty$ demak $x=-2$ → vertikal asimptota.

$$k = \lim_{x\to\infty} \frac{f(x)}{x} = \lim_{x\to\infty} \frac{x^2-2x+3}{x(x+2)} = 1$$

$$b = \lim_{x\to\infty} (f(x)-kx) = \lim_{x\to\infty} \left(\frac{x^2-2x+3}{x+2} - x\right) =$$

$$= \lim_{x\to\infty} \frac{-4x+3}{x+2} = -4 \qquad y = x-4 \text{ → og'ma asimptota}$$

Ehtimollar nazariyasi.

80-savol. Kombinatorika elementlari. Orinlashtirish, orin almashtirish. Guruhlash.

Kombinatsiyalar 3 xil:

1. Orin almashtirish (permutation - перестановка)
2. Orinlashtirish (arrengent - размещения)
3. Guruhlash (combination - сочетания)

Orin almashtirish.

n ta elementli orin almashtirishlar deb bir-biridan faqat elementlarining tartibi bilan farq qiladigan n ta elementli birikmalarga aytiladi.

Masalan, 3 ta A, B, C elementlardan 6 ta orin almashtirish bajarish mumkin:
ABC, ACB, BAC, CBA, BCA, CAB.

$$P_n = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot (n-1) \cdot n = n!$$

Orinlashtirishlar.

n ta elementdan m tadan orinlashtirishlar deb, har birida berilgan n ta elementdan t tasi olingan shunday birikmalarga aytiladiki, ularning har biri hech bolmaganda bitta elementi bilan yoki faqat ularning joylashish tartibi bilan farq qiladi.

Masalan 3 element A, B, C dan 2 elementli 6 ta orinlashtirish mavjud:
AB, AC, BC, BA, CA, CB.

$$A_n^m = \frac{n!}{(n-m)!} \qquad 0 \le m \le n$$

O'rin almashtirishga misol:
1, 2, 3 raqamlardan ularning har biri faqat bir marta uchraydigan nechta 3 xonali son tuzish mumkin?

$$P_3 = 3! = 6$$

O'rinlashtirishga misol:
Tijorat banki boshqarmasi turli lavozimlarga 10 ta nomzoddan 3 tasini tanlamoqda. Har bir nomzod bir xil imkoniyatga ega. 10 ta nomzoddan 3 kishidan iborat nechta guruh tuzish mumkin?

$n=10 \quad m=3$

$$A_n^m = \frac{n!}{(n-m)!} = \frac{10!}{7!} = 10 \cdot 9 \cdot 8 = 720$$

Guruhlash.

n element orasidan t ta elementdan tuzilgan guruhlar deb har biriga berilgan n ta elementdan m tasi olingan, shunday birikmalarga aytiladiki, ularning har biri hech bo'lmaganda bitta elementi bilan farq qiladi. Misol uchun, 3 element A, B, C dan ikkita elementli 3 ta moslik mavjud: AB, AC, BC

$$\left[C_n^m = \frac{n!}{m!(n-m)!}\right] \quad (0 \le m \le n)$$

Uning xossalari:

1. $C_n^0 = C_0^0 = 1$
2. $C_n^1 = n$
3. $C_n^m = C_n^{n-m} \quad \left(m > \frac{n}{2}\right)$
4. $C_n^0 + C_n^1 + \dots + C_n^n = 2^n$
5. $C_n^m + C_n^{m+1} = C_{n+1}^{m+1}$ – reccurrent formula.

Misol: Tijorat banki boshqarmasi bir xil lavozimlarga 10 ta nomzoddan 3 tasini tanlamoqda. Har bir nomzod bir xil imkoniyatga ega. 10 ta nomzoddan 3 kishidan iborat nechta guruh tuzish mumkin?

Yechish: $C_{10}^{3} = \frac{10!}{3! \cdot (10-3)!} = 120$

Takrorlanishli o'rin almashtirishlar

Aytaylik, n ta A, B, ..., C elementlar mavjud bo'lib, ularning ichida A element α marta, B element β marta va h.k. hamda C element γ marta takrorlansin va $n = \alpha + \beta + \ldots + \gamma$ bo'lsin. U holda takrorlanishli o'rin almashtirishlar quyidagi formula bo'n topiladi:

$$\boxed{P_{takr} = \frac{n!}{\alpha! \cdot \beta! \cdot \ldots \cdot \gamma!}}$$

Misol: Aytaylik, 4 ta element mavjud bo'lib, ularning ikkitasi bir xil bo'lsin: A, A, B, C. Ulardan mumkin bo'lgan barcha o'rin almashtirishlar quyidagicha:

AABC	ABAC	ACBA
AACB	ABCA	ACAB
BAAC	BCAA	CABA
BACA	CBAA	CAAB

$$P_{takr} = \frac{n!}{\alpha! \cdot \beta! \cdot \ldots \cdot \gamma!} = \frac{4!}{2! \cdot 1! \cdot 1!} = 12$$

Takrorlanishli o'rinlashtirishlar

n ta elementdan m tadan takrorlanishli o'rinlashtirishlarda ($m \leq n$) ixtiyoriy element dan m martagacha uchrashi yoki umuman uchramasligi mumkin, ya'ni har bir n ta elementdan m tadan takrorlanishli o'rinlashtirish nafaqat turli elementlardan, balki t ta ixtiyoriy ravishda takrorlanuvchi ixtiyoriy elementlardan tashkil topgan, hech bolmaganda elementlarining joylashish tartibi bn farq qiluvchi guruhlar har xil guruh hisoblanadi.

Masalan 3 A, B C elementdan 2 tadan takrorlanishli o'rinlashtirishlar quyidagicha: AA, BB, CC, AC, Bc, cA, CB, BA, AB

$$\left[A_{n\ takr}^{m} = n^{m}\right]$$

Misol: Seyfning shifrli kodi 6 xonali sondan iborat. Kodlashtirilganda nechta turli kombinatsiya tuzish mumkin?

Yechish: Bu misolda $n=10$, $m=6$

$$A_{n\ takr}^{m} = 10^{6} = 1000000.$$

Takrorlanishli guruhlashlar.

p ta elementdan t tadan element bolgan takrorlanishli guruhlashda element 1 dan m martagacha uchrashi yoki umuman uchramashligi mumkin, ya'ni har bir n ta elementdan m tadan takrorlanishli o'rinlashtirish nafaqat turli elementlardan, balki t ta ixtiyoriy ravishda takrorlanuvchi ixtiyoriy elementlardan tashkil topishi m-n. Tartibi bir xil bolib, faqat elementlari-

ning tartibi bilan farq qiluvchi guruhlar farq qilinmaydi, ya'ni faqat elementlarning joylashish tartibi bilangina farq qiluvchi guruhlar bir xil guruh hisoblanadi.

Masalan, uch A, B, C elementdan 2 tadan takrorlanishli mosliklar quyidagicha:

AA, BB, CC, AC, BC, AB.

$$C_{n\,takr}^{m} = C_{n+m-1}^{m} = \frac{(n+m-1)!}{m!\cdot(n-1)!}$$

Misol: Qandolat dokonidagi 4 xil shirinlikdan 6 donasini necha xil usul bilan tanlash mumkin?

Yechish: $n=4 \quad m=6$

$$C_{4\,takr}^{6} = C_{4+6-1}^{6} = C_{9}^{6} = \frac{9!}{6!\cdot 3!} = 84.$$

Misollar:

① 3, 4, 5, 6, 8, 9 raqamlari yordamida hammasi bolib a) raqamlar takrorlanmasa, b) raqamlar takrorlanishi mumkin bolsa, nechta 3 xonali son tuzish mumkin?

a) Orinlashtirish: $n=6, \quad m=3.$

$$A_n^m = \frac{n!}{(n-m)!} = \frac{6!}{3!} = 4\cdot 5\cdot 6 = 120$$

b) Takrorlanishli orinlashtirish: $n=6, \quad m=3$

$$A_{n\,takr}^{m} = n^m = 6^3 = 216$$

② 5 nafar o'quvchidan 2 nafarini, Bilimlar bellashuvida qatnashish uchun tanlab olish kerak. Buni necha xil usulda bajarish kerak?

Yechish: Guruhlash: $n=5$ $m=2$.

$$C_n^m = \frac{n!}{m!(n-m)!} = \frac{5!}{2!\cdot 3!} = 10$$

③ 4 ta a, b, c, d elementdan 2 tadan olib tuzilgan har xil guruhlar soni nechta?

Guruhlash: $n=4$ $m=2$.

$$C_n^m = C_4^2 = \frac{4!}{2!\cdot 2!} = 6$$

④ 4 ta talaba 2 ta stulga necha xil usul bilan o'tirishi mumkin? (2tasi o'tirishi, 2tasi tik turishi k-k).

O'rinlashtirish: $n=4$ $m=2$.

$$A_n^m = \frac{n!}{(n-m)!} \Rightarrow A_4^2 = \frac{4!}{2!} = 12$$

⑤ 7 yigit va 4 qizdan iborat o'quvchilar guruhidan 6 ta o'quvchini shunday tanlab olish kerakki, ularning ichida qizlar soni ikkitadan kam bo'lmasin. Buni necha xil usul bilan amalga oshirish mumkin.

Yechish: 2 ta qiz, 4 ta yigit → $C_4^2 \cdot C_7^4$
3 ta qiz, 3 ta yigit → $C_4^3 \cdot C_7^3$
4 ta qiz, 2 ta yigit → $C_4^4 \cdot C_7^2$

$C_4^2 \cdot C_7^4 + C_4^3 \cdot C_7^3 + C_4^4 \cdot C_7^2 = 371$ ta

⑥ 2, 4, 7, 9 raqamlaridan ularni takrorlamasdan nechta 4 xonali son tuzish m-n? Ularning nechtasi 2 ga, 4 ga, 11 ga bolinadi?

a) $P_4 = 4! = 24$.

b) Oxirgi raqami 2 va 4 bolgan sonlar barcha sonlarning yarmini tashkil qiladi, demak 2 ga bolinadigan sonlar 12 ta.

c) Oxirgi 2 ta raqami 24 bolgan sonlar 4 ga bolinadi, ularning soni 2 ta.

d) 2□9□ → 2 ta.
9□2□ → 2 ta. 8 ta
□9□2 → 2 ta
□2□9 → 2 ta

1	3	2	4
4	2	1	1

$4 \cdot 2 = 8$ ta

⑦ Zalda 2 ta bosh joy bor, 3 nafar kishidan 2 tasini shu joyga necha xil usulda otqazish m-n?

Guruhlash: $n = 3$ $m = 2$

$$A_n^m = A_3^2 = \frac{3!}{(3-2)!} = 6.$$

⑧ 1, 2, 3, ..., 8 raqamlaridan ularni takrorlamay tuzilgan 8 xonali sonlar ichida 1 va 8 raqamlari yonma-yon turadiganlari nechta?

J: $2 \cdot 7 \cdot 6! = 2 \cdot 7!$

⑨ 5ga bo'linadigan 6 xonali sonlar nechta?

9	10	10	10	10	2

→ yoki 5

$9 \cdot 10^4$ ta

⑩ Futbol jamoasidagi 11 kishi orasidan jamoa sardori va uning yordamchisini necha xil usulda tanlab olish mumkin?

~~Guruhlash~~: $n=11$ $m=2$. O'rinlashtirish:

$$A_{11}^2 = \frac{11!}{(11-2)!} = 10 \cdot 11 = 110.$$

⑪ Matematika to'garagida faol qatnashuvchi 10 ta o'quvchidan 4 tasini xalqaro matematika olimpiadasiga yuborish uchun ularni necha xil usulda tanlasa bo'ladi?

Guruhlash: $n=10$ $m=4$.

$$C_{10}^4 = \frac{10!}{4! \cdot 6!} = \frac{7 \cdot 8 \cdot 9 \cdot 10}{24} = 210.$$

53-savol. Ehtimollikning klassik ta'rifi. Shartli ehtimollik

54-savol. Tasodifiy miqdorlar va ularning sonli xarakteristikalari.

Hodisalar.

1. Tasodifiy hodisalar
2. Muqarrar hodisa U
3. Mumkin bolmagan hodisa V

Tasodifiy hodisalarga misollar:

1. A, B komandalar

A_1 = {A komandaning yutishi}
A_2 = {B komandaning yutishi}
A_3 = {Durang natija}

2. Tanga

A = {Gerb tushishi}
B = {Raqam tushishi}

Ehtimollikning klassik ta'rifi.

A hodisaning P(A) ehtimoli (roy berish ehtimoli) A hodisa roy berishiga qulaylik tugdiradigan (yaratadigan) elementar hodisalar m ning barcha elementar hodisalar soni n ga nisbatiga teng.

$$\left[P(A)=\frac{m}{n}\right]$$

Barcha elementar hodisalar teng imkoniyatli bolgan hol klassik hol deb ataladi. shuning uchun $P(A)=m/n$ ehtimollikni kopincha klassik ehtimollik deb ataladi

Misol. ① Raqamlangan kubik 2 marta tashlanganda chiqqan raqamlar yigindisi 5 dan kichik bolishi ehtimolini toping.

$\square + \square < 5$

$2 \to (1;1)$
$3 \to (1;2), (2;1)$ $\quad m=6$
$4 \to (1;3), (2;2), (3;1)$

$\square \quad \square$
$6 \cdot 6 = 36 \to n=36$

$P(A) = \frac{m}{n} = \frac{6}{36} = \boxed{\frac{1}{6}}$

② 2 ta tanga tashlanyapti. Kamida 1 ta tangada raqam chiqish ehtimolligini toping.

Yechish: RR, RG, GR, GG $\to n=4$

$A=\{RR, RG, GR\} \to P(A) = \frac{m}{n} = \frac{3}{4}$
$m=3$

③ Qutida 5 ta oq va 4 ta yashil sharlar bor. Tavakkaliga 2 ta shar olinganda.
a) Ikkala sharning ham oq bolish $=A$
b) Turli rangda bolish $=B$
ehtimolini toping.

Yechish: a) $n = C_9^2$ $\quad m = C_5^2$

$$P(A) = \frac{m}{n} = \frac{C_5^2}{C_9^2} = \frac{\frac{5!}{2!\cdot 3!}}{\frac{9!}{2!\cdot 7!}} = \frac{5!\cdot 7!}{3!\cdot 9!} = \frac{4\cdot 5}{8\cdot 9} = \frac{5}{18}$$

b) $m = \underset{C_5^1}{5} \cdot \underset{C_4^1}{4} = 20$ $\quad n = C_9^2$

$$P(B) = \frac{20}{C_9^2} = \frac{20}{\frac{9!}{2!\cdot 7!}} = \frac{20\cdot 2}{8\cdot 9} = \frac{5}{9}$$

④ O'quvchi tavakkaliga doskaga 2 ta 2 xonali son yozdi. Bu sonlar turli bolish ehtimolligini toping.

Yechish: 2 xonali sonlar 90 ta

$n = 90 \cdot 90 = 90 \cdot 90$

2 ta 2 xonali son bir xil chiqish holatlari:
(10;10), (11;11), (12;12), ..., (99,99) → 90 ta.

turli xil bolish holatlari:

$m = 90\cdot 90 - 90$

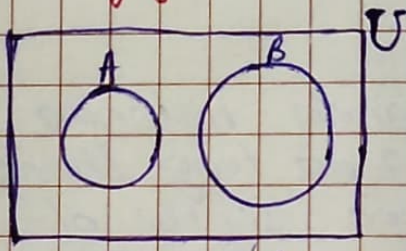
$$P(A) = \frac{m}{n} = \frac{90\cdot 90 - 90}{90\cdot 90} = \frac{90-1}{90} = \frac{89}{90}$$

⑤ 15 ta lampochkadan 4 tasi yaroqsiz. Tasakkaliga 2 ta lampochka tanlanganda ikkala lampochkaning yaroqsiz bolish ehtimolligini toping.

Yechish: $n = C_{15}^2$ $m = C_4^2$

$$P(A) = \frac{m}{n} = \frac{C_4^2}{C_{15}^2} = \frac{\frac{4!}{2!\cdot 2!}}{\frac{15!}{2!\cdot 13!}} = \frac{3\cdot 4}{14\cdot 15} = \frac{2}{35}$$

Birgalikda bolmagan hodisalar yigindisi ehtimolligi.



$$P(A+B) = P(A) + P(B)$$

$x \in A,\ x \notin B$
$x \in B,\ x \notin A$

Misollar:

① Qutida 3 ta oq, 2 ta qora, 5 ta yashil sharlar bor. Tasodifan 1 ta shar olinganda uning qora yoki yashil bolish ehtimolini toping.

Yechish: 1 ta shar olganda bu shar bir vaqtda ham qora, ham yashil bolmaydi. Shuning uchun bu hodisalar birgalikda emas.

$$P(Q) = \frac{2}{10} = 0{,}2 \leftarrow \quad n = 10 \quad m = 2$$

$$P(Y) = \frac{5}{10} = 0{,}5 \leftarrow \quad n = 10 \quad m = 5$$

$$P = P(Q) + P(Y) = 0{,}2 + 0{,}5 = 0{,}7.$$

② Oquvchiga ixtiyoriy bir raqamni yozish taklif qilindi. Uning 3 dan kichik yoki 7 dan katta raqam yozish ehtimolini toping.

Yechish: $\underbrace{n<3}_{A}$ yoki $\underbrace{n>7}_{B}$

$P(A)=\frac{3}{10}=0,3$ $\quad P(B)=\frac{2}{10}=0,2$

$P(A+B)=P(A)+P(B)=0,3+0,2=0,5$

Qarama-qarshi hodisalar.

Misollar:

① Kamonchi 1 marta otganda nishonga tegiza olish ehtimolligi 0,7 ga teng. Uning bir marta otganda nishonga tegiza olmaslik ehtimolligini toping.

Yechish: $P(A)=0,7$

$P(\bar{A})=1-P(A)=1-0,7=0,3$

② 2 ta raqamlangan kubik tashlanganda chiqqan raqamlar yig'indisi 3 dan katta bo'lish ehtimolligini toping.

A. $m+n>3$ $\quad \boxed{6}\cdot\boxed{6}=36$

$\bar{A}$ $m+n\leq 3$

(1;1), (1;2), (2;1)

$P(\bar{A})=\frac{3}{36}=\frac{1}{12}$ $\quad P(A)=1-P(\bar{A})=1-\frac{1}{12}=\frac{11}{12}$

③ 1, 2, 3, ..., 100 sonlaridan tasodifan bittasi tanlanganida raqamlari turli 2 xonali son chiqish ehtimolligini toping.

Y: 11, 22, 33, ..., 99 → 9 ta

1, 2, 3, ..., 9, 100 → 10 ta $\quad P(\bar{A})=19/100=0,19$

$P(A)=1-P(\bar{A})=1-0,19=0,81$

④ Guruhda 10 ta fizik va 7 ta matematik bor. Guruhdan tasodifan 5 kishi tanlanganda ularning ichida kamida bitta matematik bor bolish ehtimolligini toping.

Yechish: A: m,f,f,f,f
m,m,f,f,f
.....

$\bar{A}$: f,f,f,f,f

$m = C_{10}^5 \quad n = C_{17}^5$

$$P(\bar{A}) = \frac{k}{n} = \frac{C_{10}^5}{C_{17}^5} = \frac{\frac{10!}{5!\cdot 5!}}{\frac{17!}{12!\cdot 5!}} = \frac{10!\cdot 12!}{5!\cdot 17!} = \frac{6\cdot 7\cdot 8\cdot 9\cdot 10}{13\cdot 14\cdot 15\cdot 16\cdot 17} = \frac{9}{221}$$

$$P(A) = 1 - P(\bar{A}) = 1 - \frac{9}{221} = \frac{212}{221}$$

Bog'liq bo'lmagan hodisalar

$$P(AB) = P(A)\cdot P(B)$$

Misollar:

① Tanga 2 marta tashlanganda birinchisida gerb, ikkinchisida raqam chiqish ehtimolligini toping.

Yechish: A – 1-G
B – 2-R

$$P(AB) = P(A)\cdot P(B) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$$

② 2 ta raqamlangan kub tashlanganda 1-kubda 3 dan kichik, 2-kubda 4 dan kichik raqam chiqish ehtimolligini toping.

Yechish: A. $m<3$ (1, 2)
B. $m<4$ (1, 2, 3)

$P(A)=\frac{2}{6}=\frac{1}{3}$

$P(B)=\frac{3}{6}=\frac{1}{2}$ $P(AB)=P(A)\cdot P(B)=\frac{1}{3}\cdot\frac{1}{2}=\frac{1}{6}$

③ 1-merganning nishonga tekkizish ehtimoli 0,9 ga, 2-siniki esa 0,8 ga teng. Har bir mergan bir martadan o'q otganda nishonga
a) ikkala o'q tegishi
b) ikkala o'q tegmasligi
c) kamida bitta o'q tegishi ehtimolligini toping.

Yechish:
a) $P(A)=0,9$ $P(B)=0,8$ $P(AB)=0,9\cdot 0,8=0,72$

b) $P(\bar{A})=1-P(A)=1-0,9=0,1$
$P(\bar{B})=1-P(B)=1-0,8=0,2$
$P(\bar{A}\bar{B})=P(\bar{A})\cdot P(\bar{B})=0,1\cdot 0,2=0,02$

c) $P(A\bar{B}+\bar{A}B+AB)=1-P(\bar{A}\bar{B})=1-0,02=0,98$

④ Merganning nishonga tekkizish ehtimolligi 0,8 ga teng. U nishonga 3 marta o'q uzganda barcha o'qlari nishonga tegishining ehtimolligini toping.

Yechish: $P(AAA)=P(A)\cdot P(A)\cdot P(A)=0,8\,0,8\cdot 0,8=0,512$

⑤ Har birining nishonga tekkizish ehtimolligi mos ravishda 0,8; 0,7 va 0,6 ga teng. Nishonga faqat 1- va 2-merган-

larning ōglari tegishi ehtimolligini toping.

Yechish: $P(A_1A_2\bar{A}_3) = P(A_1)\cdot P(A_2)\cdot P(\bar{A}_3) = 0,8\cdot 0,7\cdot(1-0,6) =$
$= 0,56\cdot 0,4 = 0,224.$

Hodisalar yig'indisi ehtimolligi

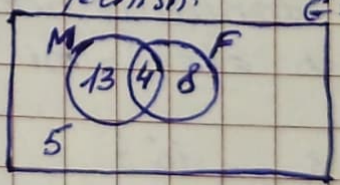
$$P(A+B) = P(A) + P(B) - P(AB) \quad (1)$$

Biz avval $P(A+B) = P(A) + P(B)$ (2) formulani ko'rgandik. (2) formula A va B hodisalar birgalikda bo'lmagan holat uchun, ya'ni $P(AB)=0$. (2) formula (1) formulaning xususiy holi.

Misollar.

① Guruhda 30 ta o'quvchi bor. Ularning 17 tasi matematikaga, 12 tasi fizikaga, 4 tasi ikkala fanga ham qiziqadi. Tasodifan tanlangan o'quvchi tanlanganda matematikaga yoki fizikaga qiziqish ehtimolini toping.

Yechish:



$$P(M+F) = P(M) + P(F) - P(MF)$$

$$P(M) = \frac{17}{30} \quad P(F) = \frac{12}{30} \quad P(MF) = \frac{4}{30}$$

$$P(M+F) = \frac{17}{30} + \frac{12}{30} - \frac{4}{30} = \frac{25}{30} = \frac{5}{6}$$

② Merganning nishonga tekkizish ehtimoli 0,8 ga teng. U nishonga 2 marta o'q uzganda kamida 1 ta o'q nishonga tegish ehtimolligini toping.

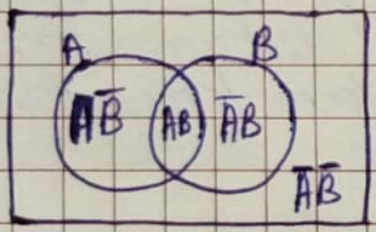
Yechish: a) A – 1-marta otganda nishonga tegish ehtimolligi.
B – 2-marta ...

$$P(A+B) = P(A) + P(B) - P(AB)$$

A va B hodisalar bog'liq bo'lmagan hodisolar bo'lganligi uchun $P(AB) = P(A) \cdot P(B)$

$$P(A+B) = 0{,}8 + 0{,}8 - 0{,}64 = \boxed{0{,}96}$$

b)



$$P(A\bar{B} + AB + \bar{A}B) = 1 - P(\bar{A}\bar{B}) =$$

$$= 1 - P(\bar{A}) \cdot P(\bar{B}) = 1 - 0{,}2 \cdot 0{,}2 = \boxed{0{,}96}$$

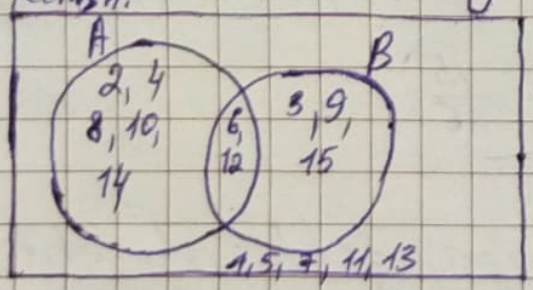
Shartli ehtimollik

$$P(A/B) = \frac{P(AB)}{P(B)}$$

Misollar:

① 1 dan 15 gacha bo'lgan natural sonlar ichidan, 3 ga karrali son tanlanganda, bu sonning 2 ga karrali bo'lish ehtimolini toping.

Yechish:



$U = \{1, 2, 3, \ldots, 15\}$
A – 2 ga karrali
B – 3 ga karrali

$P(A/B) - ?$

$$P(AB) = \frac{2}{15} \qquad P(B) = \frac{5}{15} \qquad P(A/B) = \frac{2}{15} : \frac{5}{15} = \boxed{\frac{2}{5}}$$

② 1 dan 6 gacha raqamlangan kub tashlanganda toq raqam tushgan bolsa, 5 chiqish ehtimolligini toping.

Yechish: $A=\{5\}$
$B=\{toq\}$

$$P(A/B)=\frac{P(AB)}{P(B)}$$

$P(AB)$ – kubning ham toq, ham 5 tushish ehtimolligi bolgani uchun $P(AB)=P(A)=\frac{1}{6}$ boladi.

$P(B)=\frac{1}{2}$

$$P(A/B)=\frac{\frac{1}{6}}{\frac{1}{2}}=\frac{1}{3}$$

③ Qopda 2 ta qora va 3 ta oq sharlar bor. Tasodifan ketma-ket qaytarib qoyilmay 2 ta shar olinganda 1-shar oq va 2-shar qora bolish ehtimolligini toping

Yechish: $P(A/B)=\frac{P(AB)}{P(B)} \rightarrow P(AB)=P(A/B)\cdot P(B)$

$B \rightarrow$ 1-shar oq bolish ~~chi~~
$A \rightarrow$ 2-shar qora

$P(B)=\frac{3}{5}$ $\quad P(A/B)=\frac{2}{5-1}=\frac{2}{4}=\frac{1}{2}$

$$P(AB)=\frac{1}{2}\cdot\frac{3}{5}=\frac{3}{10}=0,3$$

Bernulli formulasi.

Biror hodisaning roy berish ehtimolligi $P(A)=p$ uning roy bermaslik ehtimolligi $P(\bar{A})=1-p=q$ bo'lsin, u holda ushbu hodisani n marta bajarganimizda bu hodisaning m marta roy berish ehtimolligi quyidagi Bernulli formulasi orqali topiladi:

$$[P(n,m)=C_n^m \cdot p^m \cdot q^{n-m}]$$

Misollar:

① Merganning bir marta o'q uzganda nishonga tegiza olish ehtimolligi 0,8 ga teng. Mergan 4 marta o'q uzganida uning nishonga 3 marta tekkizish ehtimolligini toping.

$P(A)=0,4=p$ $\quad n=4$

$P(\bar{A})=1-0,4=0,6=q$ $\quad m=3$

$$P(n,m)=C_n^m \cdot p^m \cdot q^{n-m}=C_4^3 \cdot 0,4^3 \cdot 0,6^1=$$

$$=\frac{4!}{1!\cdot 3!}\cdot\frac{8}{125}\cdot\frac{3}{5}=\boxed{\frac{96}{625}}$$

② Tanga 7 marta tashlanganda 5 marta gerb o'q 2 marta raqam tomoni tushishining ehtimolligini toping.

Yechish: $n=7 \quad m=5$

$$P(G)=\frac{1}{2}=p \rightarrow q=\frac{1}{2}$$

$$P(7,5)=C_7^5 \cdot p^5 \cdot q=\frac{7!}{2!\cdot 5!}\cdot\left(\frac{1}{2}\right)^5\cdot\frac{1}{2}=\frac{3\cdot 7}{2^7}=\boxed{\frac{21}{128}}$$

③ Kub 5 marta tashlanganda 3 marta 4 dan katta raqam tushish ehtimolligini toping.

Yechish: $P(A) = \frac{2}{6} = \frac{1}{3} = p \quad (5,6 > 4)$

$P(\bar{A}) = \frac{4}{6} = \frac{2}{3} = q.$

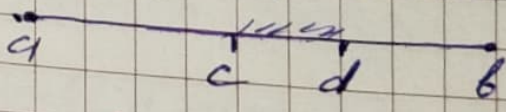
$n = 5 \qquad m = 3$

$P(5,3) = C_5^3 \cdot p^3 \cdot q^2 = \frac{5!}{3! \cdot 2!} \cdot \frac{1}{3^3} \cdot \frac{2^2}{3^2} =$

$= \frac{2 \cdot 5 \cdot 4}{3^5} = \boxed{\frac{40}{243}}$

Ehtimollikning geometrik ta'rifi.

To'g'ri chiziqda $[a, b]$ oraliqdan olingan nuqtaning $[c; d]$ oraliqqa tegishli bo'lish ehtimolligi: $(a \le c \le d \le b)$



$$\boxed{P(A) = \frac{d-c}{b-a}}$$

Misollar.

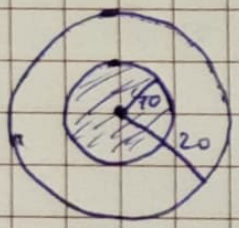
④ $[0; 10]$ oraliqdan tasodifan tanlangan nuqtaning $[5; 7]$ oraliqqa tegishli bo'lish ehtimolligini toping.

Yechish: $P(A) = \frac{7-5}{10-0} = \frac{2}{10} = \boxed{0,2}$

⑤ $ABCD$ kvadratdan tasodifan tanlangan nuqtaning ABC uchburchakka tegishli bo'lish ehtimolligini toping.

Yechish: $P(E) = \frac{S_{\Delta ABC}}{S_{ABCD}} = \frac{\frac{1}{2}a^2}{a^2} = \boxed{\frac{1}{2}}$

③ Otilgan nayza markazi O nuqtada va radiusi 20 sm bolgan doira shaklidagi nishonning nuqtalariga teng imkoniyatda tushadi. Nayzaning O markazli va 10 sm radiusli doiraga tushish ehtimolligini toping.



$$P(A) = \frac{\pi \cdot 10^2}{\pi \cdot 20^2} = \boxed{\frac{1}{4}}$$

Sonlar nazariyasi.

61-savol. EKUB. Yevklid algoritmi

gcd - greatest common divisor

a, b → butun nomanfiy sonlar.

Algoritm:

$$gcd(a,b) = \begin{cases} a, & \text{if } b=0 \\ gcd(b, a \bmod b), & \text{otherwise (if } b\neq 0) \end{cases}$$

C++:

```
int gcd(int a, int b){
    return b? gcd(b, a%b):a;
}
```

Ishlash tezligi.

Yevklid algoritmining ishlash tezligi Lame teoremasiga asoscan baholanadi. Bu teoremada Yevklid algoritmi va Fibonachchi ketma-ketligidagi ajoyib bog'liqlik aks etgan:

Agar $a > b \geq 1$ va $b < F_n$ $(n \in N)$ bo'lsa, u holda Yevklid algoritmi $n-2$ dan oshmagan sanoqda amal bajaradi.

Agar $a = F_n$, $b = F_{n-1}$ bo'lsa $n-2$ ta amal bajaradi. Bu Yevklid algoritmining eng yomon holatda ishlashi.

Fibonachchi ketma-ketligining eksponensial o'sishini hisobga olsak, Yevklid algoritmining ishlash tezligi $O(\log \min(a,b))$.

Algoritm

EKUB (a; b) = EKUB (b%a; a) ekanligi ma'lum

shuning uchun:

$$\begin{cases} a \cdot x + b \cdot y = gcd(a;b) & (1) \\ (b\%a)x_1 + ay_1 = gcd(a;b) & (2) \end{cases}$$

$g = gcd(a;b)$ deb olamiz

$(2) \rightarrow b\%a = b - \left[\frac{b}{a}\right] \cdot a$

$$\left(b - \left[\frac{b}{a}\right] \cdot a\right) \cdot x_1 + ay_1 = g$$

$$a \cdot \left(y_1 - \left[\frac{b}{a}\right] \cdot x_1\right) + bx_1 = g$$

demak: $\begin{cases} x = y_1 - \left[\frac{b}{a}\right] \cdot x_1 \\ y = x_1 \end{cases}$

Dastur (C++)

```
int gcd (int a, int b, int &x, int &y){
    if(a==0){
        x=0; y=1;
        return b;
    }
    int x1, y1;
    int d=gcd(b%a, a, x1, y1);
    x=y1-(b/a)*x1;
    y=x1;
    return d;
}
```

Misollar:

① 453 va 17 sonlari EKUB; va oldidagi koeffitsiyentlarini toping. $(453x+17y=1)$

1-qadam

$453 = 17 \cdot 26 + 11$
$17 = 11 \cdot 1 + 6$
$11 = 6 \cdot 1 + 5$
$6 = 5 \cdot 1 + 1$

2-qadam

$11 = 453 - 17 \cdot 26$
$6 = 17 - 11$
$5 = 11 - 6$
$1 = 6 - 5$

3-qadam.

$1 = 6 - 5 = 6 - (11-6) = 2 \cdot 6 - 11 = 2 \cdot (17-11) - 11 =$
$= 2 \cdot 17 - 3 \cdot 11 = 2 \cdot 17 - 3 \cdot (453 - 17 \cdot 26) = 2 \cdot 17 - 3 \cdot 453 +$
$+ 78 \cdot 17 = 80 \cdot 17 - 3 \cdot 453$

$x = -3; \quad y = 80$

② 25 va 15. $(25x+15y=5)$

1-qadam

$25 = 15 \cdot 1 + 10$
$15 = 10 \cdot 1 + 5$ ekub
$10 = 5 \cdot 2 + 0$

2-qadam

$10 = 25 - 15$
$5 = 15 - 10$

3-qadam.

$5 = 15 - 10 = 15 - (25-15) = 2 \cdot 15 - 25$

$x = -1 \quad y = 2$

③ 24 va 42.

$42 = 24 \cdot 1 + 18$
$24 = 18 \cdot 1 + 6$
$18 = 6 \cdot 3 + 0$

$18 = 42 - 24$
$6 = 24 - 18$

$6 = 24 - 18 = 24 - (42-24) = 2 \cdot 24 - 42$

$x = -2, \quad y = +1$

63-savol. Diofant tenglamasi.

Diofant tenglamasining umumiy ko'rinishi:

$$f(x_1, x_2, \dots, x_n) = 0$$

Bu yerda f ifoda $x_1, x_2, \dots, x_n$ o'zgaruvchilari butun sonlar bo'lganida butun qiymatlar qabul qiladi.

Diofant tenglamalari 2 xil:
- chiziqli
- yuqori darajali.

Chiziqli diofant tenglamalari:

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

Misollar:

① $7x + 13y = 113$ $\quad x, y - ?$ $\quad (x, y \in N)$

$7x + 14y - y = 113$

$7(x+2y) = y + 113$

$y + 113 = y + 1 + 112 = 7k$

$y + 1 = 7(k - 16) = 7n$

$y = 7n - 1$

$7x + 13(7n-1) = 113$

$7x + 91n - 13 = 113$

$7x = 126 - 91n$

$x = 18 - 13n \quad n \in Z$

$n = 1 \rightarrow x = 5, \ y = 6$

$n \geq 2 \rightarrow x < 0$ bo'ladi. demak yagona natural yechim: $x = 5, \ y = 6$

② $127x - 52y + 1 = 0$ tenglamaning natural sonlardagi yechimini toping.

$4 \cdot (32x - 13y) = x - 1$ $\quad x - 1 = 4k$

$x \in N \to k \in N$.

$\boxed{x = 4k + 1}$

$32 \cdot (4k + 1) - 13y = k$

$128k + 32 - 13y = k$.

$13y = 127k + 32$

$\boxed{y = \frac{127k + 32}{13}} = \frac{130k + 26 - 3k + 6}{13} = 10k + 2 + \frac{3(2 - k)}{13}$

$k = 1 \to y \notin N$

$k = 2 \to x = 9;\ y = 22$

$k = 15 \to x = 61;\ y = 149$

.....

2-darajali diofant tenglamalari.

2-darajali 2 noma'lumli tenglamalarning umumiy ko'rinishi

$$\boxed{ax^2 + bxy + cy^2 + dx + ey + f = 0}$$

Misollar:

④ $3x^2 - 4y^2 = 13$ tenglamaning natural sonlardagi yechimini toping.

$4x^2 - 4y^2 - 12 = 1 + x^2$

$4(x^2 - y^2 - 3) = 1 + x^2$ (1)

Tenglikning chap tomoni 4 ga karrali. O'ng tomoni haqida so'z yuritamiz.

Sonning kvadratini 4 ga bo'lganda qoldiq 0 yoki 1 chiqadi, demak $1+x^2$ ni 4 ga bo'lganda 1 yoki 2 qoldiq chiqadi. Demak tenglama natural yechimga ega emas.

② $x^3-y^3=91$ $\quad x,y\in N$.

$(x-y)(x^2+xy+y^2)=91$

$91=(\pm1)\cdot(\pm91)=(\pm7)\cdot(\pm13)=(\pm13)\cdot(\pm7)$.

$$\begin{cases} x-y=1 \\ x^2+xy+y^2=91 \end{cases} \rightarrow \begin{cases} x=6 \\ y=5 \end{cases}$$

③ $x+y=2xy$ $\quad x,y\in Z$

$$x=\frac{y}{2y-1} \rightarrow 2x=\frac{2y}{2y-1}=\frac{2y-1+1}{2y-1}$$

$$2x=1+\frac{1}{2y-1}$$

~~$y=-1 \rightarrow y=1 \rightarrow x=1$~~

$y=0 \rightarrow x=0$

$y=1 \rightarrow x=2$

$(0;0),(2;1)$

④ $(x-y)^2+(y-z)^2+(z-x)^2=30$.

tenglama butun sonlarda yechimga ega emasligini isbotlang.

~~Yechish $(x-y)^2+(y-z)^2=(x-y+y-z)^2-2(x-y)(y-z)=$~~

~~$=(x-z)^2-2(x-y)(y-z)$~~

$x+y+y-z+z-x=0$.
demak 3 ta butun sonning yigindisi 0 va
kvadratlarining yigindisi 30 ga teng
$(\pm1)^2+(\pm2)^2+(\pm5)^2=30$ lekin
$\pm1+(\pm2)+(\pm5)\neq0$ demak
butun sonlarda yechimga ega emas.

64-sabak. Taqqoslama.

Ta'rif: Agar 2 ta a va b sonni m – natural songa bolganda hosil bolgan qoldiqlar ozaro teng bolsa, u holda a va b sonlarini m modul boyicha teng qoldiqli sonlar yoki m modul boyicha taqqoslanuvchi sonlar deyiladi.

$$[a \equiv b \pmod m]$$

Masalan: $22 \equiv 27 \pmod 5$

Taqqoslama xossalari:

1) $a \equiv b \pmod n \rightarrow a \equiv b \pm n \cdot k \pmod n$
$a \pm n \cdot k \equiv b \pmod n$

2) $a \equiv b \pmod n$, $c \equiv d \pmod n \rightarrow$
a) $a + c \equiv b + d \pmod n$
b) $a - c \equiv b - d \pmod n$
c) $a \cdot c = b \cdot d \pmod n$
d) $a^m = b^m \pmod n$

Misollar:

① $9^{1000} : 7 = ?$

$9 \equiv 2 \pmod 7$
$9^3 \equiv 2^3 \pmod 7$
$9^3 \equiv 8 \pmod 7$
$9^3 \equiv 1 \pmod 7$
$(9^3)^{333} \equiv 1^{333} \pmod 7$
$9^{999} \equiv 1 \pmod 7$
$9^{999} \cdot 9 \equiv 1 \cdot 2 \pmod 7$
$9^{1000} \equiv 2 \pmod 7$

② $35^{1935} : 6$

$35 \equiv 5 \pmod 6$
$35 \equiv -1 \pmod 6$
$35^{1935} \equiv (-1)^{1935} \pmod 6$
$35^{1935} \equiv -1 \pmod 6$ /+6
$35^{1935} \equiv 5 \pmod 6$

③ $3^{4n+2} : 10$

$3^{4n+2} = 9^{2n+1}$

$3 \equiv 3 \pmod{10}$
$9 \equiv 9 \pmod{10}$
$9 \equiv -1 \pmod{10}$
$9^{2n+1} \equiv -1 \pmod{10}$
$3^{4n+2} \equiv -1 \pmod{10}$ /+10
$3^{4n+2} \equiv 9 \pmod{10}$

④ $7^{122} : 11$

$7 \equiv 7 \pmod{11}$
$7 \equiv -4 \pmod{11}$
$7^{122} = 4^{122} \pmod{11} \equiv 2^{244} \pmod{11}$

$2^5 \equiv 32 \pmod{11}$
$2^5 \equiv (-1) \pmod{11}$
$2^{240} \equiv 1 \pmod{11}$
$2^4 \equiv 5 \pmod{11}$

$2^{244} \equiv 5 \pmod{11}$

⑤ $4528^{632} : 5$

$4528 \equiv 3 \pmod 5$
$4528^{632} \equiv 3^{632} \pmod 5$

$3^2 \equiv 9 \pmod 5$
$3^2 \equiv (-1) \pmod 5$
$3^{632} = (-1)^{316} \pmod 5$
$3^{632} = 1 \pmod 5$

⑥ $2542661^{45145} : 100$

$\ldots 61^{45} = \ldots 01$ ← $(6 \cdot 5 = 30)$

71-savol. Ferma kichik teoremasi. Eyler funksiyasi.

Eyler funksiyasi:

n musbat butun soni bilan o'zaro tub bo'lgan $[1;n]$ oraliqdagi sonlar sonini ma'no qo'yuvchi funksiya Eyler funksiyasi d-di. $[\varphi(n)]$

Eyler funksiyasining 3 ta xossasi:

1) p-tub son: $\boxed{\varphi(p) = p-1}$

2) p-tub son, a-natural son;

$$\boxed{\varphi(p^a) = p^a - p^{a-1}}$$

3) a va b o'zaro tub son:

$$\boxed{\varphi(ab) = \varphi(a)\cdot\varphi(b)}$$

$n = p_1^{q_1}\cdot p_2^{q_2}\cdot\ldots\cdot p_k^{q_k}$ bo'lsa, Eyler funksiyasi quyidagi formula b-n topiladi:

$$\boxed{\varphi(n) = n\cdot\left(1-\frac{1}{p_1}\right)\cdot\left(1-\frac{1}{p_2}\right)\cdot\ldots\cdot\left(1-\frac{1}{p_k}\right)}$$

Time complexity: $\boxed{O(\sqrt{n})}$

Realizatsiya:

```
int phi(int n){
    int result = n;
    for(int i=2; i*i<=n; ++i)
        if(n%i==0){
            while(n%i==0)
                n/=i;
            result-=result/i;
        }
    if(n>1)
        result-=result/n;
    return result;
}
```

Eyler funksiyasi n sonining faktorizatsiyasini topishda kerak bo'ladi. ($O(\sqrt{n})$)

Eyler teoremasi:
Agar a va m o'zaro tub bo'lsa:

$$[a^{\varphi(m)} \equiv 1 \pmod m]$$

Fermaning kichik teoremasi:

Agar Eyler teoremasida m - tub son bo'lsa:

$$[a^{m-1} \equiv 1 \pmod m]$$

yoki

$$[a^m \equiv a \pmod m]$$

20-savol. Berilgan modulga ko'ra teskari elementni topish.

Ta'rif: a soniga m modul bo'yicha teskari son deb shunday b songa aytiladiki: $[a \cdot b \equiv 1 \pmod m]$ tenglik o'rinli bo'ladi. $(b = a^{-1})$

- Agar a va m sonlari o'zaro tub bo'lsagina b mavjud bo'ladi.

Teskari element topishni Binar darajadan foydalanilgan usuli.

Eyler va Fermaning kichik teoremasidan foydalaniladi:

$$a^{\varphi(m)} \overset{\text{Eyler}}{\equiv} 1 \pmod m \qquad a^{m-1} \overset{\text{Ferma kichik}}{\equiv} 1 \pmod m$$

$$\downarrow \qquad\qquad \downarrow$$

$$\left[a^{\varphi(m)-1} = a^{-1} \pmod m\right] \qquad \left[a^{m-2} \equiv a^{-1} \pmod m\right]$$

65-savol. Qoldiqlar haqida Xitoy teoremasi.

Quyidagi taqqoslamalar sistemasini qaraymiz:

$$\begin{cases} x \equiv b_1 \pmod{m_1} \\ x \equiv b_2 \pmod{m_2} \\ \dots\dots \\ x \equiv b_k \pmod{m_k} \end{cases}$$

bu yerda $m_1, m_2, m_3, \dots, m_k$ sonlar juft-juft bilan o'zaro tub sonlar ya'ni

$\forall i, j \in \overline{1,k}$ va $i \neq j$ lar uchun $EKUB(m_i; m_j) = 1$.

Bu sistema bir noma'lumli chiziqli taqqoslamalar sistemasi deyiladi.

Qoldiqlar haqida Xitoy teoremasi.
Bir noma'lumli chiziqli taqqoslamalar sistemasi berilgan bo'lib, M_s va M_s^* sonlari quyidagicha aniqlangan bo'lsin:

$$\left[m_1 \cdot m_2 \cdot \dots \cdot m_s = M_s \cdot m_s\right] \text{ va } \left[M_s \cdot M_s^* \equiv 1 \pmod{m_s}\right]$$

M_s^* sonini shunday tanlab olamiz:

x_0 sonini ~~shunday~~ quyidagicha aniqlaymiz

$$x_0 = M_1 \cdot M_1^* \cdot b_1 + M_2 \cdot M_2^* \cdot b_2 + \dots + M_k \cdot M_k^* \cdot b_k$$

U holda $x \equiv x_0 \pmod{m_1 \cdot m_2 \cdot \dots \cdot m_k}$ taqqoslamani qanoatlantiruvchi barcha x lar to'plami berilgan sistemaning yechimlar

koʻpaytmiga teng boʻladi.

Misol: ① Quyidagi sistemani yeching.

$$\begin{cases} x \equiv b_1 \pmod 4 \\ x \equiv b_2 \pmod 5 \\ x \equiv b_3 \pmod 7 \end{cases}$$

bu yerda: $M_1 = 35$, $M_2 = 28$, $M_3 = 20$

$M_1^* = 3 \rightarrow (M_1^* \cdot M_1 = 3 \cdot 35 \equiv 1 \pmod 4)$

$M_2^* = 2 \rightarrow (M_2 \cdot M_2^* = 28 \cdot 2 \equiv 1 \pmod 5)$

$M_3^* = 6 \rightarrow (M_3 \cdot M_3^* = 20 \cdot 6 \equiv 1 \pmod 7)$

Demak:

$$x_0 = 35 \cdot 3 \cdot b_1 + 28 \cdot 2 \cdot b_2 + 20 \cdot 6 \cdot b_3 = 105 b_1 + 56 b_2 + 120 b_3.$$

$$x \equiv (105 \cdot b_1 + 56 b_2 + 120 b_3) \pmod{(140)}$$

$b_1 = 1$, $b_2 = 3$, $b_3 = 2$ boʻlsa:

$$x \equiv (105 \cdot 1 + 56 \cdot 3 + 120 \cdot 2) \pmod{140}$$

$$x \equiv 93 \pmod{140}$$

$$x = 140k + 93$$

② Fermada ördak, tovuq va qoʻylar boʻlib, ördak va qoʻylarning oyoqlari soni 170 ta, ördaklarning oyoqlari va tovuqlarning panjalari soni esa 32 ga teng boʻlsa, ördaklar, tovuqlar va qoʻylarning sonini toping.

(Tovuqning panjalari soni 3 ta)

Yechish: $\begin{cases} 2x+4y=70 \\ 2x+3z=32 \end{cases} \Rightarrow \begin{cases} x+2y=35 \\ 2x+3z=32 \end{cases}$

$\begin{cases} x=-2y+35 \\ 2x=-3z+32 \end{cases} \Rightarrow \begin{matrix} x \equiv 35 \pmod 2 \\ 2x \equiv 32 \pmod 3 \end{matrix} \Rightarrow$

$\Rightarrow \begin{cases} x \equiv 1 \pmod 2 \\ 2x \equiv 2 \pmod 3 \end{cases} \Rightarrow \begin{cases} x \equiv 1 \pmod 2 \\ 4x \equiv 4 \pmod 3 \end{cases} \Rightarrow \begin{cases} x \equiv 1 \pmod 2 \\ 4x \equiv 1 \pmod 3 \end{cases} \Rightarrow$

$\Rightarrow \begin{cases} x \equiv 1 \pmod 2 \\ x \equiv 1 \pmod 3 \end{cases}$ $\quad b_1 = b_2 = 1$

$m_1 = 2 \quad m_2 = 3$

$M_1 = 3 \quad M_2 = 2$

$M_1^* = 1 \rightarrow (M_1 \cdot M_1^* = 3 \cdot 1 \equiv 1 \pmod 2)$ (yoki $3^{2-2} = 1$)

$M_2^* = 2 \rightarrow (M_2 \cdot M_2^* = 2 \cdot 2 \equiv 1 \pmod 3)$ (yoki $2^{3-2} = 2$).

$x_0 = 3 \cdot 1 \cdot 1 + 2 \cdot 2 \cdot 1 = 7.$

$x \equiv x_0 \pmod{m_1 \cdot m_2}$

$x \equiv 7 \pmod 6 \rightarrow \boxed{x = 6k+7}$

$2y = 35 - x = \boxed{28-6k} \rightarrow \boxed{y = 14-3k}$

$3z = 32 - 2x = 32 - 2(6k+7) = 18 - 12k$

$\boxed{z = 6-4k}$

$\begin{cases} x = 6k+7 \\ y = 14-3k \\ z = 6-4k \end{cases}$ faqat $k=0$ va $k=1$ da yechimga ega.

Misollar:

① (Eyler funksiyasi) 2031^{2042} sonning oxirgi 2 ta raqamini toping.

Yechish: $2031^{2042} \equiv x \pmod{100}$

$EKUB(2031; 100) = 1$

$\varphi(100) = 100 \cdot \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{5}\right) = 40$

$2031^{40} \equiv 1 \pmod{100}$.

$(2031^{40})^{51} \equiv 1 \pmod{100}$

$2031^{2040} \equiv 1 \pmod{100}$

$2031^{2042} \equiv 2031^{2040} \cdot 2031^2 \equiv 1 \cdot 2031^2 \equiv 31^2 \equiv 961 \equiv$

$\equiv 61 \pmod{100}$ → $\boxed{x = 61}$

68-69-saodlar. Lejandr va Yakobi simvoli, ularning xossalari.

Lejandr simvoli.

Lejandr simvoli Yakobi simvolining xususiy holi hi-di, Yakobi simvoli esa Kroneker-Yakobi simvolining xususiy holidir.

Ta'rif:

Agar a - butun son, p - 2 dan farqli tub son bolsa, $\left(\frac{a}{p}\right)$ Lejandr simvoli quyidagicha aniqlanadi.

- $\left(\frac{a}{p}\right)=0$, agar a soni p ga bolinsa.
- $\left(\frac{a}{p}\right)=1$, agar a soni p modul boyicha kvadratik chegirma bolsa, (ya'ni shunday x butun son borki, $x^2 \equiv a \pmod p$), lekin bu holda a p ga bolinmaydi.
- $\left(\frac{a}{p}\right)=-1$, agar a soni p modul boyicha kvadratik chegirma bolmasa.

Xususiyatlari.

→ Multiplikativlik

a) $\left(\frac{ab}{p}\right)=\left(\frac{a}{p}\right)\cdot\left(\frac{b}{p}\right)$

b) Agar a soni p ga bolinmasa $\left(\frac{a^2}{p}\right)=1$

c) Agar a sonning tub ko'paytuvchilarga kanonik yoyilmasi quyidagicha bo'lsa:

$$\left[a = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot \dots \cdot p_k^{\alpha_k}\right]$$

$$\left[\left(\frac{a}{p}\right) = \left(\frac{p_1}{p}\right)^{\alpha_1 \pmod 2} \cdot \left(\frac{p_2}{p}\right)^{\alpha_2 \pmod 2} \cdot \dots \cdot \left(\frac{p_k}{p}\right)^{\alpha_k \pmod 2}\right]$$

→ Agar $a \equiv b \pmod p$ bo'lsa:

$$\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$$

→ $\left(\frac{1}{p}\right) = 1$

→ Eyler kriteriyasi:

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod p$$

→ $p \neq 2$ bo'lsa:

a) $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$ (Eyler kriteriyasining xususiy holi)

b) $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$

Misol: $x^2 \equiv 426 \pmod{491}$ taqqoslama yechimga egami.

Yechish: $\left(\frac{426}{491}\right) \equiv \left(\frac{2}{491}\right) \cdot \left(\frac{3}{491}\right) \cdot \left(\frac{71}{491}\right)$

a) $\left(\frac{2}{491}\right) \equiv (-1)^{(491^2-1)/8} \equiv -1$ $\quad (491 \equiv 3 \pmod 8)$

2) $\left(\frac{3}{491}\right) = -\left(\frac{491}{3}\right) = -\left(\frac{2}{3}\right) = -(-1)^{(3^2-1)/8} = 1$

3) $\left(\frac{71}{491}\right) = -\left(\frac{491}{71}\right) = -\left(\frac{65}{71}\right) = -\left(\frac{5}{71}\right)\cdot\left(\frac{13}{71}\right) = -\left(\frac{71}{5}\right)\cdot\left(\frac{71}{13}\right) =$

$= -\left(\frac{1}{5}\right)\cdot\left(\frac{6}{13}\right) = -\left(\frac{2}{13}\right)\cdot\left(\frac{3}{13}\right) = -(-1)^{(13^2-1)/8}\cdot\left(-\left(\frac{13}{3}\right)\right) =$

$= -\left(\frac{1}{3}\right) = -1$

$\left(\frac{426}{491}\right) = (-1)\cdot 1\cdot(-1) = 1$.

Javob: yechimga ega.

Yakobi simvoli va uning xossalari.

Yakobi simvoli Lejandr simvolini 1 dan farqli barcha toq sonlar ustida birlashtiradi, kroneker - Yakobi simvoli esa barcha butun sonlarda Yakobi simvolini umumlashtiradi.

Yakobi simoli: Agar P - toq son va 1 dan farqli bolsa, va $P = p_1 p_2 \dots p_n$ tub koeffitsientlarga ajralsa ($p_1, p_2, \dots, p_n$ lar ichida ozaro teng sonlar bolishi mumkin), u holda ixtiyoriy a butun son uchun quyidagi Yakobi simboli orqali boladi:

$$\left(\frac{a}{P}\right) = \left(\frac{a}{p_1}\right)\cdot\left(\frac{a}{p_2}\right)\cdot\dots\cdot\left(\frac{a}{p_n}\right)$$

$\left(\frac{a}{p_i}\right)$ - Lejandr simvoli

bu yerda agar uchun $\left(\frac{a}{1}\right) = 1$ orinli.

Xususiyatlari:

Multiplikativlik: $\left(\frac{ab}{P}\right) = \left(\frac{a}{P}\right) \cdot \left(\frac{b}{P}\right)$

- Xususiy holda: $\left(\frac{a^2 b}{P}\right) = \left(\frac{b}{P}\right)$

Davriylik: Agar $a \equiv b \pmod P$ bolsa, u holda

$$\left[\left(\frac{a}{P}\right) = \left(\frac{b}{P}\right)\right]$$

- $\left(\frac{1}{P}\right) = 1$
- $\left(\frac{-1}{P}\right) = (-1)^{\frac{P-1}{2}}$
- $\left(\frac{2}{P}\right) = (-1)^{\frac{P^2-1}{8}}$
- Agar Q - toq natural son va P bilan ozaro tub bolsa;

$$\left(\frac{Q}{P}\right) \cdot \left(\frac{P}{Q}\right) = (-1)^{\frac{P-1}{2} \cdot \frac{Q-1}{2}}$$

- Xususiy holda Q va P ozaro tub va toq sonlar bolsa:

$$\left(\frac{Q}{P}\right) = (-1)^{\frac{P-1}{2} \cdot \frac{Q-1}{2}} \cdot \left(\frac{P}{Q}\right)$$